

UNIVERSITY OF CALIFORNIA

Santa Barbara

Development of Polymeric Coatings for Antifouling Applications

A dissertation submitted in partial satisfaction of the
requirements for the degree Doctor of Philosophy
in Chemical Engineering

by

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ACKNOWLEDGEMENTS

I would like to dedicate this dissertation to my late advisor Professor Ed J. Kramer. His passion for science and dedication to mentorship was the greatest inspiration I could have ever hoped for. I thank my advisors Prof. Glenn Fredrickson and Prof. Craig Hawker, as well as the members of the Fredrickson, Hawker and Kramer research groups for all their help. The collaborative and supportive nature of these groups and the UCSB community has created for me a unique and stimulating environment. The MRL staff has been incredibly helpful in all aspects of my time as a graduate student at UCSB, and I am extremely grateful for that. I also would like to acknowledge my loving family, Eric, Trini and Elena; their unconditional support helped me through this journey.

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ABSTRACT

Development of Polymeric Coatings for Antifouling Applications

by

Edward Philip Toumayan

Fouling, or the deposition of unwanted material onto a surface, is a serious problem that can impair the function of submerged structures, such as marine-going vessels and underwater equipment. Water filtration membranes are particularly susceptible to fouling due to their microstructure and high water pressure operating conditions. For this reason, there has been considerable interest in developing fouling-resistant, or “antifouling” coatings for membranes, specifically coatings that mitigate fouling propensity while maintain high water flux. Polymer coatings have garnered significant interest in antifouling literature, due to their synthetic versatility and variety, and their promising resistance to a wide range of foulants. However, antifouling research has yet to establish a consistent framework for polymer coating synthesis and fouling evaluation, making it difficult or impossible to compare previously established methodologies. To this end, this work establishes a standardized methodology for synthesizing and evaluating polymer antifouling coatings. Specifically, antifouling coatings are synthesized using a grafting-from polymerization and fouling propensity is evaluated by quartz crystal microbalance with dissipation (QCM-D). Using this framework, a number of different surface functionalization strategies are compared, including grafting-to and grafting-from polymerization.

A number of different surface functionalization strategies, including grafting-to and grafting-from, were investigated and the fouling performance of these films was evaluated. Primarily, sulfobetaine methacrylate, and poly(ethylene oxide) methacrylate monomers were investigated, among others. Grafting-to, while advantageous from a characterization standpoint, was ultimately limited to low grafting densities, which did not afford a significant improvement in fouling resistance. However, the higher grafting densities achievable by grafting-from did indicate improved fouling resistance. A grafting-from strategy based on polydopamine (PDA)-initiated surface polymerization was developed as a facile method capable of deposition on various substrates. In addition, the PDA method demonstrated polymerization of various monomer classes, enabling versatility in surface chemistry modification in a reproducible fashion. QCM-D then enabled the deposition of arbitrarily selected foulants onto these polymer-modified substrates. Protein fouling was employed as the basis for evaluating antifouling performance, though other foulants were also demonstrated. Improved resistance to protein fouling was corroborated by QCM-D experiments for hydrophilic and zwitterionic polymer coatings including poly(ethylene oxide) and poly(sulfobetaine methacrylate). The combination of PDA-initiated polymer films and QCM-D based fouling experiments provides a versatile and reproducible platform for antifouling research.

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LIST OF ACRONYMS

Acronym	Definition
ACVA	4,4'-azobis(4-cyanovaleric acid)
AFM	atomic force microscopy
AIBN	azobisisobutyronitrile
BSA	bovine serum albumin
DEGE	diisopropyl ethanolamine glycidyl ether
DLS	dynamic light scattering
dopa	dopamine
FITC-BSA	fluorescein isothiocyanate conjugate bovine serum albumin
hopo	2-methyl-3-hydroxy-4-pyridinone ethylamine
LCST	lower critical solution temperature
MMA	methyl methacrylate
MPC	2-Methacryloyloxyethyl phosphorylcholine
ndopa	6-nitrodopamine
NIPAM	N-isopropylacrylamide
PDA	polydopamine
PEG	poly(ethylene glycol), also referred to as PEO
PEGMA	poly(ethylene glycol) methacrylate
PEO	poly(ethylene oxide)
P(S)	poly(styrene)
RO	reverse osmosis
SEM	scanning electron microscopy
SBMA	sulfobetaine methacrylate, also [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide
SVA	succinimidyl valerate
TEA	triethylamine
TEM	transmission electron microscopy
XRR	x-ray reflectivity

I. Introduction

A. Fouling and its Impact

The production and distribution of clean, potable water to a steadily growing global population is an increasing difficult challenge. This is especially difficult in remote locations, where access to clean water sources may be limited or non-existent. In addition to not having access to clean water, many times these locations do not even have access to the electrical grid. As a result, there is considerable interest in developing cheap and simple filtration solutions that can be implemented in locations where access to clean water is scarce. Lower water filtration costs are also beneficial for heavily populated locations where clean water is consumed liberally. The state of California serves as a perfect example of this. Afflicted by periodically occurring droughts, but by far the country's leading consumer of water resources, California is experiencing a dire water crisis.¹ In order to cope with ever scarcer water resources California is turning to many desalination solutions, such as the desalination plants opened in 2015 in Carlsbad and Santa Barbara, CA. One of the most common and cost-effective desalination methods is by reverse osmosis (RO) filtration. RO relies on filtering an influent water stream through a semi-permeable membrane, resulting in a desalinated effluent water stream.

One major concern for filtration equipment, and indeed most submerged and underwater equipment, is the potential for fouling. Fouling is the adsorption of unwanted species onto a surface that can lead to damage or failure. Detrimental fouling of submerged structures can therefore contribute considerably to operating and maintenance costs. A prime example is the impact of fouling on the commercial shipping industry. Fouling of the hulls of shipping

vessels contributes an estimated \$30 billion in fuel costs.² Membranes used for water filtration applications are particularly susceptible to fouling as they are operated using large pressure gradients to drive water past the selective membrane. The porous network of these membranes can become impeded, leading to lower flux across the membrane. Specifically, the costs related to the reduction of flux in membranes are estimated at over \$1 billion per year.² Consequently, there have been many efforts to minimize the fouling propensity of susceptible surfaces.

B. Antifouling

For most applications where surfaces are in contact with water for extended periods of time fouling is inevitable. This reality necessitates periodic cleaning and treatment of afflicted surfaces, leading to costly down-time and maintenance. However, the use of so-called antifouling coatings offers the ability to decrease the frequency of these cleaning cycles, and potentially lengthen the effective lifetime of the equipment. In the 1960s and 1970s tributyltin (TBT) was employed as a toxic, but effective solution to fouling on the underside of marine vessels.³⁻⁵ The extremely detrimental effects that TBT had on the environment ultimately led to its complete ban by 2003.³ This created a strong impetus for the discovery and development of environmentally benign alternatives for antifouling. Most notably, the use of organic, polymeric coatings emerged as the most common approach in the antifouling community. With a plethora of methods for synthesis and functionalization, polymer coatings offer a huge range of chemical and material diversity, allowing for customization of desired properties for different antifouling applications. Specifically, polymer properties can be manipulated by changing the chemical compositions and by controlling the structure of the polymer itself.

In terms of polymer chemistry, coatings with a hydrophilic character have been well documented as exhibiting fouling resistance. The strong affinity of these coatings for water creates a hydrated polymer layer, with highly structured water molecules in proximity to the polymers.⁶ Resistance to the deposition of foulants is derived from this hydration layer, and the extended conformation of the polymers.^{7–10} Within this class of hydrophilic materials, zwitterionic polymers have also shown particular success in fouling resistance, attributed to their very hydrophilic character.

Another common way to manipulate and impart fouling resistance is by controlling the structure of polymers. Specifically the conformation that surface-tethered polymers take on plays a pivotal role in the ultimate surface properties and fouling resistance. Polymer brushes (polymers that are covalently attached to surfaces) can span a range of grafting densities; a metric for the number of polymers per unit area. Generally, low grafting densities results in collapsed polymers that offer low fouling resistance, while high grafting densities yield extended polymer conformations that aid in repelling foulants. The grafting density is largely dictated by the method of surface deposition, notably, grafting-to and grafting-from, where polymers are either tethered-to or grown-from a suitable surface, respectively.

C. Studying Fouling Resistance Using Polymers

Various methods of polymer synthesis, attachment and modification are explored in the context of creating surfaces relevant for antifouling. The objective is to explore and understand how surface properties play a role in fouling resistance. Polymer coatings provide an ideal platform from which to generate structure-property relationship due to their versatility in synthesis and conformational manipulation. Herein, several deposition

methods, including grafting-to and grafting-from, are explored and the resultant polymer films are evaluated for fouling resistance. These polymer-functionalized surfaces are evaluated for their fouling resistance via protein adsorption studies. Their performance is then correlated to surface properties such as film thickness, composition, water contact angle, among others.

II. Background

A. Polymer mechanics

Polymers are long chains of molecules composed of smaller repeating units, or monomers. Typically these monomers are organic compounds that are linked end-to-end. For these linked molecules to be considered polymers, they typically have more than 20-30 repeat units, and otherwise are considered oligomers. Polymers can have hundreds, thousands, and in some cases, millions of repeating units per chain. These macromolecules exhibit unique properties in their liquid and solid state.

The simplest approach to understanding the behavior of polymer fluids is through the use of ideal chain models. Polymers can be considered as highly flexible chains of monomer units that are free to coil up when in liquid form. In order to describe the coiled conformation of a polymer chain, a coarse-grained model can be employed, where the individual monomer units are represented as beads connected along a flexible backbone, commonly referred to as the “bead-spring” model. The interactions between individual beads can be described by bonded and non-bonded potentials that ultimately dictate the spatial configuration of the chain. Figure 1 below is a depiction of a polymer represented by a coarse-grained, “bead-spring” model.

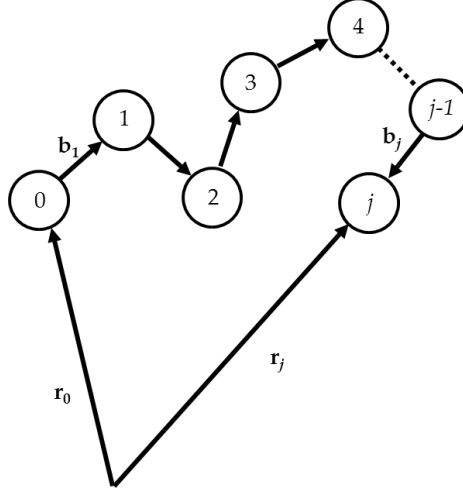


Figure 1: Bead-spring model for a polymer chain. The individual beads are connected by the vectors b_j while the positions of the beads are defined in space by the position vector r_j from some arbitrary point in space.

The ideal chain model can be used to approximate the average size of the polymer based on statistical mechanics. In particular, the size may be written as $R = \sqrt{\langle \mathbf{R} \cdot \mathbf{R} \rangle}$ where $\mathbf{R}$ is the end-to-end distance as depicted in Figure 1. The $\langle \dots \rangle$ operator implies an average computed over all possible conformations of beads. Using this simplified model an expression for the size of the folded polymer and how it relates to the number of repeat units can be derived.

Ultimately, the polymer coil size follows a universal scaling relationship which can be written as:¹¹

$$R \sim bN^\nu, \quad N \rightarrow \infty \quad (1)$$

In this case, R is the coil size in nm, b is the Kuhn length in nm, N is the degree of polymerization and ν is the scaling exponent. For the case of ideal chain statistics, $\nu=1/2$ regardless of the chemical details of the polymer. Ideal chain statistics occur for polymers

in the melt (free of solvent) or for polymers under “theta solvent” conditions. Under these conditions, the polymer chain takes on a self-avoiding, random-walk conformation.

The scaling exponent can take on different values depending on the interactions of the polymer with the surrounding solvent. The “universality” is manifested in the fact that dependence of N on the size of the polymer is completely independent of the chemical composition of the polymer and is only a function of the solvent “quality.” The Kuhn length is the term that accounts for the chemical structure of the polymer and its interaction with the solvent. For example, polymer coils in a good solvent are swollen compared to the melt, and in this case $\nu=3/5$ (or $\nu=0.588$ in three dimensions). Conversely, if the polymer is in a bad solvent the coils will contract and occupy less space in order to minimize interactions with the solvent, resulting in $\nu=1/3$. Of course, these relationships describe the scaling behavior in the limit as N approaches infinity.

The polymer can also be described by its radius of gyration, R_g , which is computed as the root mean square of the positions of the individual repeat units in the polymer. It also scales linearly with the coil size, R , in Equation (1) above. There are several expressions for R_g , which differ only by the assumptions made when approximating the polymer conformation in solution. For the purposes of this report, R_g was computed using Equation (2) below.

$$R_g = \left(\frac{Na \cos(37.5)}{6} \right)^{0.5} \quad (2)$$

Another important concept relevant to polymer mechanics is the concept of excluded volume. Each monomer occupies some space, which is the basis for coarse-graining models

for polymer chains. The polymer coils up in solution as a result of interactions between the solvent and the chain, but there are additional interactions between chains. The chains generally do not want to occupy the same space as other polymers in solution and this is usually described as an excluded volume interaction. The excluded volume interaction depends on a few factors, including the solvent quality and the polymer concentration. For polymers in a good solvent the excluded volume interactions are stronger, as the polymer prefers the solvent. The opposite is true for polymers in a poor solvent. Similarly, at low concentration of polymers in solution, the excluded volume interactions are more prominent, while they get suppressed as the concentration increases.

Polymers can be synthesized using a variety of techniques, which will not be discussed in the interest of brevity. It is important to note however that these different techniques can be used in clever ways to construct polymers with a wide range of chemical composition and “architecture.” In this context, “architecture” refers to the branched nature of the polymer backbone. For example, many polymers are simply linear in nature, meaning that all monomers in the macromolecule are arranged end-to-end in a linear fashion. Some polymers, however, can be branched, resulting in more complex structures. Branching may occur naturally as a consequence of the polymerization method being used, resulting in suppressed crystallinity of the resulting material. Other polymers may be intentionally synthesized to have multiple long braches or “arms,” also referred to as mikto-arm structures¹².

In addition to variations in the polymer architecture, one can also imagine the incorporation of multiple chemical functionalities within the same macromolecule; also affecting the polymer behavior. While these are all intriguing variations on synthetic

polymers, they can add confounding variables which are beyond the scope of the present work and will not be discussed in depth herein. Instead this dissertation focuses on linear polymers composed primarily of a single monomer.

B. Polymer brushes

Polymer brushes are polymer chains that are physically constrained at one end. Common examples are chains tethered to a surface, or block copolymers ordered across an interface. In the interest of brevity, the following discussion of polymer brushes is limited to polymers chains that are end-tethered on a solid substrate. The simple fact of being constrained at one end has a profound impact on the conformation of the polymer chain.¹³ As discussed previously, polymer chains prefer to take on a globular form of radius R_g for entropic arguments. The ultimate conformation of brushes is therefore a result of competing forces: steric repulsions from neighboring polymers and the substrate, which cause the polymer to extend, and an entropic force which drives the polymer to coil up. Typically end-tethered polymers are characterized by their grafting density, which is a measure of the number to chains per unit surface area. At low grafting densities the polymer chains are isolated from one another on the surface of the substrate. Consequently, the polymers occupy a hemispherical shape on the surface, commonly referred to as a “mushroom” conformation. At high grafting densities the chains are much too closely packed to occupy a hemispherical space. Indeed, the excluded volume interactions between neighboring polymers causes the chains to extended away from the surface. Intermediate grafting densities will exhibit a conformation somewhere between these two extremes as depicted in Figure 2 below.

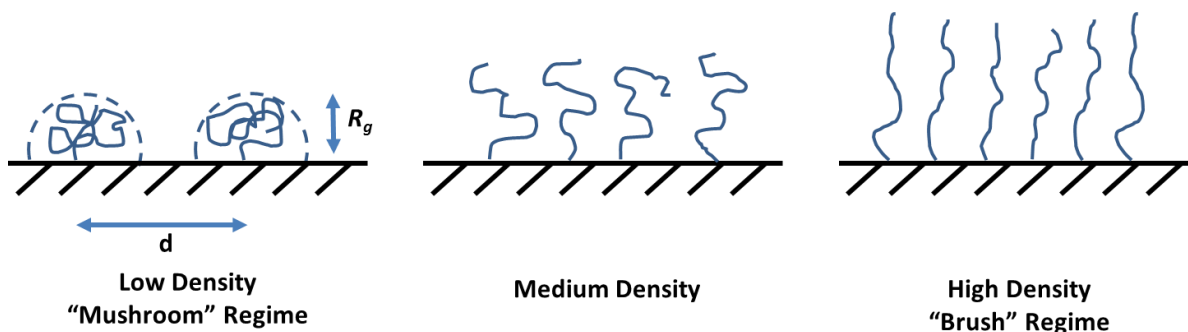


Figure 2: Polymer brushes at different grafting densities. At low grafting densities the polymer chains essentially do not interact on the surface and are considered in the “mushroom” regime. At higher densities the polymers will be forced to extend away from the surface as adjacent chains repel due to excluded volume interactions. The extended conformation is referred to as the “brush” regime. Medium densities take on an intermediate conformation.

The threshold between mushroom and brush occurs when the distance between grafting sites on the surface, d , is less than the radius of gyration, R_g . This marks a change in the polymer conformation as it must extend into the bulk solution above the surface. The repulsive forces between polymers that arise due to excluded volume interactions overcomes the loss of entropy associated with the extension of the polymer chain.

Surface-immobilized polymers can be made through various techniques that can be generally grouped into two categories: *grafting-to*, and *grafting-from*.¹³ Each approach to creating a polymer brush is slightly different and they are illustrated in **Figure 3** below. Broadly, grafting-to entails attaching previously synthesized polymers to a surface while grafting-from relies on growing polymers from initiation sites on a functionalized surface.

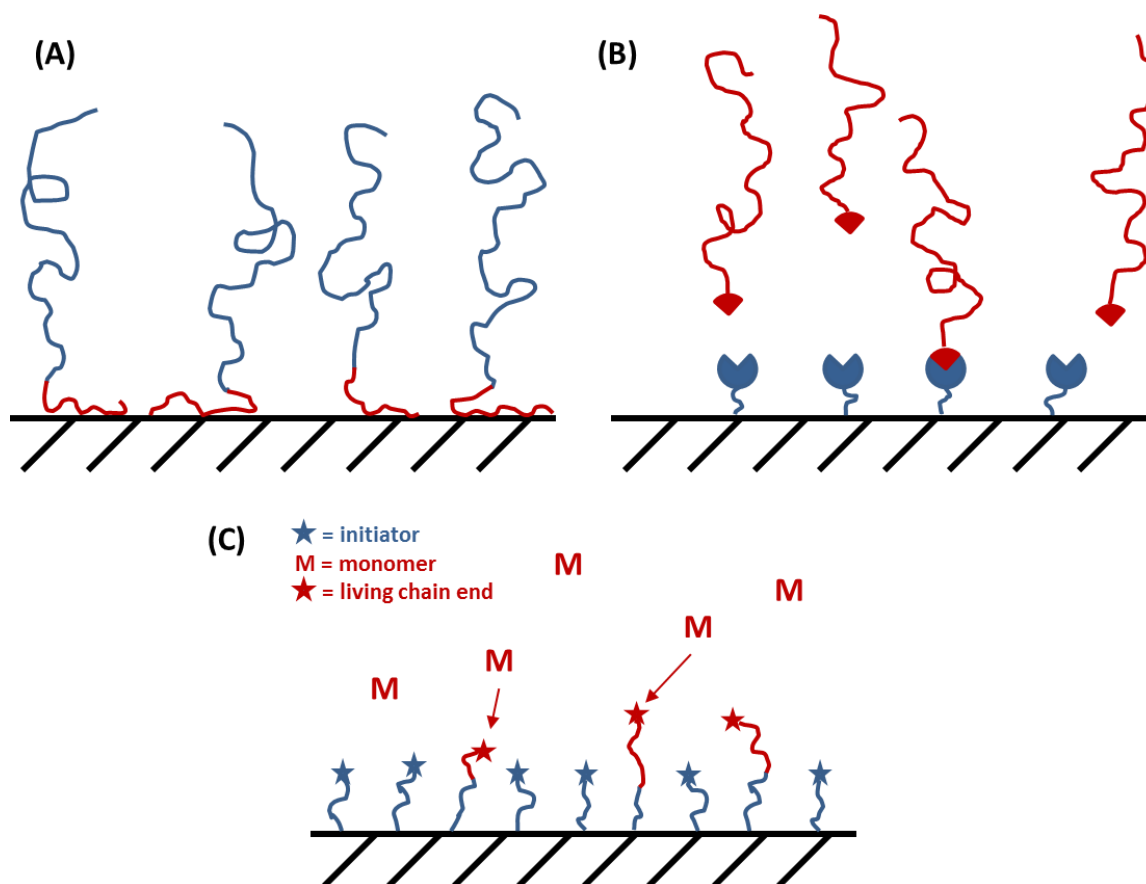


Figure 3: Different methods for creating surface-bound polymer brushes. The top two images depict polymer brushes made by grafting-to. In left depiction, a block copolymer is adsorbed through favorable interactions between the substrate and one of the blocks (A). The depiction on the right shows polymers with a reactive terminus which is coupled to a surface chemically modified with a complementary moiety (B). The bottom image depicts grafting-from in which polymer chains are grown from an initiator-functionalized surface (C).

As depicted above, grafting-to techniques tether previously synthesized polymers to a surface using a chemical linkage. In this approach the polymers either have a chain terminus or a portion of polymer chain (such as block copolymers, as in Figure 3A) that will ultimately attach to the surface. This can be done equally by physisorption or covalent

attachment. Physisorption refers to molecules that are physically adsorbed onto a surface by means of non-covalent binding. This might be due to electrostatic attraction or hydrophobic interactions between the substrate and one end of the polymer. In some cases block copolymers are designed in such a way that one block prefers the surface while the other block is swollen by the surrounding solvent. With covalent attachment the only distinction lies in the mechanism and nature of the chemical bond between surface and polymer. In this case polymers are end-functionalized with a chemical group that can be reacted with the surface, or undergoes specific binding with moieties present at the surface. The distinction being that a covalent chemical linkage is generated.

In both cases, the polymers are synthesized prior to adsorption onto the surface, which enables very good control over the polydispersity of the polymers used. In addition, grafting-to allows for the use of a wide range of polymer chemistries and can easily be adapted for different surfaces. However, this approach suffers from low grafting density due to the nature of the deposition process. The key for this technique is the selection of the adhesive group. Commonly, chemical tethers that bind strongly to specific surface groups are chosen. Examples of chemistry pairings used for grafting-to include: thiol-gold, thiol-ene, thiol-yne, amine-epoxide, amine-acid, biotin-streptavidin, silane-silanol, and catechol-metal among others. Polymers and surface are functionalized with one of the two complimentary moieties, and the surface coupling reaction results in a grafted-to polymer brush.¹⁴⁻²²

In this configuration, achieving high grafting densities is difficult due to excluded volume interactions that effectively block the approach of additional polymers beyond some critical value. The grafting density that can be obtained is therefore dependent on the

solvent quality, the polymer concentration during deposition, and the attachment mechanism to the surface. Despite these limitations, grafting-to it is still a popular technique for creating polymer brushes. In fact, there have been some recent experimental efforts that demonstrate the ability to achieve higher density brushes using a grafting-to approach.^{14,23} These approaches involve depositing polymers from a higher concentration solution in order to suppress excluded volume interactions between polymer chains and reach higher densities.

Contrastingly, grafting-from techniques rely on growing polymer chains directly from a previously functionalized surface. In doing so, the excluded volume considerations that would otherwise limit grafting density are circumvented. The most common approach is to activate a living radical polymerization from the surface. In order to do this, discrete initiation sites must be present on the surface from which the polymerization can be activated. This approach provides good control over grafting density, but can result in a higher polydispersity. It is also limited to monomers that are amenable to the compatible polymerization strategies. Some examples of techniques used for grafting-from include: Reversible Addition–Fragmentation chain Transfer (RAFT), Atom transfer radical polymerization (ATRP), polydopamine (PDA)-initiated polymerization, and benzophenone initiated polymerization.^{17,24–29} ATRP is perhaps the most common method used in these so-called surface-initiated polymerizations. In ATRP the initiator is typically an alkyl halide with the analogous copper halide as the catalyst. Consequently, the surface must be functionalized with alkyl halides in order for the polymer brush to grow. Typically, the ultimate grafting density is determined not by the polymers, but by the initial density of initiators on the surface. This approach is more complex, making the monitoring and

characterizing of the polymerization quite difficult. There have been some reports demonstrating the ability to “degraft” polymers from a surface to characterize the molecular weight, and grafting density of grafting-from films.^{30,31} However, this approach requires uniform polymer coating over a large substrate area in order to acquire enough material for analysis, and can be tedious to execute. In general, this approach is not practical as a method of screening grafting densities for a large sample size.

C. Polymer Brush Synthesis

There are several strategies used for polymerization that are discussed within this dissertation, each of them presented in detail in the relevant sections. Grafting-to and grafting-from strategies were both used for different applications.

Grafting-to was conducted using primarily polyethers, as they provided a simple synthetic route for chain-end functionalization. Specifically, the hydroxyl terminus of polyethers was converted to an epoxide, and later conjugated to surface-tethered amines in order to generate polymer brushes. Moreover, all the polyethers synthesized for grafting-to were made by anionic polymerization. Anionic is a polymerization technique in which the propagating chain end is an anion that reacts with the monomers to grow the chain end. Specifically, ring-opening anionic polymerization can be initiated with an oxygen anion performing nucleophilic attack of the epoxide rings present in the monomers to make polyethers. Of particular interest for this project is the synthesis of polyethylene oxide (PEO), poly(diisopropyl ethanolamine glycidyl ether) [P(DEGE)], and copolymers of the two together. The polymerization of EO and DEGE monomers can be initiated using primary or secondary alcohols. The reactions need to be done in a dry and oxygen-free environment, so they are typically carried out in specifically designed, air-free reaction

vessels. A schlenk line, maintained at high-vacuum with access to argon, is used to keep reaction conditions oxygen-free.

All the chemicals required were purchased from Sigma Aldrich. Benzyl alcohol is a primary alcohol used as the initiator for anionic polymerization. Potassium naphthalenide is made from naphthalene and potassium metal and is used as a strong, non-nucleophilic base to deprotonate the initiator ethylene oxide (EO) and diisopropyl ethanolamine glycidyl ether (DEGE) are used as monomers. Methanol is introduced after polymerization as a source of protons to terminate the chain ends. The chain ends can also be terminated with other functional groups to introduce different termini, and will be discussed in future directions. Tetrahydrofuran (THF) is used as a solvent when necessary, and hexanes or diethyl ether are used as a precipitation medium.

The glassware is cleaned thoroughly and allowed to dry in an oven for at least an hour prior to use. It is then removed from the oven and allowed to cool slightly before sealing the ports on the reaction flask with threaded caps. One of the caps has a septa-sealed injection port, which allows for the introduction of the various reactants, and another port is attached to the schlenk line. The dry flask is purged with argon and pumped down three times. A propane torch is used to flame-dry the exterior of the empty flask to expedite the evacuation of the vessel under vacuum. Once the empty flask has reached the lowest possible pressure (~20 mbar) after three purging cycles, the initiator is added along with solvent if the reaction is to be carried out in solution. Typically, polymerizations with only DEGE as the monomer can be done in the melt, free of solvent, but if EO is to be used at all, a few mL of THF are added. Then the potassium naphthalenide is added drop-wise while stirring until a green color persists in the mixture. The green color suggests that the strong base has been added

in sufficient excess to completely deprotonate all the initiator. At this point the monomers are introduced slowly to the reaction flask. If a copolymer is desired, both EO and DEGE can be added in the ratio desired for the final polymer. The reaction is typically conducted either at room temperature or at 40 °C using a warm water bath. Depending on the desired molecular weight, the temperature, and the amount of initiator added, the reaction may be conducted for anywhere between 24 hours and several days. After the reaction has finished, methanol is added in order to protonate the chain ends and terminate the polymerization. The product mixture is then poured slowly into a large volume of hexanes to promote the precipitation of the polymer out of solution. The precipitate was then dried carefully, first on a rotovap, and then on high-vacuum to ensure removal of all volatile components. The overall reaction pathway is shown below.

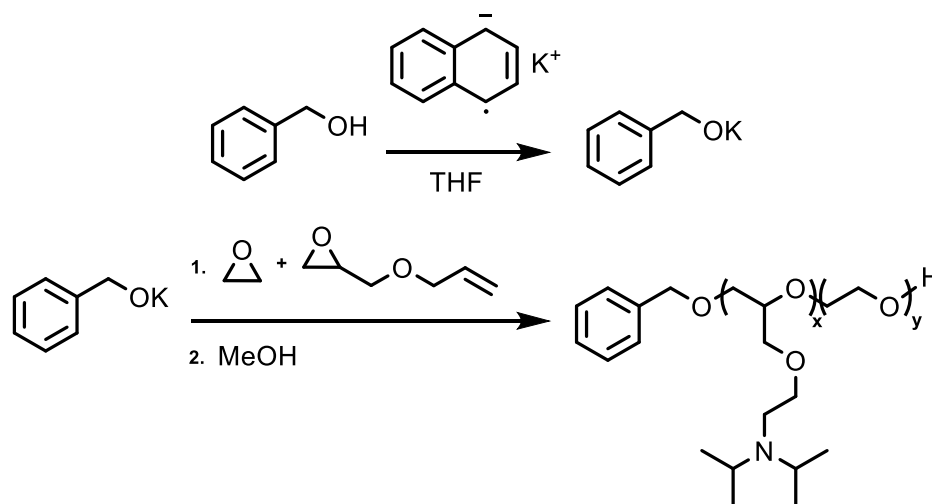


Figure 4: Anionic polymerization of EO and DEGE, using benzyl alcohol as an initiator. Potassium naphthalenide is used to deprotonate the initiator in THF, and then the monomers are introduced in their desired ratio. Polymerization is terminated by introducing methanol in excess.

Anionic polymerization affords good control over the molecular weight and polydispersity. In particular, nuclear magnetic resonance (NMR) and gel permeation chromatography (GPC) are used to characterize the resulting polymer. The molecular weight was computed using the integrations of the NMR peaks corresponding to each repeat unit in the product. The polydispersity index (PDI) was computed from the GPC trace using the formula $PDI = M_w/M_n$, where M_w and M_n are the weight-average and the number-average molecular weights, respectively.

Moreover, the hydroxyl terminus of the polyethers synthesized by anionic polymerization can later be converted to reactive groups useful for surface deposition. For example, both epoxide and catechol groups are employed as surface-tethering moieties in the present work. The synthesis of these specific molecules is described in more detail in the chapters to follow.

In addition, grafting-from methods were also used to create polymer brushes. Specifically, both controlled and uncontrolled radical polymerization strategies were employed in order to compare their efficacies in creating reproducible polymer brushes. Controlled radical polymerization was conducted in the form of surface-initiated atom transfer radical polymerization (SI-ATRP). Briefly, alkyl bromide initiator species were functionalized on silicon wafers using silane chemistry. These initiator-coated substrates were then submitted to ATRP conditions using Cu catalyst, a suitable ligand, and a monomer. The resulting polymer films exhibit predictable polymerization kinetics, making the tuning of polymer film thickness straightforward. However, SI-ATRP requires air- and moisture-free conditions, requiring rigorous and meticulous reaction setup. This hinders the throughput of surfaces that can realistically be synthesized in this fashion. For this reason

free-radical polymerizations were investigated, because despite being inherently uncontrolled, they can be conducted under less rigorous conditions enabling higher throughput. These surfaces were grown using polydopamine (PDA) as an initiation layer. This method affords a number of beneficial features. Primarily, the PDA layer is capable of coating a variety of surface chemistries using a simple and scalable deposition procedure. Briefly, PDA is deposited from a buffered aqueous solution of pH 8.5, using UV irradiation to expedite the process. Subsequently, the PDA-coated substrate can be further irradiated in the presence of a monomer to generate the desired surface-grown polymer.

D. Fouling Types and Considerations

Fouling is the detrimental deposition of unwanted material, or foulants, onto a surface. This process occurs under normal operation in a variety of applications. Most notably, applications where surfaces are in contact with water for extended periods of time are particularly susceptible to fouling. The application and environment play a critical role in the classes of foulants that a surface is exposed to, each with different mechanisms of deposition; each requiring different anti-fouling strategies.

In particular, ship hulls, water filtration systems, and other subaquatic structures are greatly affected by the deposition of marine foulants. Shipping vessels for example, are plagued by the buildup of biological and inorganic materials their hulls, increasing drag in the water, leading to higher fuel consumption. In extreme cases, moving parts on the underside of ships can become impeded by foulants and cause damage to critical components. In 2008, additional fuel costs associated with biofouling on ships were estimated at \$30 billion per annum for the commercial shipping industry.³² Other

underwater equipment can be affected in much the same way, resulting in higher operating costs and/or irreversible detrimental effects.

The impact of fouling on water filtration systems in particular can be dramatic. Water filtration membranes are composed of an interconnected network of pores that are designed to reject unwanted solutes while allowing water to diffuse through its pores. Typically these membranes are made of polymeric or inorganic materials that are processed to form the desired pore sizes. A pressure gradient is applied across the membrane to drive water molecules through the membrane pores, while the unwanted solutes are rejected by the selective membrane. The pore sizes define which solutes are rejected, as illustrated graphically below.

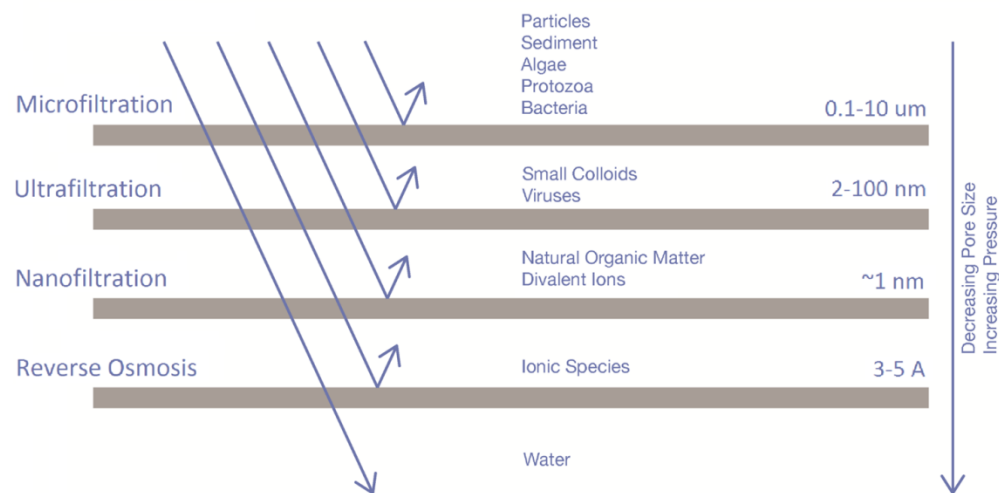


Figure 5: Pore sizes for filtration membranes. As the pore sizes get smaller, the pressure required for filtration increases.

The porous network of the membrane is susceptible to fouling, as dissolved foulants in the water may accumulate on the surface and block the pores. Pore blockage leads to lower flux across the membrane, and leads to higher pressures in order to maintain filtration. Specifically, the costs related to the reduction of flux in membranes are estimated at over \$1

billion per year.² Over time, foulant accumulation decreases the flux to the point where the membrane can no longer operate without thorough cleaning. This cleaning procedure requires the use of harsh chemical treatments that decrease the effective lifetime of the membrane. Moreover, cleaning routines can imply long downtimes for the membrane modules.

Membranes with different pore sizes, ranging from microfiltration to reverse osmosis, are implemented for different applications. In addition, there are many different membrane types that are employed for various applications, each composed of a unique materials and structure.³³ These different membrane types are themselves subjected to a wide range of foulants. The behavior of foulants can vary widely; therefore it is difficult to design a single solution to deter all of the potential foulant species from all membrane classes. There are several types of fouling, such as: scaling,³⁴ protein fouling,^{35,36} humic acid fouling,^{20,37,38} surfactant fouling,^{39,40} microbial/bacterial fouling,^{41,42} and other organic/inorganic fouling,^{43–45} which can occur either independently or in combination.

As a result, there have been numerous academic and industrial efforts over the last forty to fifty years examining “antifouling” surfaces and their integration into systems that have a propensity for fouling.^{46,47} A general overview of the antifouling literature is presented below.

E. Antifouling Literature

There are many kinds of foulants; with different foulants being more prevalent in certain applications. For subaquatic structures, the growth of biofilms and the accumulation of marine organisms are the most prevalent form of fouling. These organisms excrete an extracellular protein that first forms a biofilm, and then allows for further colonization of the

surface. This is a fundamentally different mechanism than inorganic foulants. Mineral fouling, or scaling, occurs when dissolved minerals, such as those found in groundwater, build up on surfaces. This fouling can significantly hinder the equipment in contact with the water. As a result, different strategies are being developed to deter the various types of foulants. One metric that is commonly used to gauge antifouling properties is to measure the protein resistance of a surface.^{17,36,48,49} This is a good first approximation at resistance to biofouling as many organisms excrete extracellular proteins substances (EPS) as a precursor to attachment.⁴² Therefore, generating resistance to protein fouling is an effective method for resisting more complex foulants.

In the 1960s and 1970s this most common solution to resist fouling on the hulls of ships was the application of tributyltin (TBT) containing paints.³⁻⁵ While these coating were quite effective they also had extremely detrimental effects on the environment and were completely banned by 2003.³ As a result the antifouling field shifted focus to environmentally benign alternatives. Copper emerged as an early contender to replace the tin-containing coatings, although the use of metallic additives still posed some fundamental problems.⁵ More recently the use of organic, polymeric coatings has taken the spotlight in the antifouling community. With a variety of different methods for synthesis and functionalization, polymer coatings offer a huge range of chemical and material diversity, allowing for modification and customization for different antifouling conditions. They also provide a convenient platform to investigate the way in which surface properties affect the fouling behavior of interfaces.

There are a number of polymer chemistries that have been identified and investigated with regards to their promising antifouling character. Broadly there are three main

categories of fouling resistant materials that have garnered considerable interest in the antifouling literature: hydrophilic, fluorinated and zwitterionic materials. While there are other classes of materials used for antifouling, these three categories are generally accepted as the most promising.^{50,51}

Hydrophilic polymers have long been the champions of organic antifouling materials. The prevailing explanation for their efficacy is their tendency to remain swollen (by water) in an aqueous environment. For foulants to deposit onto a surface coated by hydrophilic polymer would mean compressing the hydrated polymer, resulting in loss of entropy, and additional enthalpic work to expel the water molecules swelling the film. This combination of properties effectively repels many hydrophobic foulants that may be in suspension. Hydrophilic monomers come in a variety of flavors, but there are several motifs that are used prevalently for antifouling applications. Perhaps the most common antifouling polymer is polyethylene oxide or polyethylene glycol (PEO or PEG).^{14,28,52–57} The polyether structure provides hydrogen-bond accepting capability throughout the backbone, imparting it with a high affinity for water. Other hydrophilic monomers include amine-containing monomers such as: 2-dimethylaminoethyl methacrylate (DMAEMA), 2-diethylaminoethyl methacrylate (DEAEMA), 2-(Diisopropylamino)ethyl methacrylate (DIPAEMA), N-isopropyl acrylamide (NIPAM), and N-[2-(dimethylamino)ethyl]acrylamide (DMAEAM), to name a few. These secondary and tertiary amines are typically attractive due to their typical hydrophilicity combined with their pH and temperature responsive nature. The ability to manipulate the hydrophilicity and swelling behavior using external stimulus provides key advantages for fouling resistance. Specifically, that behavior has been shown to be effective for foulant release, whereby

foulants deposited atop these materials are driven off the surface through the switching of the polymer properties.^{4,58–60} The pH and temperature response of these materials affords two potential handles by which to impart this effect.

While the trend of hydrophilic materials being the most effective for fouling resistance holds true for most materials, one anomaly to that theory is fluorinated hydrocarbon monomers. Fluorinated monomers are by and large hydrophobic in nature, however, they exhibit very low surface energy. That is to say that fluorinated monomers prefer to segregate to the interface, making the adhesion of foulant to the surface energetically unfavorable.^{61–65} In addition, many fluorinated compounds can form crystalline-like microstructures at surfaces, further adding to their fouling resistance.^{61,66} Many fluorinated and perfluorinated monomers have been explored in the context of antifouling, and have shown promising performance in resisting biological foulants.^{52,62,64,66}

The final class of antifouling materials discussed herein are zwitterionic species. Zwitterionic materials themselves are also generally hydrophilic in nature, however, their polyampholytic nature puts them in a class of their own. Polyelectrolytes have previously been used a great deal in antifouling research, however the net charge (positive or negative) in these materials typically results in detrimental attraction of oppositely charged foulants. In order to achieve charge balance and keep oppositely charged groups in proximity, zwitterions are used. A number of monomers are common within this class of materials: sulfobetaine methacrylate (SBMA), 2-methacryloyloxyethyl phosphorylcholine (MPC), carboxybetaine methacrylate (CBMA), and others, following a similar motif.⁶⁷ The ionic nature of these materials renders them even more hydrophilic than the previously discussed hydrophilic polymers, and therefore imparts them increased fouling resistance.^{68–70}

Specifically, their excellent protein resistance has been demonstrated for both monolayers of small molecules, and for polymer films.^{71–75}

In addition to the various polymer chemistries, another layer of complexity is introduced when considering the conformation and architecture of the polymers themselves. For each of the monomer classes described above, polymers of these can be synthesized as linear macromolecules or as more complex architectures. Some diverse architectures include: branched chains,^{76–79} mixed brushes,^{69,80} and multi-arm structures.^{48,76,78,81–83} These modifications in the polymer structure have shown promising improvement in fouling resistance.^{81,84} While there is still not a complete explanation for this improvement in fouling resistance, it is generally attributed to a more complete surface coverage that can be achieved with more complex architectures. However, this dependence on surface coverage is extremely dependent on the deposition method.

There has also been a considerable effort in improving fouling resistance through the implementation of surface roughness and surface patterning. Specifically, surfaces with spatial and chemical heterogeneities can be strategically designed such that they deter fouling. This effect is best illustrated by the success of Sharklet AFTM coatings. Sharklet relies on shark placoid-inspired micro-structures to create an ambiguous surface that resists fouling.^{50,51,60,85} There has been considerable evidence to show that surfaces that present roughness on the length scales commensurate with the size of the bacteria or microorganisms that approach the surface they effectively “confuse” the would-be foulants and resist adhesion.^{50,86} Similar to this topological roughness, fouling resistance can also be achieved with chemical heterogeneity on a surface.⁸⁷ For example, surfaces that alternate

hydrophilic and hydrophobic domains on length scales commensurate with the dimensions of the relevant foulants can effectively impede surface deposition.^{17,51,88,89}

All of these different factors have been used in different contexts to achieve varying levels of fouling resistance. If the previous sentence seems vague it's because the conclusions drawn from this vast field are themselves vague. Antifouling, as a field of study, is composed of two main components: the synthesis of surface coatings, and their subsequent fouling evaluation. The synthesis of surface coatings itself can be quite varied and complex, as described above, but the determination of fouling performance can be equally nuanced and challenging. In fact, one of the major limitations of the antifouling literature is the lack of a consistent metric for evaluating fouling resistance. The different methods include using surface plasmon resonance (SPR), QCM-D, nuclear magnetic resonance (NMR), fluorescent microscopy, high pressure liquid chromatography (HPLC), and other techniques to either directly or indirectly measure the amount of fouling on a surface.

In many cases antifouling surfaces are evaluated on membranes directly, and in this case most fouling evaluation is conducted by measuring flux and permeability under various fouling conditions.^{46,90–95} There isn't however a standard accepted method for evaluating the fouling performance, so most studies are not easily comparable across the field of antifouling. The following study aims to reduce the complexity of antifouling studies and focus on a small number of surfaces, and evaluate their performance with a single approach.

F. Our Approach

The difficulty with many of the current antifouling efforts in the literature is the variety of strategies used. Many groups use different polymers entirely, and they create antifouling

films in very different ways. This makes drawing conclusions that combine the findings in different studies very difficult. For this reason, it is useful to establish a consistent framework that can be used to investigate the different effects that have been reported in the literature. Specifically, a system that can compare the effect of polymer chemistry and polymer architecture independently will help elucidate many of the puzzling results in antifouling literature. Leveraging polymer synthesis and functionalization enables the development of a platform to investigate how surface chemistry and polymer structure can be manipulated to increase the fouling resistance of a surface. Specifically, carefully designed polymer surfaces can result in targeted chemistry and architecture of the resulting brush.

In order to make the challenge of studying antifouling materials more tractable only a few variables were selected and investigated. As mentioned above, zwitterions show the most promising fouling resistance based on current antifouling research, so these materials became one main thrust for this thesis. In addition to zwitterions, other materials were sought out to serve as controls and comparisons to these ampholytic polymers. For this purpose, polyethylene oxide (PEO), and styrene were used throughout these studies. A variety of different synthetic approaches for making polymers by grafting-to and grafting-from alike were explored and evaluated. Each method was evaluated independently for their antifouling performance. The fouling resistance was determined primarily by protein adsorption, most notably using bovine serum albumin (BSA).

III. Materials and Methods

A. Surface Modification with Polymers

Modifying surfaces with polymers is a common and powerful way of manipulating and controlling surface properties. Indeed, surface functionalization of this sort is widely used in fields ranging from biomedical applications to microelectronics.^{33,96,97} Polymers themselves serve as an ideal candidate for modifying surface properties since their chemical composition, physical conformation, and material properties can all be varied in predictable ways. As described previously, there are generally two methods for attaching polymer brushes to surfaces: grafting-to and grafting-from. While grafting-to provides greater control over polymer dispersity, composition and structure, this approach suffers from lower grafting densities compared to grafting-from. Grafting-from avoids the grafting density limitation by growing the polymers from the surface; however, this may cause precise control of dispersity, and the range of compositions accessible to be sacrificed. This thesis explores both synthetic approaches, and will be discussed in further detail. The following sections describe the various methods for polymer brush synthesis utilized herein.

1. Grafting-To: Catechol-Mediated Polymer Deposition

Biomimetic adhesives have recently gained considerable interest in materials and chemistry research; of particular interest has been the mussel-foot protein mimetics.^{98–103} The exact mechanism for the strong underwater adhesion is still an active area of research, but it is generally accepted that the catechol-like moieties within the mussel foot protein are major contributors to their robust adhesive properties.^{104,105} Catechol units are capable of bidentate binding as well as coordination to various metal-oxides, permitting binding to

substrates of different chemical composition.^{99,106} Inspired by the versatility in surface adhesion offered by mussel foot proteins, the catechol group was leveraged in order to attach polymers to surfaces. One of the major limitations of the catechol moiety however is its susceptibility to oxidation to dopaquinone, which greatly impairs its adhesion to SiO₂ and other metal oxides.¹⁰⁶ However, modification of the catechol moiety can afford some resistance to oxidation and improve its binding ability as well as increasing its effective pH range.^{107–109} As a result the adhesive ability of three catechol analogues were investigated in the context of polymer grafting: dopamine (dopa), 6-nitrodopamine (ndopa), and 2-methyl-3-hydroxy-4-pyridinone ethylamine (hopo). Essentially, the addition of electron-withdrawing groups reduced the pK_a of the phenols (pK_{a1} hopo = 3.6, pK_{a1} ndopa = 6.6, pK_{a1} dopa = 9.2),¹¹⁰ effectively improving oxidation resistance.¹¹¹ A series of varying molecular weight polyethylene glycols (PEG) were end-functionalized with the three catechol derivatives, dopa, ndopa, and hopo. Their adhesion to Al₂O₃ and Ti substrates and their subsequent fouling resistance were studied by QCM-D.

A. Materials and Methods:

Dopamine hydrochloride, triethylamine, dimethylformamide (DMF), dichloromethane (DCM), and diethyl ether were purchased from Sigma Aldrich. The mPEG-succinimidyl valerate (mPEG-SVA) of three molecular weights (2, 5, and 10 kDa) was purchased from Laysan Bio, Inc. (Arab, AL).

The three catechol derivatives, dopa, ndopa and hopo (Figure 6 below) were the key elements to this study. These catechol derivatives were synthesized by a graduate student in the Hawker group as part of a joint collaboration. The synthesis of 6-nitrodopamine

(ndopa),¹¹² and 2-methyl-3-hydroxy-4-pyridinone ethylamine (hopo)¹¹³ are described elsewhere and were followed exactly.

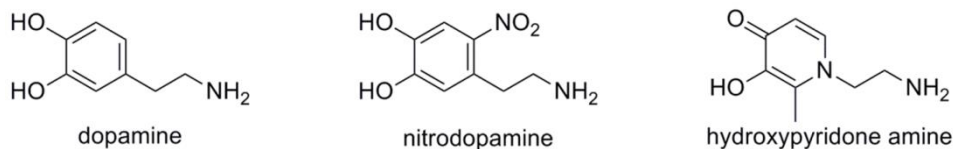


Figure 6: Dopamine derivatives utilized to end-tether polymers to a surface.

Once synthesized these catechol derivatives were reacted with an activated-ester-terminated PEG to get an end-functionalized polymer (Figure 7 below). As an illustrative example, in a 25 mL round bottom flask, 142 mg of dopamine hydrochloride, and 83 mg of triethylamine were dissolved in 5 mL of DMF, and stirred at room temperature for 10 minutes. To this 300 mg of mPEG-SVA (2k) was added, and this mixture was stirred for 24 hours at room temperature. Upon completion the reaction vessel was concentrated under vacuum, re-dispersed in DCM, filtered through a 0.2 μ m syringe filter, and the final product was precipitated into a large volume (40-50 mL) of cold ether. The polymer was then used without further purification.

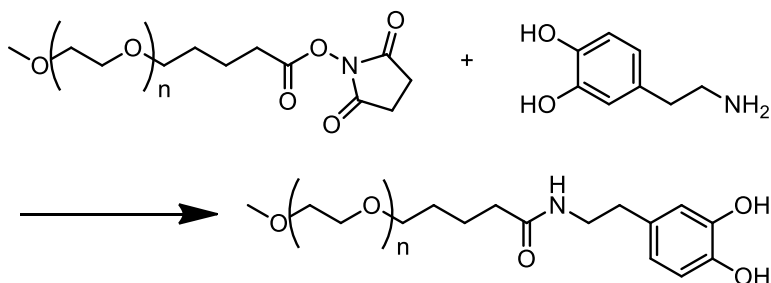


Figure 7: Synthesis of PEG-dopa (identical for other dopa-analogues). Three molecular weights of PEG, 2, 5 and 10 kDa, corresponding to $n \approx 45$, 114, and 227 respectively, were used.

This basic synthetic strategy (1 eq. PEG-SVA, 5 eq. dopa (or analogue), 5.5 eq. triethylamine, 10 % by mass in DMF) was employed for all the dopa derivatives and the different PEG molecular weights. The dried polymers were stored under nitrogen at -4 °C.

To conduct surface deposition of these dopa-functionalized PEGs the desired polymer was dissolved in an aqueous buffer solution of a prescribed pH.

2. Grafting-To: Amine-Epoxy Surface Coupling

Another common strategy for grafting-to with polymers relies on the facile reactivity between amines and epoxide moieties ¹¹⁴ Specifically, amines perform a nucleophilic ring-opening of epoxides very efficiently. Typically an epoxide-terminated polymer of interest is synthesized alongside an amine-functionalized surface which is then used as a platform onto which the end-functionalized polymers are grafted to.

A. Materials and Methods:

(3-Aminopropyl)triethoxysilane (APTES), NaH, epichlorohydrin, tetrahydrofuran (THF), methanol, acetone, isopropanol, dry toluene, and polyethylene oxide (PEO) of different molecular weights, were all purchased from Sigma –Aldrich and used without further purification. Silicon wafers were purchased from Silicon Quest International.

Silicon wafers with 100 nm of thermal wet oxide were cleaved into 1.5 x 1.5 cm squares using a diamond scribe. The wafer squares were sonicated for 10 minutes each in acetone then isopropanol, then dried in a nitrogen stream. Then UV-Oxone treatment was conducted for 20 minutes to remove residual organic films and to generate silanols at the oxide surface. The samples were then immersed in a solution of 1 % APTES in dry toluene overnight (at

least 16 hours), rinsed with toluene, and nitrogen dried. The APTES-coated samples were then annealed at 110 °C under vacuum for 30 minutes. Once annealed the APTES-functionalized substrates were then employed for grafting-to.

B. Epoxidation of hydroxyl-terminated polymers

The epoxidation reactions were conducted on polyethers that naturally have a hydroxyl terminus. Ostensibly this procedure can be conducted on any class of polymer that has reactive alcohols or hydroxyls present. First 300 mg of 20 kDa PEO was placed into a dried round-bottom flask with a magnetic stir bar, purged with argon and evacuated 3 times. Then 15 mL of dry toluene were dispensed from a dry solvent still, and added to the septa-capped flask. A large excess of NaH (15 mg in this case) was added to the polymer solution, and sparged with argon for 5 minutes. The mixture was heated to 30 °C then 1 mL epichlorohydrin was added and the reaction mixture was stirred at 30 °C for 24 hours, at which point the reaction was terminated with 5 mL methanol.

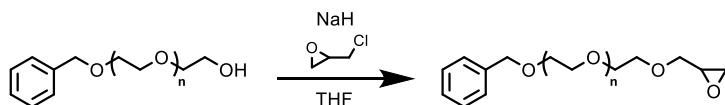


Figure 8: Epoxidation of hydroxyl-terminated PEO by epichlorohydrin.

The dried, epoxide-terminated polymers were then applied onto the amine-coated substrates and heated to 80 °C overnight. Typically a few milligrams of polymer were sufficient to cover the entire surface upon melting. After grafting-to the excess polymer was rinsed off with a good solvent (water for most of the polyethers used) and dried with nitrogen. Various molecular weights of PEO were prepared in a similar manner.

3. Grafting-From: PDA-Initiated Surface Polymerizations

Grafting-from overcomes the limitations of grafting-to by enabling polymer brushes with high grafting density; ensuring the surface-tethered polymers are in an extended conformation. To perform grafting-from, an initiation layer must first be functionalized onto the surface of interest, from which polymerizations can subsequently be conducted. One such approach involves functionalizing surfaces with polydopamine (PDA) and submitting the PDA-coated substrate to polymerization conditions. This technique was motivated by previously reported conditions in the literature.^{16,115,116}

A. Materials

Tris(hydroxymethyl)aminomethane (tris), dopamine hydrochloride, hydrochloric acid, [2-(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (sulfobetaine methacrylate or SBMA), styrene, basic alumina (Sigma-Aldrich, St. Louis, MO), silicon wafers (Silicon Quest International, San Jose, CA) and silica microspheres (Bangs Laboratories, Fishers, IN) were all used as received without further purification (except for styrene which was passed through a packed basic alumina column prior to use).

Two different light sources are employed for different portions of the polymerization procedure: a 15 W, 365 nm UV lamp (XX-15L, UV bench lamp, UVP, Upland, CA) and a collimated 405 nm 665 mW LED lamp (M405L3-C1, Thorlabs Inc., Newton, New Jersey). Oxygen plasma was conducted using a commercial plasma cleaner (PDC-32G, Harrick Plasma, Ithaca, NY) equipped with an O₂ cylinder.

A custom built “polymerization dish” was made from glass in order to allow substrates to be enclosed in a fixed volume, sealed, purged and irradiated during polymerizations.

Inside the polymerization dishes a small petri dish was used to hold the substrates and monomer solution.



Figure 9: Glass polymerization dish, with septa-sealed outlets and cover glass.

The custom-made polymerization dish is designed with barbed outlets that are sealed with rubber septa in order to permit the chamber volume to be purged with inert gas.

B. Experimental

Polymerizations were primarily conducted from silicon wafers in order to begin with a standardized, smooth, solid substrate. To begin, a 4 inch silicon wafer with 100 nm of wet thermal oxide grown was cleaved into 1.5 x 1.5 cm square samples. The smaller samples were rinsed with isopropanol, dried with nitrogen gas, and treated with oxygen plasma (100 mW, 300 mTorr) for 60 seconds. The cleaned silicon substrates were then immersed in large petri dish by 20 mL of a 2 mg/mL solution of dopamine hydrochloride in 10 mM tris buffer at pH 8.5. The tris buffer solution was prepared beforehand using 605 mg of tris dissolved in 500 mL of H₂O and then adjusted to pH 8.5 with ~1 mL of 1 M HCl. The petri dish was covered with a lid and then placed approximately 20 cm beneath the 365 nm UV lamp for 3 hours. After irradiation the polydopamine-coated silicon samples were rinsed,

sonicated in H₂O for 5 minutes and further rinsed with H₂O before drying with nitrogen. This general procedure was also used to functionalize other substrates with PDA.

The PDA-coated substrates were then utilized for polymerization after 48 hours at ambient temperature and atmosphere. Polymerizations from the PDA were conducted with either Styrene or sulfobetaine methacrylate (SBMA) monomers. Styrene was passed through a packed column of basic alumina powder to remove the inhibitor prior to use, though SBMA was used directly. The PDA surfaces were then placed in a small glass petri dish and immersed in approximately 3 mL of monomer. Styrene was applied neat, without solvent, but SBMA was dissolved in H₂O at a concentration of 0.5 g/mL prior to polymerization. The petri dish was then placed inside the glass polymerization dish and a cover glass sealed on top using vacuum grease. The polymerization dish was then purged with nitrogen for 5 minutes before beginning irradiation using the collimated LED light source for the prescribed amount of time. After polymerization the samples were removed from the petri dish, rinsed and allowed to soak in solvent for a minimum of 24 hours, changing the solution a minimum of 5 times. The coated silicon substrates were measured using Filmetrics to determine thickness before and after polymerization.

4. Grafting-From: Surface-Initiated ATRP

Surface initiated atom transfer radical polymerization (SI-ATRP) is one of the most common controlled radical polymerizations from a surface. It can be conducted almost identically to solution-based ATRP, the only difference being that the radical initiators reside at a surface instead of as dissolved molecules. This strategy has been used extensively throughout the literature and is compatible with a range of monomers.^{28,117–119}

Herein two different varieties of SI-ATRP are employed to create polymer surfaces: traditional Cu-catalyzed ATRP and phenothiazine (PTH)-based, photo-catalyzed ATRP. Both techniques rely on functionalizing a substrate with an initiator layer from which polymerization can be conducted. Specifically, silane chemistry was used to introduce an alkyl bromide initiator on silicon wafers.

A. Materials:

10-undecen-1-ol, 2-bromoisobutryl bromide, HCl, hexane, toluene, methanol, MgSO₄, trichlorosilane, Karstedt's catalyst (2% Pt in xylenes), anhydrous trimethylamine, dimethylacetamide, 2,2'-bipyridine (bpy), 1,1,4,7,10,10-Hexamethyltriethylenetetramine (HMTETA), CuBr, CuBr₂, poly(ethylene glycol) methyl ether methacrylate, (PEGMA, M_n = 475 Da), and sulfobetaine methacrylate (SBMA) were all purchased from Sigma-Aldrich (St. Louis, MO). Standard laboratory schlenk glassware was employed for all the reaction steps described herein.

Septa-capped 20 mL glass vials (CG-4908-01 Chemglass Life Sciences, Vineland, NJ) were employed to conduct all air-free reactions for Cu-catalyzed ATRP. Custom machined Teflon inserts were designed to fit within the vials allowing enough space for a stir bar and enabling two silicon wafer samples to be held upright as shown in Figure 10 below.

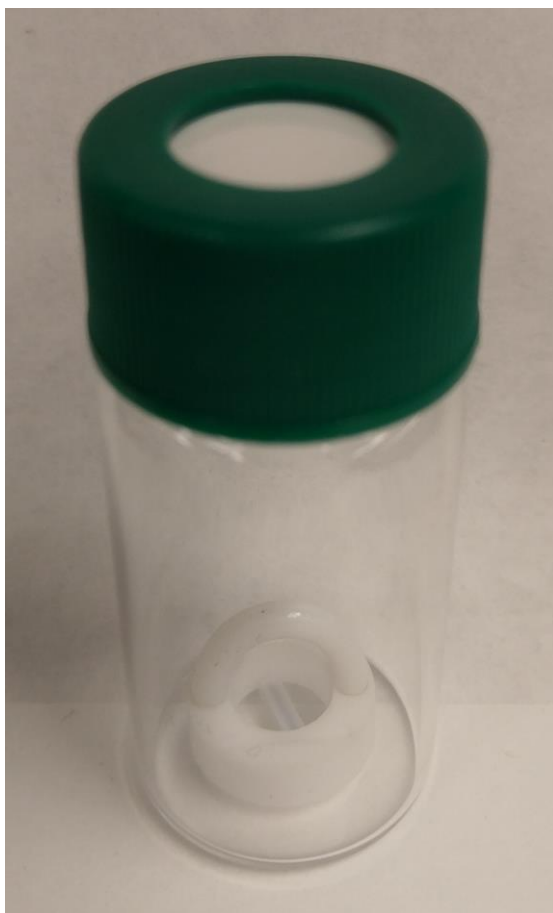


Figure 10: Teflon holder and stir bar inside 20 mL septa-capped vial

The Teflon holder has two slits (at the front and back of the holder as imaged above) that allow for wafer samples to rest upright. The micro stirbar fits inside the circular cutout in the middle, allowing for continuous stirring during polymerization.

B. Initiator Functionalization

In order to functionalize silicon wafers with a radical initiator a small molecule with trichlorosilane and alkyl bromide at either terminus was designed and synthesized. The target structure is 11-(Trichlorosilyl)undecyl 2-bromo-2-methylpropanoate, whose synthesis has been described in detail previously.^{120,121} First a schlenk flask with a magnetic stir bar was evacuated and purged with nitrogen three times before introducing 1.1 mL of 10-

undecen-1-ol, 525 μL of pyridine in 25 mL of dry toluene under inert atmosphere. Then 775 μL of 2-bromoisobutyryl bromide was added dropwise over the course of 10 min while stirring. The reaction was permitted to proceed overnight at room temperature, after which the reaction was terminated by dilution with hexane. The reaction mixture was washed with 1 M HCl, dried over MgSO_4 and further desiccated under vacuum. Purification by silica gel (25:1 hexanes : ethyl acetate) chromatography yielded the desired compound as a colorless oil. The isolated product conformed to the expected ^1H NMR spectroscopy.¹¹⁸ The synthetic scheme is illustrated in Figure 11 and Figure 12 below.

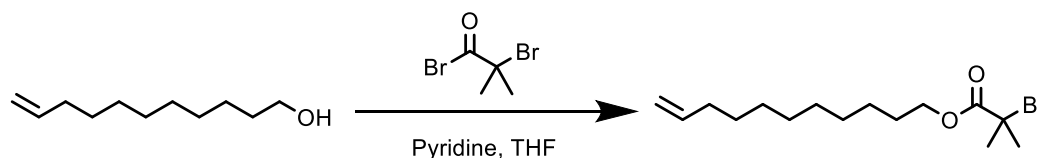


Figure 11: Synthesis of 10-undecen-1-ol 2-bromo-2methylpropanoate.

Once again, a schlenk flask equipped with a stir bar, then sealed, evacuated and purged with nitrogen three times. The isolated product from the first reaction (1.35 g of 10-undecen-1-ol 2-bromo-2methylpropanoate) was combined with 4.2 mL trichlorosilane and 5 μL Karstedt's catalyst into the purged flask. The reaction was stirred for 3 hours at which point the mixture was concentrated under vacuum to give the desired product as a clear oil. The final product was used without further purification.

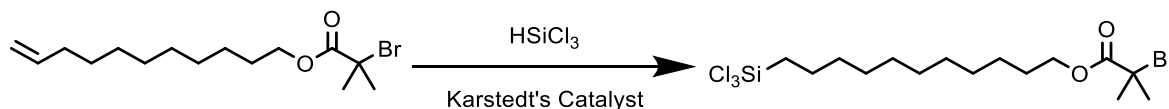


Figure 12: Synthesis of 11-(trichlorosilyl)undecyl 2-bromo-2-methylpropanoate

The synthesized alkyl bromide initiator was then applied to SiO₂ substrates to complete the sample functionalization. Silicon wafers with 100 nm of thermal wet oxide (Silicon Quest International, San Jose, CA) were cleaved into 1.5 x 1.5 cm squares. The samples were rinsed with isopropanol, dried with nitrogen gas, and treated with oxygen plasma (100 mW, 300 mTorr) for 60 seconds. A solution was prepared with 125 μ L of 11-(trichlorosilyl)undecyl 2-bromo-2-methylpropanoate, 250 mL of dry toluene, and 50 μ L of dry triethylamine. In a large glass petri dish the silicon wafer samples were immersed in the aforementioned solution and covered with a blanket of nitrogen. The petri dish was covered and the reaction was allowed to proceed overnight. The initiator-functionalized silicon samples were then rinsed with toluene followed by ethanol prior to being dried with nitrogen. The coated samples were stored in ambient conditions and subsequently used for polymerization without additional preparation. Two variations of ATRP were conducted using these initiator-functionalized wafers: traditional Cu-catalyzed ATRP^{117,118,122} and photo-catalyzed ATRP^{123,124} as described below.

C. Cu-Catalyzed ATRP

In traditional ATRP, Cu serves as a catalyst for polymer chain growth in which there exists an equilibrium between chain propagation (driven by Cu(I)) and chain deactivation (driven by Cu(II)). This equilibrium, and therefore the rate of chain growth, can be partially manipulated by controlling the starting concentrations of Cu(I) and Cu(II) when polymerization is initiated.^{119,125,126} As these polymerizations have already been extensively researched and reported in the literature, previously demonstrated polymerization conditions were leveraged as a starting point for these experiments. Specifically, surface-tethered polymer brushes composed of SBMA and PEGMA (M_n =475 Da) were synthesized.

Reaction conditions were adapted from previously published results for both SBMA¹²² and PEGMA.¹¹⁷ Traditional Cu ATRP must be conducted in air-free conditions so septa-sealed vials were utilized to prepare and conduct the SI-ATRP reactions.

For SBMA surface polymerizations, SBMA, bpy, CuBr and CuBr₂ were used in a 30 : 1.5 : 1 : 0.1 ratio and dissolved in an 2 : 1 methanol/water mixture. First 3.75 g SBMA was dissolved in 15 mL methanol/water (2 : 1 mixture) and sparged with nitrogen gas for 15 minutes while stirring in a 20 mL septa-sealed vial. In a separate vial with a stirbar, 210 mg bpy, 77 mg CuBr and 12 mg CuBr₂ were combined and the vial subsequently evacuated and purged with nitrogen three times. The SBMA solution was then transferred to the vial with CuBr using a purged syringe and needle. The combined mixture is also stirred and sparged with nitrogen for 5 minutes. In a third vial, a Teflon insert with two initiator-coated silicon samples and a stir bar (see Figure 10) were evacuated and purged 5 times. Then the sparged SBMA and CuBr solution was transferred using a purged syringe and needle into the vial with the silicon substrates. The polymerization was then stirred for a prescribed amount of time prior to termination by exposure to oxygen and dilution with methanol. The now coated substrates were subsequently rinsed, allowed to soak in H₂O for 24 hours (changing the water 5 times) and rinsed a final time prior to drying with nitrogen.

For PEGMA polymerizations, PEGMA, HMTETA, CuBr, and CuBr₂ were used in a 35 : 1.5 : 0.5 : 1 ratio and dissolved in deionized H₂O. PEGMA (M_n=475 Da) was passed through a plug of basic alumina to remove the inhibitor and then 2.4 mL of the filtered PEGMA and 40 μ L of HMTETA are dissolved in 1.5 mL of H₂O in a 20 mL septa capped vial. This solution was sparged with nitrogen and stirred for 15 minutes. In a separate vial with a stirbar, 10.4 mg of CuBr and 3.2 mg of CuBr₂ were weighed out, evacuated and

purged with nitrogen 3 times. The PEGMA solution was then transferred to the CuBr vial by means of a nitrogen purged syringe and needle. Once combined, the liquid in the vial was sparged with nitrogen for 5 minutes while stirring. In a third vial the Teflon holder with two initiator-coated silicon samples were loaded along with a stirbar and evacuated and purged with nitrogen 5 times. The sparged reaction solution was then transferred to the vial with the substrates using a nitrogen purged syringe and needle. The polymerization was stirred at 40 °C for a prescribed amount of time. The polymerizations were terminated by exposing the solution to oxygen and diluting the reaction with water. The polymer-coated substrates were then rinsed and soaked in H₂O for 24 hours, changing the solution 5 times, and finally rinsed again prior to drying with nitrogen.

D. PTH-Catalyzed ATRP

While Cu-catalyzed ATRP is the most traditional strategy for surface polymerizations, the metal catalyst has its drawbacks. For many applications the presence of trace metals is undesirable or even intolerable.^{127,128} For this reason a metal-free radical polymerization recently developed in the Hawker group was employed.^{123,124} Not only does this approach avoid metal catalysts, but it also affords spatial and temporal control due to the nature of the organic photocatalyst (10-phenylphenothiazine [PTH]). In the previously published work PTH was used to polymerize a number of acrylate and methacrylate monomers. Briefly, Discekici, et al. conducted surface polymerizations by taking an ATRP-initiator functionalized SiO₂ substrate, applying a few drops (50-100 µL) of monomer solution (containing between 0.1 - 1 mol% PTH) sandwiched on the wafer sample beneath a glass cover slip. The initiator-coated substrate and solution were placed in a polymerization dish

(see Figure 9), purged with nitrogen gas and irradiated using a collimated LED lamp (405 nm, $1.1 \mu\text{W}/\text{cm}^2$).

For the present work SBMA and styrene polymer brushes were desired; monomers that were not previously demonstrated. Drawing from the previous work, reactions conditions were explored in order to identify optimized conditions that yielded reproducible brush thicknesses. To polymerize styrene a solution containing 250 μL styrene, 2 mg PTH, and 50 μL dimethylacetamide was sparged with nitrogen for 5 minutes and applied dropwise onto an initiator-coated substrate. A glass cover slip was placed onto the substrate/solution combination and the entire setup placed inside a polymerization dish, purged with nitrogen gas for 5 minutes and irradiated for 4 hours.

SBMA however could not be synthesized using PTH as those two do not share a common solvent. Instead, a post-doc in the Hawker group synthesized a new analogue of phenothiazine with a zwitterionic moiety in place of the phenyl group in the PTH. The synthesis and structure of this novel photocatalyst are confidential and remain unpublished, so they will not be discussed herein, instead this compound will be referred to as zPTH for simplicity. The zPTH catalyst is water soluble, enabling the polymerization of SBMA brushes. In order to grow these SBMA films, a solution containing 250 mg SBMA, 2 mg zPTH, 50 μL dimethylacetamide, and 250 μL H_2O were mixed, sparged with nitrogen for 5 minutes and applied dropwise onto an initiator-coated substrate. A glass cover slip was placed onto the substrate/solution combination and the entire setup placed inside a polymerization dish, purged with nitrogen gas for 5 minutes and irradiated for 7 hours.

5. Determination of Polymer Film Thickness:

When growing polymer films, thickness is one of the most important parameters that can be controlled for. In the case of polymer brushes, especially, the thickness gives some information about the conformation of the macromolecules on the surface, indicating whether they are in a brush or mushroom conformation. In the present work three methods for determining film thickness were employed: interferometry, ellipsometry, and x-ray reflectivity. All three are optical techniques that rely on the partial reflection and transmission of electromagnetic waves that incident on an interface between materials of dissimilar optical properties.

A. Interferometry

Interferometry is a simple method for measuring film thicknesses based on the behavior of light as it interacts with interfaces of two materials of different indices of refraction according to the Fresnel equations.^{129,130} Interference fringes are generated based on the constructive and destructive interference of light beams travelling different distances. For multi-layer films, such as polymer-coated substrates, partial transmission and partial reflection at interfaces between dissimilar materials creates two distinct light paths which subsequently generate these interference fringes. This serves as a non-destructive method for interrogation of film thickness.

For the purposes of these studies a Filmetrics F20 interferometer (Filmetrics, Inc., San Diego, CA) and the included software was employed. The instrument is setup such that the desired samples are irradiated perpendicular to the substrate, and the reflected light is collected by the detector, and the resulting interference pattern is fit using the software. Using the FILMeasure software several fitting parameters can be manipulated. Both the

index of refraction and the thickness of the films can be manipulated independently and fitted for. First, 20 measurements across the surface where refractive index and thickness are both allowed to vary are conducted. The average refractive index was calculated and set as a fixed value for 20 more measurements where only the thickness is allowed to vary. The average thickness for the fixed refractive index measurements is then reported as the thickness of the film.

B. Ellipsometry

Similar to interferometry, ellipsometry relies on the interference caused in the reflected light from a surface due to interfaces between dissimilar materials. At each such interface there is partial reflection and partial transmission of the incident light.

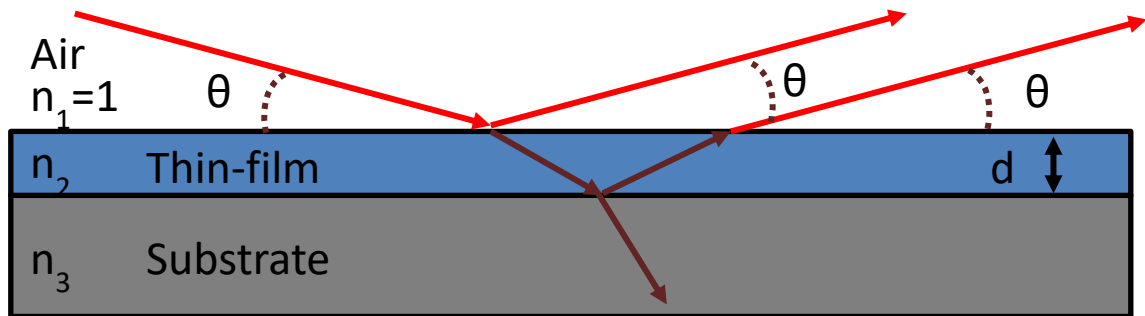


Figure 13: Cross-section of a thin-film-coated substrate showing partial reflection and partial transmission of an incident beam at the interfaces within the film. This serves as the basis for ellipsometry.

Specifically, ellipsometry employs elliptically-polarized light and using a rotating detector is able to collect light intensity as a function of detector angle.¹³¹ With this information, two functions psi and delta (which relate to the relative amplitude and phase

change respectively) are computed as a function of wavelength. These two parameters are then fit using different models that account for the index of refraction, and the film thickness.¹³²

A commercial ellipsometer (M2000DI VASE, Woollam Co., Lincoln, NE) was employed for thickness measurements of the films synthesized herein. A three layer model was used to fit the data: substrate = Si, Layer #1 = SiO₂ (t = 100 nm), Layer #2 = Cauchy Film. The Cauchy film model approximates the refractive index and its dependence on wavelength using the follow relationship.

$$n = A + B/\lambda^2 + C/\lambda^4 \quad (3)$$

Where n is the refractive index, λ is the wavelength, and A , B , and C are Cauchy parameters. These parameters as well as the thickness are varied in order to fit to the psi and delta values as a function of wavelength. For the most part the thickness was the important parameter of interest, so while the Cauchy parameters were monitored closely to ensure sensible fits, only the thickness values were recorded. In addition to ellipsometric measurements in air, a “5 mL Heated Liquid Cell” (J.A. Woollam Co., Lincoln, NE) was also utilized to conduct measurements in a liquid environment.

C. X-Ray Reflectivity

For x-ray reflectivity (XRR) the same general irradiation geometry is present as in ellipsometry (Figure 13 above), but instead of using polarized light, x-rays serve as the incident irradiation. In direct analogue to visible light irradiation, instead of indices of refraction, different materials have differing scattering length densities (SLDs).¹³³

Fitting of XRR data was conducted using Motofit, a plugin package for Igor plotting software. The fitting package enabled a multi-layered film stack to be modelled, with each

layer having an SLD, a thickness and a roughness. The resulting SLD distribution as a function of film depth was generated and used to fit to the raw XRR data. The thickness values for the layers of interest were recorded for the appropriate samples.

1. Calculating Grafting Density

For films that are generated by grafting-to, the molecular weight of the polymers used was known exactly, making it possible to compute grafting density of the resultant polymer brush. Using the thicknesses measured by XRR, grafting density, σ , and coverage, Ω , were computed according to equations 4 and 5 below.

$$\sigma = \frac{\rho_{PEO} h_{dry} N_A}{M_w} \quad (4)$$

$$\Omega = \sigma \pi R_g^2 \quad (5)$$

More detailed explanations of these equations can be found in the background section. The grafting density is a metric of the number of chains per unit area that is calculated using the density of the bulk polymer, ρ_{PEO} , the dry height of the polymer, h_{dry} , Avogadro's number, N_A , and the molecular weight, M_w . Coverage is then a metric of how densely packed the film is with regard to the radius of gyration of the polymer in solution, R_g . Coverages larger than unity indicate a film that is in the “brush” regime, while coverages less than one indicate polymers in the “mushroom” state.

In the case of grafting-from, the degree of polymerization of the surface-grown polymers was unknown, making it impossible to calculate the absolute grafting density.

Instead, for these surface-initiated polymerizations, relative grafting densities could be obtained by comparing the dry and swollen thickness of polymer brushes as shown below.

$$\sigma \sim \left(\frac{h_{dry}}{h_{wet}} \right)^{\frac{3}{2}} \quad (6)$$

When a polymer brush is immersed in a good solvent it will swell to a predictable extent that depends on the grafting density, molecular weight, and thickness of the film. The derivation of this equation is expanded in detail in the background section. Therefore, it was possible to measure the thickness of polymer brushes in their dry and solvated states and use that ratio as an approximation of the grafting density. While this enables qualitative comparisons of grafting density for films made of the same polymer, this relationship cannot compare between polymers composed of different monomers.

D. Quartz Crystal Microbalance:

Quartz Crystal Microbalance (QCM) is a surface sensitive characterization method that measures the change in frequency of an oscillating piezoelectric quartz sensor as a way to quantify changes in mass per unit area.^{134,135} In addition to measuring the frequency, it is possible to also monitor the dissipation (D), defined as the inverse of the quality factor (Q) of the resonance, which is a metric for how rapidly the oscillation decays over time when the applied driving voltage is switched off.^{135,136} This method of using both frequency and dissipation to characterize changes in the quartz sensor is commonly referred to as QCM-D. The dissipation factor is particularly powerful for understanding mass changes and viscoelastic properties of material deposited on quartz when exposed to a liquid

environment.^{136–138} The present work employs a Q-Sense E4 QCM-D (Biolin Scientific, Stockholm, Sweden) in order to monitor both polymer attachment to surfaces (when applicable) as well as foulant deposition.

1. QCM-D Theory:

Piezoelectrics are a class of materials that build up an electric potential by the application of mechanical stress on the material. These materials can also be used in the opposite way by applying an electric field to impart physical deformation of the material. Quartz is a naturally occurring mineral whose crystallographic structure gives it piezoelectric properties. Quartz crystals may be cut in a number of different orientations to give different modes of piezoelectric oscillation. These are referred to as bulk acoustic wave (BAW) devices, and can be made to have different oscillatory behavior. For most QCM-D applications, AT-cut quartz with a resonant frequency of 5MHz is used. Specifically, the quartz sensor is cut into a circular disk with gold electrodes on the top and bottom sides of the disk. The AT-cut crystal results in a thickness shear mode (TSM) resonator, where the top and bottom surfaces oscillate in opposite directions from one another, creating shear waves through the thickness of the quartz.¹³⁹ This configuration is depicted below, where the arrows indicate the displacement profile at its largest magnitude for the fundamental frequency.

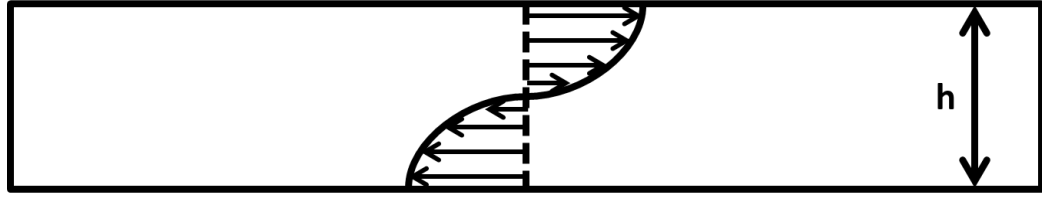


Figure 14: Cross-section of a thickness shear mode (TSM) resonator of thickness h . The top and bottom faces of the QCM sensor oscillate in opposite directions, creating a shear gradient through the thickness of the quartz.

(Note: image not drawn to scale)

For QCM applications, the quartz sensor is driven by an external electric field whose frequency is near the resonant frequency of the sensor, where the sensor's oscillations are largest in amplitude. This frequency analysis can be conducted at the resonant frequency of the crystal as well as odd-numbered multiples of that frequency. These integer multiples of the resonant frequency are called harmonics or overtones. Specifically, the QSoft software can monitor the 3rd, 5th, 7th, 9th and 11th harmonics along with the resonant frequency.

There have been many models developed over the past 60 years describing the frequency response of an oscillating piezoelectric during the adsorption of species onto its surface.^{134,139–142} The simplest of these is known as the Sauerbrey model. This model assumes that the material being deposited onto the quartz surface binds rigidly to the sensor. The model ultimately predicts the following relationship between the frequency change of the crystal versus adsorbed mass:¹⁴¹

$$\Delta m = c \frac{\Delta f_n}{n}, \quad c = -17.7 \frac{\text{ng}}{\text{Hz cm}^2} \quad (7)$$

Equation (7) shows that the frequency shift is directly proportional to the mass change of the sensor. In this case, Δm is the change of mass on the sensor in ng/cm^2 , Δf_n is the change

in frequency of the quartz crystal for the n^{th} overtone in Hz, and c is the Sauerbrey constant as shown above. The Sauerbrey constant depends on the resonant frequency for the sensor, the area between the electrodes on the crystal, the density of the quartz, and the shear modulus of quartz. For the QCM sensors used (AT-cut crystals with a resonant frequency of 5MHz), the Sauerbrey constant is $-17.7 \text{ ng/Hz}\cdot\text{cm}^2$. This suggests that as the mass of adsorbed species increases, the frequency of the crystal will decrease.

QCM-D is similar to the approach of using the oscillatory frequency of a quartz sensor to glean information about the deposition of material on the sensor. The difference in this technique is the use of the dissipation (D) to gain additional information about the behavior of the deposited species. The dissipation is measured by periodically cutting off the driving frequency being used to oscillate the QCM sensor and measuring the relaxation of the oscillations in the absence of the driving frequency. The oscillations in the quartz decay exponentially with some characteristic decay time τ in s^{-1} . Using this decay time the dissipation can be computed using Equation (8).¹⁴³

$$D = \frac{2}{\omega\tau}, \quad \omega = 2\pi f \quad (8)$$

The angular frequency, ω in s^{-1} , is proportional to the frequency of the oscillating crystal, f in Hz, as shown above. Thus, D is a dimensionless number that gives a metric for the amount of energy lost to dissipative effects due to the material deposited on the crystal. Just as multiple overtones can be recorded for the frequency, so can the dissipation of different harmonics be measured. It has been shown that for small values of dissipation (less than approximately $2\text{E-}6$), the material adsorbed onto the sensor surface is “non-dissipative”.^{135,140} This satisfies the criteria for a rigidly bound material and the Sauerbrey

equation can safely be used to estimate the mass adsorbed. Another characteristic mark of a rigidly adsorbed layer is that the frequency responses for each of the overtones are exactly the same when normalized by the harmonic number, n . However, if the dissipation is large, then the material has significant viscoelastic character that needs to be treated using more complex model. A good rule of thumb is if the dissipation is greater than about 5% of the frequency change in Hz, then the adlayer is no longer rigid. This is accompanied by variations in the frequency response for each overtone. The lower overtones become more sensitive to spatial variations on the sensor surface when the adlayer isn't rigidly bound to the piezoelectric, and therefore have larger frequency shifts than the higher overtones.

One of the most commonly used viscoelastic models employed for QCM-D analysis is the Kelvin-Voigt model, or Voigt model. This model is based on a spring and dashpot element coupled in parallel to account for the elastic and viscous responses respectively. This model can be applied to a piezoelectric sensor, with a film of adsorbed material in the presence of a bulk fluid. The geometry in question is illustrated in the figure below.

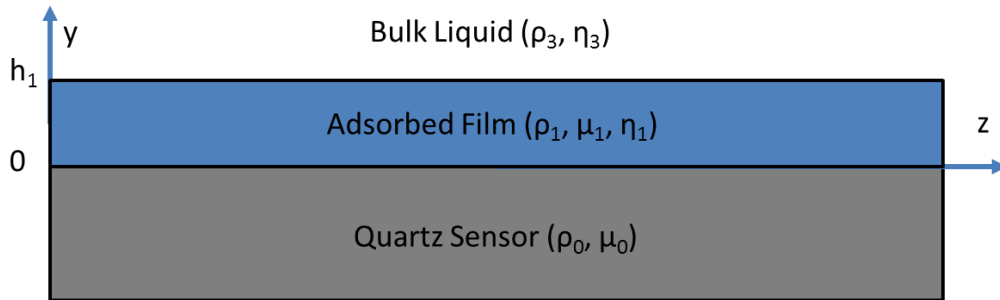


Figure 15: A QCM sensor with an adsorbed film in a liquid environment. The y-axis is normal to the sensor surface that lies in the x-z plane.

Where ρ_0 , ρ_1 , and ρ_3 are the densities of the quartz, the film, and the liquid, respectively, μ_0 and μ_1 are the elastic moduli of the quartz and the film, and η_1 and η_3 are the viscosities of the film and liquid, respectively. The adsorbed film of thickness h_1 can be treated as a

viscoelastic, Voigt element. When a shear stress is applied to this viscoelastic element, the elastic response from the spring component and the viscous response from the dashpot component can be described by the following equation.¹⁴⁰

$$\sigma_{xy} = \mu \frac{\partial u_x(y,t)}{\partial y} + \eta \frac{\partial v_x(y,t)}{\partial y} \quad (9)$$

In this equation the piezoelectric surface lies in the x-z plane, and the y-direction is normal to the surface. Where σ_{xy} is the shear stress, μ is the elastic modulus of the spring, η is the shear viscosity of the dashpot, u_x is the displacement in the x-direction, and v_x is the velocity in the x-direction. This equation can be solved in tandem with the wave equation for shear waves travelling in a viscoelastic material, shown below.

$$\mu^* \frac{\partial^2 u_x(y,t)}{\partial y^2} = -\rho \omega^2 u_x(y,t) \quad (10)$$

$$\mu^* = \mu + i\omega\eta$$

The solution of these equations with the appropriate boundary conditions and assumptions is treated thoroughly in the literature and is beyond the scope of this thesis.^{76,139,142,144,145} In the interest of brevity, the derivation leads to the follow equations to describe viscoelastic layers on a piezoelectric in a bulk liquid, keeping only terms that are linear with respect to the height, h_1 .¹⁴⁰

$$\Delta f \approx -\frac{1}{2\pi\rho_0 h_0} \left\{ \frac{\eta_3}{\delta_3} + h_1 \rho_1 \omega - 2h_1 \left(\frac{\eta_3}{\delta_3} \right)^2 \frac{\eta_1 \omega^2}{\mu_1^2 + \omega^2 \eta_1^2} \right\} \quad (11)$$

$$\Delta D \approx \frac{1}{2\pi f \rho_0 h_0} \left\{ \frac{\eta_3}{\delta_3} + 2h_1 \left(\frac{\eta_3}{\delta_3} \right)^2 \frac{\eta_1 \omega^2}{\mu_1^2 + \omega^2 \eta_1^2} \right\} \quad (12)$$

Equations(11) and (12) above describe the frequency (Δf) and dissipation (ΔD) response due to a film with density ρ_1 , shear viscosity η_1 , elastic modulus μ_1 , and thickness h_1 . The thickness and density of the quartz plate are h_0 and ρ_0 respectively. The properties of the bulk liquid are denoted by a subscript 3 such as the shear modulus, η_3 , and viscous penetration depth, δ_3 . The above expressions are approximations because they omit terms that are higher order in h_1 . These equations make up the Voigt model.

2. Data Interpretation

Once the frequency and dissipation data have been collected using QSoft, the data file can be opened with another software, QTools, for analysis. For rigid films, the application of the Sauerbrey equation is quite trivial, and can be done manually. However, implementing the Voigt model for viscoelastic films requires assistance from modeling software. Conveniently, the QTools software has a built-in Voigt model that will fit the experimentally collected frequency and dissipation responses to equations (11) and (12). The QTools Voigt model allows users to model the thickness, viscosity, and shear modulus of a thin-film, given the film density, and the bulk fluid density and viscosity. This requires making an assumption about the film density, which can generally be approximated as that of the solvent for low density brushes of polyethylene oxide (PEO).

In a typical experiment the fundamental frequency and dissipation are monitored along with the 3rd, 5th, 7th, 9th, and 11th harmonics. The various harmonics are utilized to glean more information about the viscoelastic properties of materials during deposition. As mentioned previously, there are two regimes that categorize the majority of all QCM-D data, those with low dissipation values for whom the Sauerbrey theory is applicable, and those with high levels of dissipation, indicating significant viscoelastic behavior and requiring modelling using a Voigt-Kelvin model.¹⁴⁶ The plots in Figure 16 below depict the frequency and dissipation behavior characteristic of these two regimes.

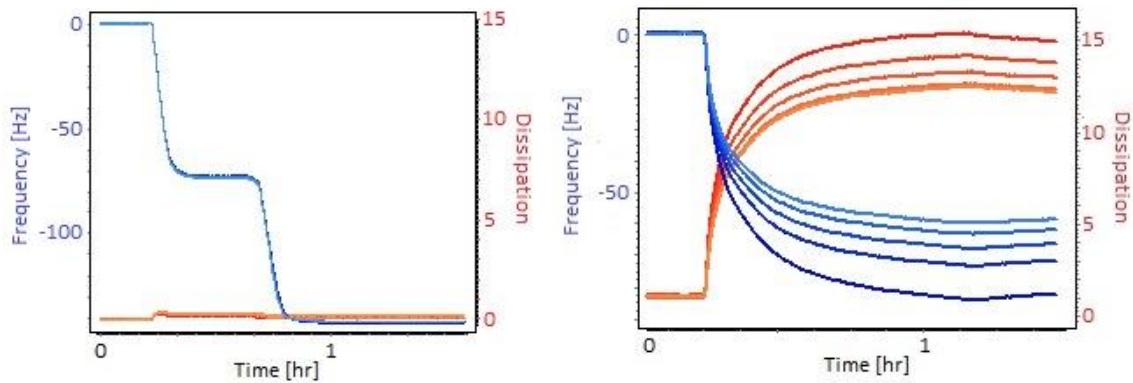


Figure 16: Frequency and dissipation (in blue and red respectively) for rigid (left) and viscoelastic films (right). Low dissipation values are characteristic of rigid films, while larger dissipation values indicate viscoelastic behavior.

The frequency and dissipation responses for rigid and viscoelastic films are quite different as indicated in the two examples above. Not only do rigid films exhibit low dissipation values, but the various harmonic frequencies all overlap, showing no dependence on the overtone number. Viscoelastic films on the other hand show higher dissipation values, but also show a characteristic splitting pattern for the different harmonics. This gives two metrics by which rigid and viscoelastic types of deposition can be delineated. For

rigid films the Sauerbrey equation is used to compute change in mass, whereas viscoelastic films need to be modelled and fitted.

3. Validation of QCM-D Fouling Procedure

Numerous initial experiments were conducted in the QCM-D, to establish a reliable and reproducible procedure. In order to validate the QCM-D methodology, protein adsorption, in the form of BSA and fibronectin (two commonly used model foulants), onto clean surfaces were studied within the QCM. At first these experiments yielded inconsistent results, sometimes varying more than 50% of the measured values between identically conducted experiments (see SI for further detail). The overwhelming feature present in the inconsistent data was a perceived “drift” in the frequency data when nothing should have been occurring to the QCM-D sensors. The degree to which the drift affected the data varied from run to run. In an attempt to reduce this drift the QCM-D sensors were left inside the QCM-D chambers for longer periods of time prior to running an experiment. It was determined that after 48 hours of “resting” within the QCM chambers the sensors no longer exhibited the perceived drift that occurred in many of the previous experiments. The current hypothesis is that this drifting is a result of stresses that are generated upon mounting the piezoelectric sensor between the two o-rings of the QCM chambers. These mounting stresses are then relaxed over time until a stable frequency response is observed.

Using the “relaxation” procedure of leaving the sensors in the QCM-D for 48 hours prior to running an experiment finally resulted in reproducible fouling data that was validated by previous studies. Specifically, protein fouling experiments were conducted on clean SiO₂ QCM-D sensors with identical conditions as used by Lord, et al.¹⁴⁷ These results are shown in Table 1.

Table 1: Measured QCM-D values for protein fouling validated against previously reported values.

QCM-D values	BSA [$\mu\text{g}/\text{cm}^2$]	fibronectin [$\mu\text{g}/\text{cm}^2$]
Calculated Values	0.560 ± 0.030	0.560 ± 0.035
Literature Values*	0.600 ± 0.020	0.600 ± 0.020
* <i>Biomaterials</i> , 2006 , 27, 4856-4862.		

The calculated protein fouling values match quite nicely (to within error) to the expected values reported previously. These values were calculated using the sauerbrey equation (equation (7)) since the dissipation values for both foulants were sufficiently low. All subsequent QCM-D experiments were conducted using this 48 hour “relaxation” period in order to ensure reproducibility.

4. Polymer and foulant deposition

In general, both grafting-to and grafting-from classes of surface modifications can be analyzed and monitored by QCM-D, however for practical considerations it is difficult to conduct grafting-from polymerizations within the QCM-D chambers. Instead, the QCM-D analysis of surface functionalization was limited to grafting-to reactions that did not require elevated temperature or other reagents to achieve surface deposition. Specifically, the deposition of catechol-terminated polymers from aqueous solutions of different pH were measured. In addition, in many cases polymer deposition and subsequent foulant absorption were conducted all within a single QCM-D experiment. Figure 17 below depicts a typical QCM-D experiment, demonstrating both the polymer grafting-to and the foulant deposition steps.

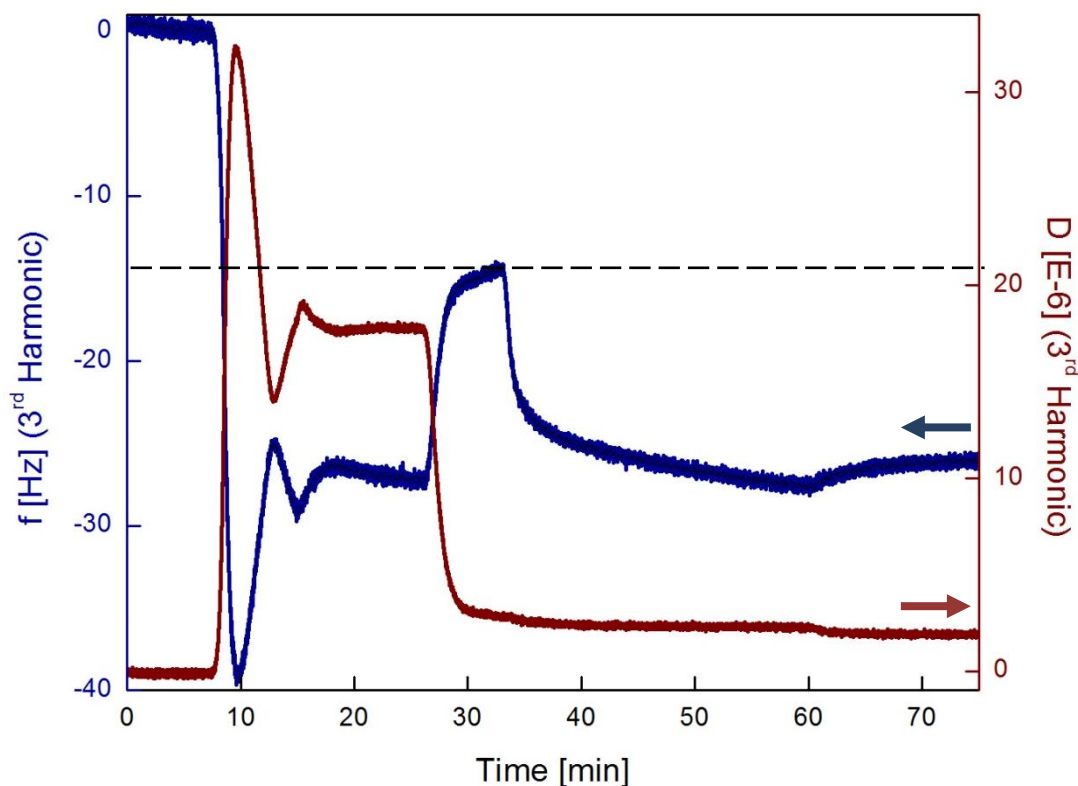


Figure 17: Characteristic plot of frequency and dissipation (shown in blue and red respectively, only the 3rd harmonic plotted for simplicity) for a multi-step deposition.

Ti QCM-D sensors are used.

The experiment above begins at time=0 with Ti QCM-D sensors in a PBS buffer, pH 7.4. After approximately 9 minutes the peristaltic pump is used to draw the solution of polymer through the flow chamber at a flow rate of 150 $\mu\text{L}/\text{min}$. In this case 10 kDa PEO-hopo was prepared in a PBS buffer, pH 7.4 at a concentration of 1 mg/mL. Once the frequency stabilized, (when Δf is roughly less than 0.2 Hz/min) the solution in the flow chamber was switched back to the PBS buffer solution to rinse away un-attached polymers. After stabilization, a foulant solution was introduced at approximately the 33 minute mark. The foulant solution remained in the flow chamber for about 25 minutes, until stabilization,

followed again by a PBS rinsing step. In this instance BSA dissolved in pH 7.5 PBS buffer was employed as a foulant solution.

There are number of key features throughout this deposition profile. The first step in the procedure shows the deposition of the polymer onto the sensor surface, accompanied by a decrease in frequency, which stabilizes after approximately 25 minutes. The dissipation value during this time increases, which indicates viscoelastic behavior of the deposited film, as is expected. The polymer solution is then rinsed away, indicated by an increase in frequency, corresponding to a loss of mass (in the form of un-tethered polymers). Importantly, the dissipation also decreases and drops below the threshold of a viscoelastic film (less than 10% of frequency value). Since a rigid film can be assumed for the remaining polymer on the surface the Sauerbrey relation is applicable, allowing for the mass to be determined directly. The frequency at which it stabilized then becomes the reference point for the fouling step (as noted by the dotted line). The foulant solution is then introduced and the frequency once again decreases, indicating the adsorption of foulants. However, the dissipation does not increase during this step, suggesting that the foulants adhere rigidly to the polymer-coated substrate. This is a key feature as it once again enables the use of the sauerbrey equation to determine the mass of adsorbed foulants.

5. Foulant deposition

Fouling experiments were also conducted without the aid of QCM-D, which required a modified procedure. The primary protein used for these experiments was bovine serum albumin (BSA). In general, fouling was conducted by immersing polymer-coated surface in a solution of BSA dissolved in PBS buffer of pH 7.4 for a prescribed amount of time. Quantifying the amount of adsorbed protein then becomes a challenge. One method for

quantification is by using fluorescently-labelled BSA (FITC-BSA was employed). The fluorescence of the polymer brush surface is measured before and after treatment with FITC-BSA, and this value is used to measure relative amounts of fouling between different surfaces. Prior to inspection by fluorescent microscopy, all samples are carefully rinsed and dried, to ensure the measurement only registers FITC-BSA that is irreversibly adsorbed to the surface. However, this approach is inherently indirect as the fluorescence of the surface is measured after rinsing and drying of the surface, which may or may not be indicative of the affinity of a surface for a given foulant.

Contrastingly, QCM-D enables the real-time, in-situ measurement of mass changes on a surface. As such QCM-D was employed to measure BSA fouling on polymer-coated QCM-D sensors. Similar to way protein adsorption was described above, in-situ monitoring of foulant deposition is possible for various coated QCM-D sensors. Generally, the QCM-D sensors are coated and prepared prior to insertion into the instrument. Again, a 48 hour resting period is required after mounting the sensors. BSA is dissolved in PBS buffer at pH 7.4, and introduced into the QCM-D module, while monitoring mass over time. The BSA solution is subsequently rinsed away when no further mass changes are occurring. The QCM-D methodology has an additional benefit that fouling can be conducted with any foulant without having to change the procedure. This enables the use of multiple foulants and surfaces, all evaluated under identical fouling conditions.

IV. Understanding Surface Polymerizations

The objective of this thesis is to further understand antifouling behavior in order to guide the rational design of future fouling-resistant surfaces. To that end fundamental relationships between materials, their surface properties and their corresponding antifouling performance were investigated. Polymer-functionalized surfaces provide a convenient and practical way to alter surface chemistry in a predictable manner.^{148–151} Polymers themselves exist in a wide range of chemical compositions and properties, but additionally, due to the various ways in which they can be polymerized and attached to surfaces, there is a rich level of diverse properties that can be realized through polymer coatings. Therefore, by carefully designing a set of desired surface functionalities, and thorough characterization of their properties, structure-property relationships can be established, in order to elucidate their antifouling potential.

In order to understand the antifouling performance of polymer-coated surfaces, first an understanding of surface functionalization methods is required. With this in mind, a variety of techniques for synthesizing polymer-coated surfaces were explored. For simplicity, only two classes of polymer attachment strategies were considered, grafting-to and grafting-from, each with its own advantages and limitations. In grafting-to, previously synthesized macromolecules are tethered to the surface, whereas, in grafting-from, polymers are grown directly from an initiator-functionalized surface. In both cases the polymers are covalently tethered to the surface at one or more points along the chain. In this chapter various strategies for achieving surface functionalization and the subsequent antifouling potential will be discussed.

The antifouling literature references hydrophilicity as being a major factor in enabling fouling resistance. Despite that, there are relatively few studies that systematically investigate this property independent of other variables.^{36,62,152,153} For this reason the follow study investigates primarily poly(ethylene oxide) [PEO] and poly(sulfobetaine methacrylate) [P(SBMA)] in an attempt to ascertain the effect of hydrophilicity.

A. Grafting-To

1. Catechol-Terminated Polyethylene Oxide

Grafting-to is the process of attaching polymers to a surface or interface by means of reacting one or more reactive groups present on the polymer with a complementary moiety expressed at the surface. One key advantage of this method is the ability to synthesize and thoroughly characterize the polymers prior to surface modification. This permits a wide range of carefully designed polymer compositions and architectures to be employed for grafting-to. However, grafting-to carries with it inherent limitations in the grafting density that can be achieved.

The first point to consider in grafting-to is the chemistry that will be utilized for surface attachment, and how to functionalize the surfaces and polymers with complementary moieties. One strategy that has recently gained considerable interest within the polymer community is a biomimetic approach inspired by the underwater attachment of mussels.^{154–158} Briefly, mussels secrete a substance containing a cocktail of proteins (broadly referred to as mussel foot proteins) that are designed to gel and robustly adhere to underwater surfaces. The mechanism for their robust attachment is quite complex and has been the subject of a large body of research.^{99,104,105,156} One important component of the mussel foot protein is the *ortho*-dihydroxybenzene, or catechol group. Catechols are capable of coordinating to

transition metals as well as binding to metal oxide surfaces, making them excellent adhesives. In the context of surface attachment using polymers, the binding of catechols to metal oxides are most relevant for this application. However, catechols are also susceptible to a two-electron, two proton oxidation from the catechol to the dopaquinone (Figure 18).

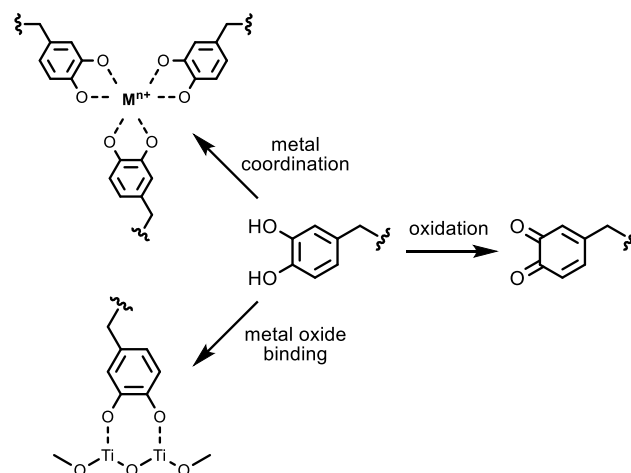


Figure 18: Catechols (middle) are capable of coordination to transition metals (upper left), as well as to metal oxide surfaces (lower left), but are also susceptible to oxidation to dopaquinone (right) which impedes their adhesive qualities.

This oxidation to the dopaquinone renders the adhesive properties of the catechol moiety unavailable. Consequently, catechols must be employed under a pH range over which oxidation is sufficiently retarded to enable the binding to metals and metal oxides. There has been significant research trying to prevent oxidation of catechols in order to leverage their versatile binding properties for various applications. In order to achieve efficient surface modification, the pH range over which catechol groups can be used for surface binding needs to be extended. Specifically, three catechol derivatives with varying pK_a

values of the hydroxy groups were attached to the terminus of poly(ethylene oxide) (PEO) polymers.^{112,113}

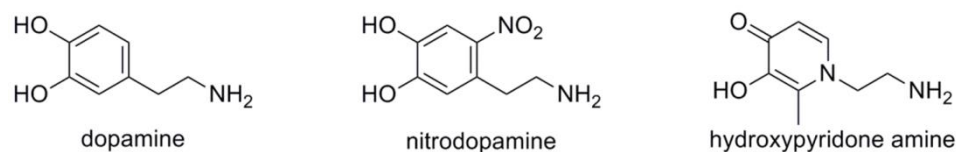


Figure 19: Dopamine derivatives utilized to end-tether polymers to a surface, referred to as “dopa”, “ndopa“, and “hopo“ respectively.

For initial studies, PEO (a well-documented, fouling-resistant polymer) was chosen in order to study catechol-mediated, grafting-to attachment (general synthesis shown in Figure 7). Three different molecular weights of PEO were employed to explore how the degree of polymerization affects deposition. Key to this experiment is that each polymer is functionalized with an adhesive moiety at a single terminus.

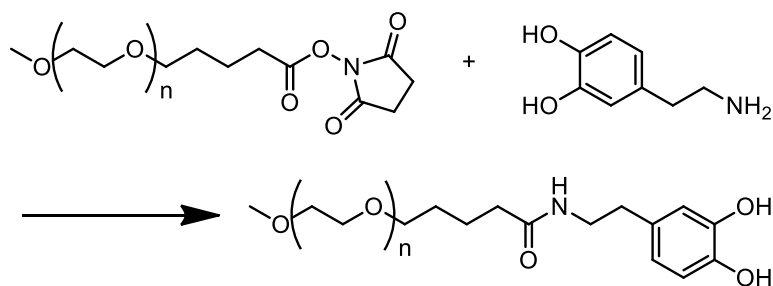


Figure 20: Synthesis of PEO-dopa (identical for other dopa-analogues). Three molecular weights of PEG, 2, 5 and 10 kDa, corresponding to $n \approx 45$, 114, and 227, respectively, were used.

Once an appropriate synthesis for dopa-functionalized PEO was identified, the deposition of these materials was analyzed. Specifically the deposition onto titanium-coated silicon wafers of five polymers: 2k PEO-ndopa ($M_w = 2000$ Da), 5k PEO-ndopa ($M_w = 5000$ Da), 10k PEO-ndopa ($M_w = 10000$ Da), 10k PEO-hopo ($M_w = 10000$ Da), and 10k PEO-

dopa ($M_w = 10000$ Da) was measured. Solutions of each of these polymers at a concentration of 3.6 mM in phosphate buffer solution (PBS) at pH 7.4 and deposited onto Ti-coated silicon wafers by dip coating, followed by rinsing and drying. The functionalized wafers were then measured using x-ray reflectivity (XRR) and the thicknesses fitted according to the procedure outlined in Materials and Methods section. Using the thicknesses measured by XRR, grafting density, σ , and surface coverage, Ω , were computed according to equations

$$\sigma = \frac{\rho_{PEO} h_{dry} N_A}{M_w} \quad (13)$$

$$\Omega = \sigma \pi R_g^2 \quad (14)$$

The grafting density is a metric of the number of chains per unit area that is calculated using the density of the bulk polymer, ρ_{PEO} , the dry height of the polymer, h_{dry} , Avogadro's number, N_A , and the molecular weight, M_w . Coverage is then a metric of how densely packed the film is with regard to the radius of gyration of the polymer in solution, R_g . Coverages larger than unity indicate a film that is in the “brush” regime, while coverages less than one indicate polymers in the “mushroom” state. The grafting densities and corresponding coverages computed for the dopa-terminated PEO polymers are shown in Table 2 below.

Table 2: Grafting densities and coverages for various polymer films deposited on Ti QCM-D sensors.

Polymer Film (on Ti)	Grafting Density, σ [nm⁻²]	Coverage, Ω
2k PEO-ndopa	0.51 ± 0.02	2.7 ± 0.1
5k PEO-ndopa	0.18 ± 0.05	2.3 ± 0.7
10k PEO-ndopa	0.14 ± 0.01	3.5 ± 0.3
10k PEO-hopo	0.11 ± 0.01	3.0 ± 0.4
10k PEO-dopa	0.11 ± 0.01	2.9 ± 0.1

While the highest grafting density achieved was for the 2 kDa molecular weight PEO, it should be noted that this did not translate into the largest coverage. In fact, the higher molecular weight 10 kDa PEO generated the highest coverage films. This is due to the balance between steric limitation of grafting density that is counteracted by the effective coverage, which depends on the square of the radius of gyration. This is consistent with previously reported behavior of grafting-to of PEO.¹⁵⁹ In comparing the different dopa-derivatives, dopa-, ndopa-, and hopo-terminated 10k PEO all behave similarly, and do not show an appreciable difference in grafting density or coverage between them. This agrees with the hypothesis that, as expected, the different catechol analogues only affected the pH behavior, and not the deposition dynamics.

The highest and most uniform coverage films correspond to the most fouling resistant films, thus, the 10k PEO series was studied in subsequent fouling experiments. The dopa-, ndopa-, and hopo-terminated 10k PEO were deposited on Al₂O₃ and Ti QCM-D sensors,

followed by subsequent fouling experiments (as described in Materials and Methods section). QCM-D surfaces coated with Al_2O_3 and Ti were also examined to serve as a comparison to the synthesized polymer coatings (Figure 21).

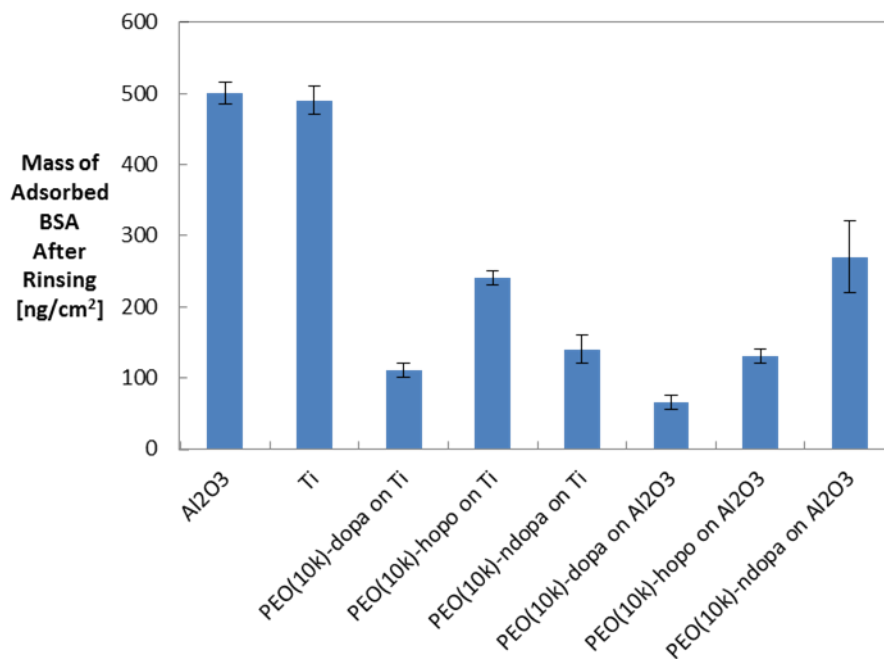


Figure 21: Protein (BSA) fouling as measured by QCM-D for various coated QCM-D sensors. Nylon, Au, SiO_2 , Al_2O_3 , and Ti were all purchased from Q-Sense, while the remaining six surfaces were prepared by grafting-to. The average and standard deviation across 4 replicate runs is shown.

All the PEO-coated samples displayed improved resistance to BSA fouling compared to the as-received, un-functionalized QCM-D sensors. Despite some variation between the various polymers, there seemed to be no clear trend between the different dopa-analogues. The substrate chemistry did not seem to play a large role in the fouling behavior.

In addition to these singly-functionalized dopa-analogues, polymers with adhesive groups at both ends were also prepared. This class of polymer that has both termini

functionalized (as with catechols) are commonly referred to as telechelic. For these studies, two telechelic polymers of different molecular weights were prepared: 10k and 20k PEO. The telechelic polymers were 10k and 20k PEO with activated ester groups (SVA) at both termini, and both were converted to dopa termini. This ostensibly would enable the polymers to form loops at the surface with both termini coordinating to the substrate. The ability to form looped structures on a surface may result in higher effective surface coverage at the same level of grafting density. This would enable fouling resistance in a synthetically accessible range, overcoming traditional limitations of grafting-to. The fouling performance of these two telechelic polymers was compared with 10k and 20k analogues with only a single reactive terminus. These singly-functionalized polymers were prepared as the dopa and ndopa derivatives. The fouling results for the comparison of these 6 different surfaces are shown in Figure 22.

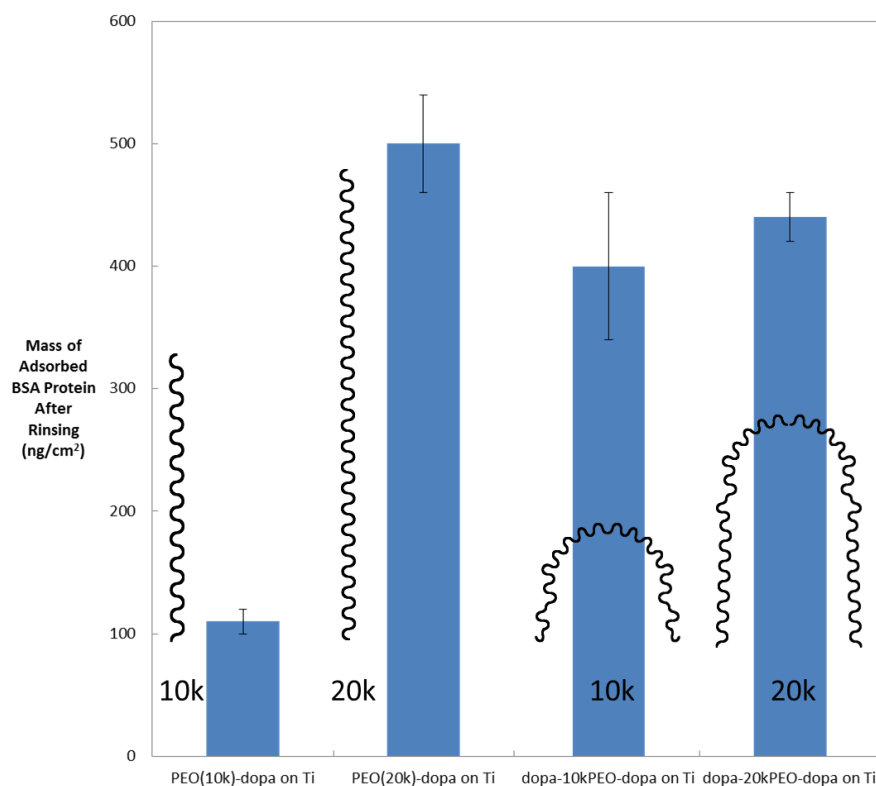


Figure 22: BSA protein fouling on various catechol-terminated PEO polymers deposited on Ti QCM-D sensors. The first two polymers are 10 kDa and 20 kDa molecular weight PEO with a single reactive terminus with dopa. The final two samples were prepared with telechelic PEO, with two reactive termini, each being dopa-functionalized. Deposition and subsequent fouling with BSA was conducted in-situ using the QCM-D flow modules. The average and standard deviation for the mass of BSA remaining on the surface after rinsing was computed for 4 replicate experiments.

These data indicate that the higher molecular weight linear brushes exhibit less fouling resistance than their lower molecular weight counterparts. These results are reminiscent of previously reported results for grafting-to of PEO.¹⁵⁹ Particularly, the authors demonstrated

that with increasing molecular weight of PEO, the grafting density and surface coverage suffer. This is due to the fact that even though the coverage increases with the square of R_g , the grafting density is negatively impacted by using higher molecular weights for grafting-to. In this case the 20k PEO is past the threshold above which lower coverage is expected; corresponding to a decrease in fouling resistance, as observed.

Furthermore, no appreciable improvement in fouling resistance was observed when employing the telechelic PEO polymers. In fact, for the 10k PEO increase in fouling propensity is observed when there are two dopa moieties, one at each polymer terminus. It is hypothesized that the polymers may not be able to deposit with both ends of the polymer tethered to the surface. This would result in pendant catechols that are potentially exposed to the surrounding solution, and may enhance the affinity for protein fouling. However, without further information about the conformation of the polymer chains at the surface to really elucidate their properties, no further conclusions could be drawn. Gaining insight into the true conformation of these surface-tethered telechelic polymers would require

quantification of the distance of each polymer terminus from the substrate. No experimental studies have provided such information, and this would require synthesis of completely novel polymer structures. Specifically, polymers that contained both an adhesive moiety and a diagnostic functionality at each terminus would be required. This could potentially enable the tracking of the individual polymer termini and measure their distance from a solid substrate. If possible, this would give insight into the polymer conformation when grafted to a surface. However, in the absence of such structures, some initial experiments on the deposition and subsequent fouling of telechelic PEO was

conducted (described in Supporting Information). Since this is not the objective of the present work, these telechelic polymers were not pursued further.

While a moderate amount of fouling resistance was observed for all PEO surfaces created, the fouling performance was still not sufficiently low to justify excellent fouling resistance ($\leq 50 \text{ ng/cm}^2$). This is likely due to the inherent limitations in grafting density that can be achieved through the grafting-to process. In order to overcome the lower grafting densities achieved by grafting-to, a grafting-from approach was employed moving forward.

B. Grafting-From: Cu-Catalyzed SI-ATRP

Grafting-from is the process of polymerizing macromolecules from surface-tethered initiators to create a polymer brush. Typically, grafting-from results in higher grafting densities than grafting-to, which in turn can be used to achieve higher coverage films and better fouling resistance in many cases. For this reason different grafting-from structures were targeted and compared with regards to their fouling performance. There are numerous strategies for conducting surface-initiated polymerizations, however traditional, copper(Cu)-catalyzed, surface-initiated ATRP (SI-ATRP) was chosen since it is the most common surface grafting method and has a plethora of previously reported results.^{21,28,117,160,161}

However, one of the limitations of grafting-from is the inability to directly characterize the polymers that are grown from the surface, making grafting density difficult to calculate. Instead, the surfaces were characterized by their thickness and swelling behavior. Thickness and swelling behavior were measured using a liquid-compatible ellipsometry stage.

1. Experimental Methods

SI-ATRP was conducted for poly(ethylene glycol) methacrylate (PEGMA) and sulfobetaine methacrylate (SBMA), two common, fouling-resistant, methacrylate monomers that are compatible with SI-ATRP. The polymerization of ethylene oxide is not amenable to ATRP reaction conditions, so while PEO cannot be studied directly, the methacrylate analog, PEGMA, can be used instead. While not identical, these two monomers behave similarly once polymerized, and can be used for comparison purposes. Polymerization conditions were inspired by previously reported conditions and adapted for the needs of this study, as described in Materials and Methods section.^{74,117,161}

Thicknesses of the films synthesized were measured by interferometry and ellipsometry. In addition to normal thickness measurements in air, ellipsometry measurements were conducted in a liquid environment using a sealed liquid cell with optically transparent windows. This allowed the hydrated thickness of films to be measured directly. The ratio of dry and wet thicknesses were then used as a proxy for grafting density (since grafting density cannot be measured directly, see Background section).

Contact angle measurements were conducted using a Dataphysics OCA 15pro (DataPhysics Instruments, Filderstadt, Germany). Advancing and receding contact angles were measured starting with a 2 μ L drop with an advancing and receding speed of 0.5 μ L/min. These measurements were conducted at three locations on each sample, and the average and standard deviation were taken.

2. Results and Discussion

Polymer surfaces of SBMA have been demonstrated to have promising antifouling properties, attributed in part to their hydrophilic nature.¹⁶² Another particularly intriguing

feature of P(SBMA) surfaces is their thickness-dependent properties as reported by Cheng et al.¹²² In particular, the authors demonstrated that above a certain thickness (approximately 200 nm) P(SBMA) brushes transition from hydrophilic to hydrophobic. They posited that this transition is driven by a competition between intrachain association and solvation by water. After a critical thickness is reached, the interchain interactions overcome the entropic penalty that is lost by expulsion of water and subsequent polymer chain collapse. This sort of thickness-dependent hydrophilicity has also been reported in P(NIPAM) and other polymer systems.^{163–165} Within the context of antifouling this property enables the unique opportunity to compare hydrophobic and hydrophilic surfaces of identical chemical composition, ostensibly permitting the isolation of the hydrophilic contribution on fouling resistance. The role of hydrophilicity in antifouling is cited frequently in the literature, however there are comparatively few studies that systematically investigate this property in isolation.^{36,62,152,153}

With the objective of investigating the effect of hydrophilicity on fouling resistance in mind, the thickness-dependent hydrophilicity of P(PSBMA) was sought to be reproduced herein. The reaction conditions were replicated exactly (described in detail in Materials and Methods section) and the resultant film thicknesses were measured as a function of polymerization time as shown in Figure 23.¹²²

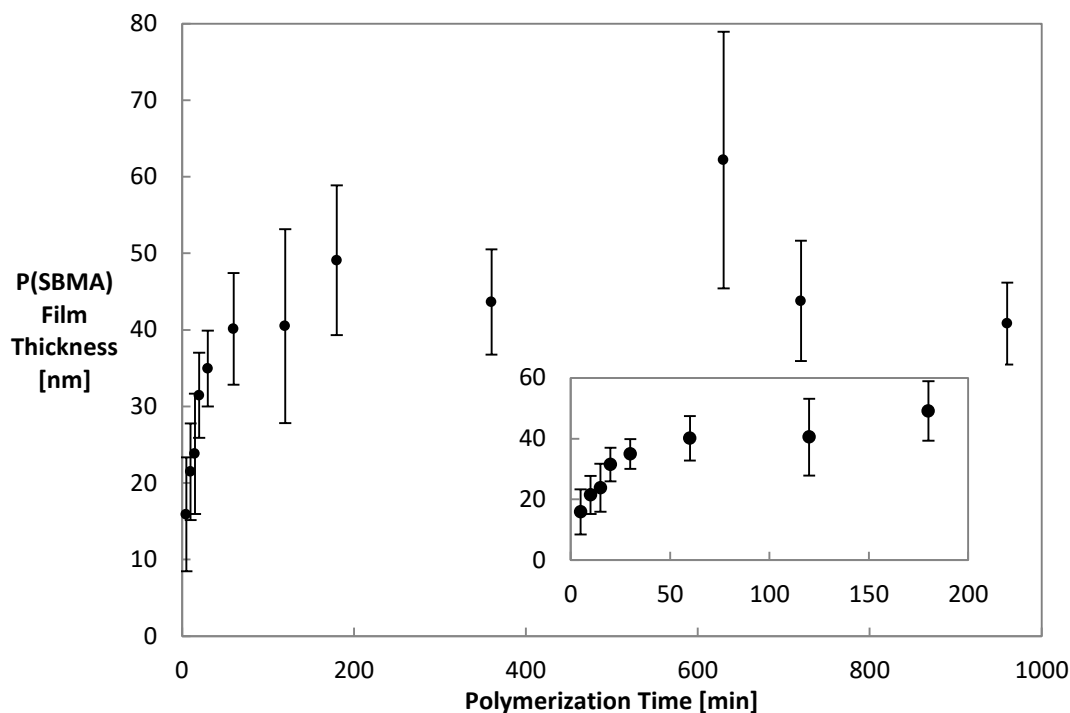


Figure 23. Films thicknesses for P(SBMA) polymerizations conducted by SI-ATRP.

Inset has same axis units as larger graph.

Despite using the reaction conditions outlined by Cheng et al., the reproduction of their results was not achieved. According to their published work, these conditions should have yielded P(SBMA) films of approximately 400 nm within 1000 minutes of polymerization time. Instead, the films plateaued between 40-60 nm after only 200 minutes. The inset graph in Figure 23 further illustrates the slowed reaction kinetics.¹

Despite not achieving the thicknesses expected the films synthesized were characterized to further understand their properties. Contact angle measurements were conducted to determine if these SBMA films exhibited similar hydrophobic behavior to those reported previously (shown in Figure 24).

¹ The corresponding author of the previous work¹²² was contacted and they indicated that they were unable to reproduce their own results as well as confirmed that the results shown in Figure 23 were consistent with their attempts to replicate their work.

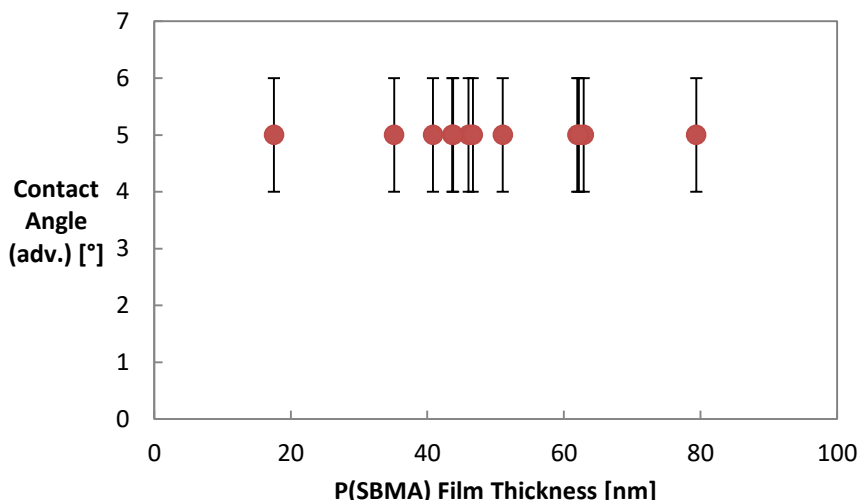


Figure 24: Advancing contact angles measured for P(SBMA) films of varying thickness. Each data point is the average of 3 drops across the sample surface.

All the P(SBMA) films were extremely hydrophilic; in fact the advancing and receding contact angles were too small to accurately measure. Therefore, the advancing and receding contact angles are reported as $5 \pm 1^\circ$; indicative of a very hydrophilic surface. These values are lower than reported by Cheng et al. but generally are in agreement with their observation that P(SBMA) at thicknesses less than 100 nm have a low contact angle and are very hydrophilic. However, as predicted, it was not possible to grow films thick enough to access the hydrophobic regime.

Instead of focusing on a single polymer, additional polymers were studied in the same context. Specifically, polymerization kinetics for poly(polyethylene glycol methacrylate) [P(PEGMA)] films, were explored, again by measuring film thickness as a function of time. The results are plotted below alongside the P(SBMA) results over the same time scale (Figure 25).

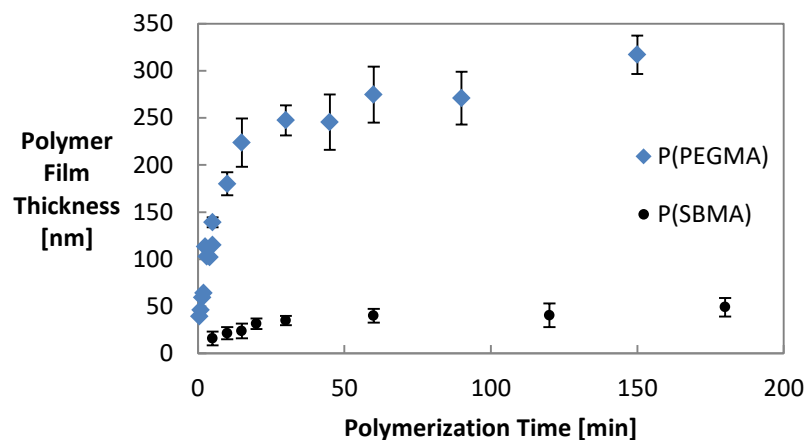


Figure 25: Polymerization kinetics for P(PEGMA) and P(SBMA). Values plotted reflect the average and standard deviation for 3 replicates polymerizations.

As evidenced in the figure above, PEGMA exhibits much faster polymerization kinetics than SBMA under the conditions investigated. Qualitatively, a plateauing of film thickness occurs over the same time-scales for both monomers. P(PEGMA) films level off around 300 nm after 100-150 minutes. These films were further characterized by measuring their contact angle (shown in Figure 26).

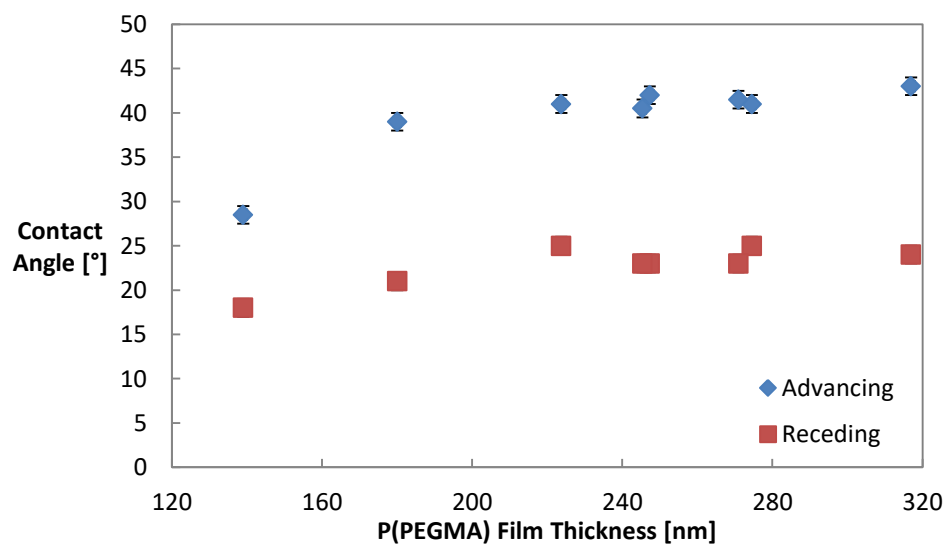


Figure 26: Contact angles measured for various P(PEGMA) film thicknesses.

All of the P(PEGMA) films were fairly hydrophilic, though not as strongly as the P(SBMA) films. Slight thickness dependence of the contact angle was observed for films thinner than 200 nm; with the contact angle essentially remaining constant for thicknesses between 220-320 nm. Even so, there was negligible variation in hydrophilicity as a function of thickness over the range of polymer thicknesses synthesized.

As mentioned previously, grafting density for these grafting-from systems is difficult and often impossible to measure directly since the degree of polymerization from the surface is not known. However, the grafting density can be approximated by studying the swelling behavior of polymer-tethered films. Specifically a relationship between grafting density and the thicknesses of polymer films in dry and solvated states can be derived. First, an expression of the solvated polymer thickness in a good solvent, h_{wet} [m] is required, which has been derived elsewhere.¹⁶⁶

$$h_{wet} = \left(\frac{12}{\pi}\right)^{\frac{1}{2}} N \sigma^{\frac{1}{3}} \left(\frac{\omega}{v}\right)^{\frac{1}{3}} \quad (15)$$

In this expression, N is the degree of polymerization, σ [chains/m²] is the grafting density, ω [m³] is the excluded volume parameter and v [m⁻²] is defined as $3/a^2$ where a [m] is the Kuhn monomer length. Equation 3 can then be combined with an expression for grafting density as shown below.^{167,168}

$$\sigma = \frac{\rho_0 h_{dry} N_A}{m_0 N} \quad (16)$$

Here, the grafting density, σ [chains/m²], is related to the mass density, ρ_0 [g/m³], the dry thickness of the polymer h_{dry} [m], Avagadro's number N_A , the monomer molecular weight m_0 [g/mol], and the degree of polymerization N . Combining equations 3 and 4 results in the following expression:

$$\sigma = \left(\frac{\rho_0 N_A}{m_0} \left(\frac{12}{\pi} \right)^{\frac{1}{2}} \left(\frac{\omega}{v} \right)^{\frac{1}{3}} \right)^{\frac{3}{2}} \left(\frac{h_{dry}}{h_{wet}} \right)^{\frac{3}{2}} \quad (17)$$

This relationship explicitly relates the ratio of dry and solvated polymer brush thicknesses to the grafting density. However, determining the excluded volume parameter and Kuhn length for any arbitrary monomer can be impractical. Instead the equation is reduced to a simple proportionality between the grafting density and the brush thickness ratio as shown below.

$$\sigma \sim \left(\frac{h_{dry}}{h_{wet}} \right)^{\frac{3}{2}} \quad (18)$$

Therefore, it was possible to measure the thickness of polymer brushes in their dry and solvated states and use that ratio as an approximation of the grafting density. While this enables qualitative comparisons of grafting density for films made of the same polymer, this relationship cannot compare between polymers composed of different monomers.

So since the absolute value for the grafting density could not be computed, equation 6 was used as a relative metric for grafting densities. The ratio of dry and wet film thicknesses (h_{dry} and h_{wet} respectively) for P(SBMA) and P(PEGMA) was measured as shown in Figure 27. This ratio is defined such that larger values correspond to higher

grafting densities. Both dry and wet thicknesses were measured using an ellipsometer equipped with a liquid cell for solvated measurements.

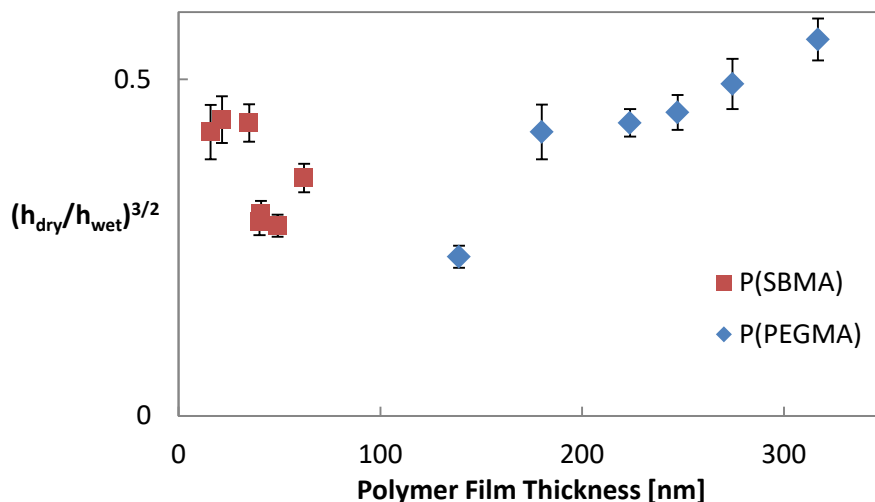


Figure 27: Ratio of dry and swollen brush heights for P(SBMA) and P(PEGMA) films of various thickness.

The P(PEGMA) films (shown in blue) show a clear trend to higher grafting densities as the thickness of the polymer layer increased. This is the expected behavior for grafting-from polymers, and shows good confirmation that the synthesized films are achieving good coverage. The P(SBMA) films on the other hand do not show the same increase in grafting density with increased thickness. Instead, little variation between the various thicknesses is observed. However, it should be noted that all the P(SBMA) films were restricted to relatively low thicknesses, which may have been the reason an increase in grafting density was not observed. Unfortunately it is not possible to compare these two polymer systems directly, as each monomer has a different Kuhn length and excluded volume parameter.

Fouling experiments were then conducted with the polymer films of P(PEGMA) and P(SBMA). For these experiments fluorescein isothiocyanate conjugate bovine serum albumin (FITC-BSA) was employed as a fluorescently-tagged protein. This enabled the

determination of protein fouling by measuring these surfaces using a fluorescent microscope (Olympus BX51 Upright Compound Microscope). Each polymer-coated surface was fouled by a solution of FITC-BSA at pH 7.4 for 24 hours and subsequently rinsed and dried. The fluorescence of the fouled surfaces was determined through images captured through the fluorescent microscope. The fluorescence intensity of each sample was then normalized to the fluorescence measured for FITC-BSA fouled SiO_2 surfaces. Figure 28 shows the fluorescence intensity relative to the bare SiO_2 surface for P(SBMA) and P(PEGMA) films of various thickness.

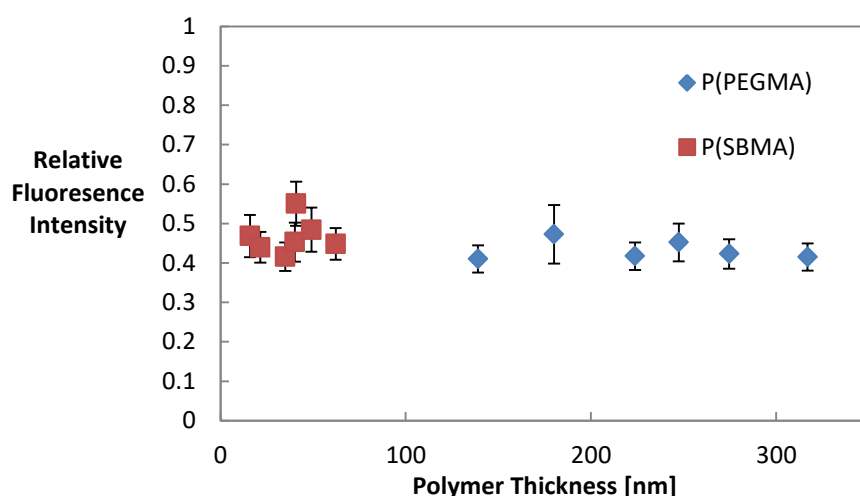


Figure 28: Fluorescence intensity for the various P(SBMA) and P(PEGMA) films fouled by FITC-BSA.

All fluorescence values were normalized to the fluorescent intensity of the FITC-BSA fouled, unmodified SiO_2 substrate. Comparable levels of fouling resistance for both P(SBMA) and P(PEGMA) films were observed. They all exhibit about a 40-50% reduction in fouling propensity compared to the unmodified SiO_2 substrate. Thickness seems not to play a large role in fouling propensity over the range of surfaces investigated.

The ultimate fouling performance of these films did not appreciably change of the range of film thicknesses accessed. In the context of antifouling coating for membranes this suggests that the thinner P(SBMA) films are preferable, as the thinner coating may permit a higher flux of water across the filtration membrane. However, these films were not pursued further, as the data did not enable more trends and relationships to emerge. In addition, the synthesis and control of these ATRP-grown films was tedious, making the synthesis of a large number of surfaces for replicate experiments prohibitively time consuming. Consequently, more synthetically simple grafting-from strategies were sought out.

C. Grafting-From: PDA-Initiated Films

Inspired by biomimetic strategies that leverage catechol moieties for various adhesion and binding applications, there has been considerable interest in utilizing dopamine as a surface coating.^{103,169–171} While there are various strategies for employing dopamine, one common and simple method involves taking advantage of the oxidation and subsequent crosslinking of dopamine under alkaline conditions, forming “polydopamine” or PDA.^{115,172–175} In addition to exhibiting a robust attachment to various substrate chemistries, PDA can also be employed as an initiation layer for grafting-from polymerizations.^{29,176} The mechanism by which this polymerization occurs is still under dispute, however, the leading hypothesis is that under UV irradiation, nitrogen-based radicals initiate polymerization in the presence of suitable monomers, and proceeds in an uncontrolled manner.^{29,177} The strategy has many benefits, most notably, the PDA layer itself attached robustly to numerous chemistries, including many metals, oxides, and even organic substrates.^{29,178} This enables polymerization of subsequent monomers from surfaces that would otherwise be challenging to functionalize with surface-tethered initiators. The PDA methodology is still quite novel

as far as surface-initiated polymerizations, so the further development of this process was the main objective of this study.

1. Experimental Methods

In order to conduct experiments on the various relevant polymerization conditions silicon oxide substrates were coated with PDA. Silicon wafers with 100 nm of silicon oxide grown on top were cleaved into 1.5 x 1.5 cm square samples. The smaller samples were rinsed with isopropanol, dried with nitrogen gas, and treated with oxygen plasma (100 mW, 300 mTorr) for 60 seconds. The cleaned silicon substrates were then immersed in large petri dish by 20 mL of a 2 mg/mL solution of dopamine hydrochloride in 10 mM tris buffer at pH 8.5. The petri dish was covered with a lid and then placed approximately 20 cm beneath the 365 nm UV lamp for 3 hours. After irradiation the polydopamine-coated silicon samples were rinsed, sonicated in H₂O for 5 minutes and further rinsed with H₂O before drying with nitrogen.

The PDA-coated substrates were then utilized for polymerization after 48 hours at ambient temperature and atmosphere. The PDA substrates were immersed in monomer solution and then irradiated with a collimated 405 nm LED lamp for 12 hours. MMA, and Styrene monomers were filtered through a plug of basic alumina and then used neat (without solvent). SBMA and MPC were used without further purification, but dissolved in H₂O at a concentration of 0.5 g/mL prior to immersion of the PDA surface. In some cases the polymerizations were irradiated for shorter or longer periods of time. The polymerized surfaces were then rinsed and soaked overnight in solvent to remove unreacted monomer, changing solutions 5 times.

For some of the polymer films polymerization was re-initiated to produce a second block. For these reactions a previously polymerized wafer sample was immersed in solution containing monomer and a thermal radical initiator. For MMA, AIBN was used as a radical initiator, and dissolved in the neat monomer at a concentration of 0.4 mol was used neat, after filtration through basic alumina, while SBMA was prepared at 0.5 g/mL in H₂O. The petri dish was then placed inside a larger polymerization dish, which was sealed and purged with N₂ for 5 minutes.

2. Results and Discussion

The first thing that needed to be validated was the formation of the PDA layer, as reported previously. The oxidative crosslinking of dopamine was performed from a pH 8.5 solution under UV irradiation. As noted by Sheng et al., UV irradiation accelerates the PDA formation, so based on their findings, a 3 hour irradiation time was selected for this study, to target a moderate PDA film thickness.²⁹ Initial experiments indicated an approximate PDA thickness of 15 ± 3 nm over the course of 10 replicates. These samples were then measured by atomic force microscopy (AFM) to further characterize the PDA layer (Figure 29 below).

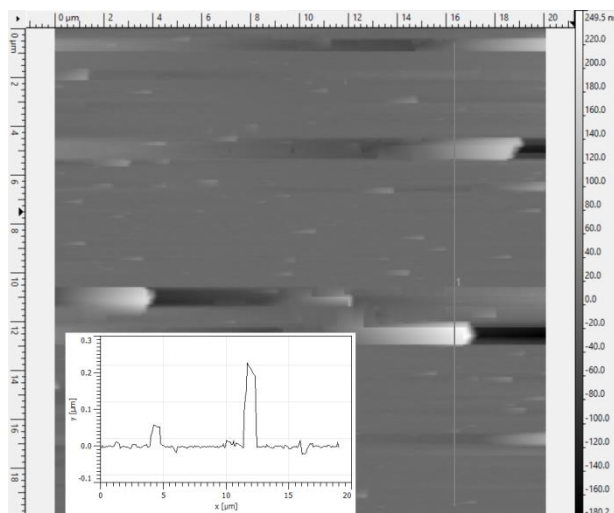


Figure 29: AFM scan of a 20 x 20 μm square from a PDA-coated SiO_2 substrate (3 hr. irradiation). The insert is a line-scan conducted along the grey line in the main figure. The RMS roughness of this sample is 24.9 nm.

This AFM image is of an as-prepared PDA-coated SiO_2 sample, showing several large aberrations on the surface. There appears to be large particles that deposited on the PDA surface up to 200 nm in size. These particles are presumably particles of polydopamine that aggregate and grow in solution and subsequently settle and deposit on the substrate. This concept is consistent with previous research.^{88,178,179} However, these large surface heterogeneities are deleterious to ultimate fouling performance, and potentially affect the surface polymerization, therefore, strategies to eliminate and/or minimize them were explored. The first strategy employed was degassing of the dopamine solution by sparging with nitrogen gas prior to surface deposition. Aside from degassing, the PDA surface was prepared identically as before, and a representative AFM image is shown below.

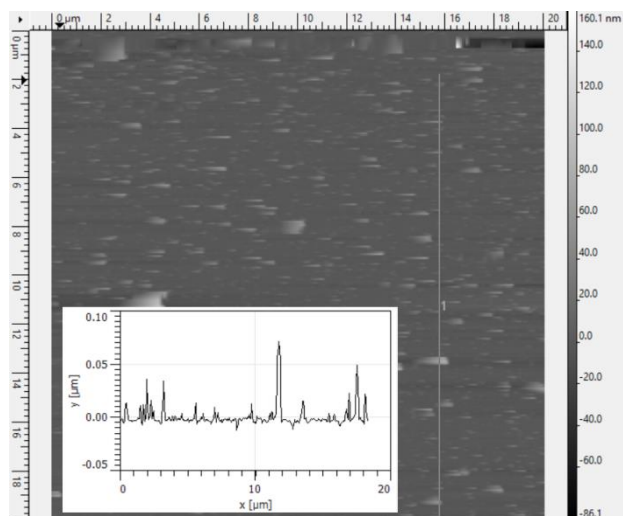


Figure 30: AFM scan of a 20 x 20 μm square from a PDA-coated SiO_2 substrate (degassed dopamine solution, 3 hr. irradiation). The insert is a line-scan conducted along the grey line in the main figure. The RMS roughness of this sample is 11.7 nm.

This procedure already showed some improvement over the previous result, with much smaller aggregates present of around 50 nm in size, and an RMS roughness of 11.7 nm compared to 24.9 nm previously. However, a surface with an RMS roughness of less than 10 nm and with minimal clusters and heterogeneities was desired. In a separate run PDA surfaces were prepared using the usual procedure (no degassing), but the substrates were submitted to 5 minutes of sonication following the 3 hours under irradiation. The resulting surfaces were imaged with AFM as shown in Figure 31.

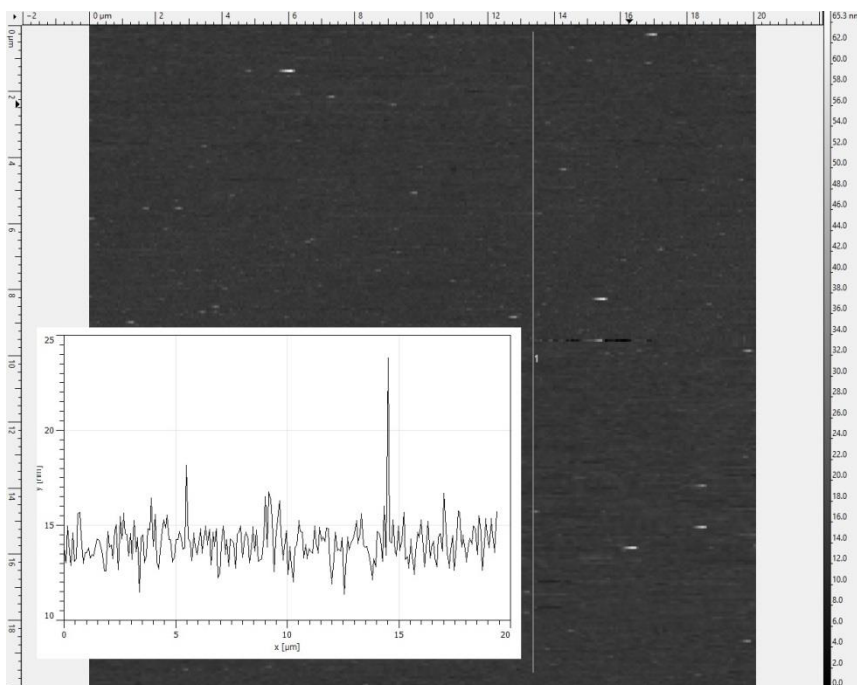
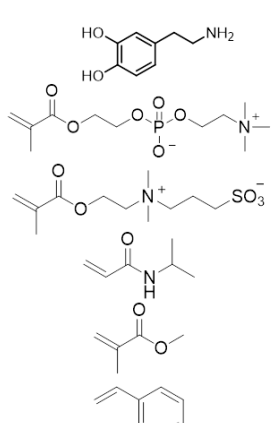


Figure 31: AFM scan of a 20 x 20 μm square from a PDA-coated SiO_2 substrate (3 hr. irradiation, followed by 5 min sonication). The RMS roughness of this sample is 1.5 nm.

This sample contains noticeably fewer particles, with a smaller size of approximately 10-20 nm. In addition, the RMS roughness is only 1.5 nm, showing a marked improvement over the previous conditions. Interestingly enough, this observation of improved surface roughness was corroborated independently a few months later by a recent publication on the topic of smooth PDA surfaces.¹⁸⁰ The authors of that work also assert that sonication of PDA surfaces greatly reduces the roughness and prevalence of particles, which is excellent validation for these independently-determined conclusions. Moving forward, the PDA surfaces were always submitted to 5 minutes of sonication following irradiation in dopamine solution.

Using this new procedure, the ability to conduct surface polymerizations from these PDA surfaces was validated. Ultimately several monomers were successfully polymerized from the PDA surfaces, as demonstrated by film thickness and corresponding contact angle changes (Figure 32 below).



Polymer	Thickness [nm]	Contact Angle (adv.) [°]
PDA	9.0 ± 0.5	49 ± 1
P(MPC)	25.1 ± 1.7	< 5
P(SBMA)	18.9 ± 1.1	15 ± 1
P(NIPAM)	15.2 ± 1.0	64 ± 1
P(MMA)	2.8 ± 1.5	74 ± 1
P(S)	6.9 ± 2.2	82 ± 1

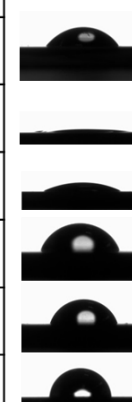


Figure 32: Films thicknesses and contact angles for polymers grown from PDA-coated SiO₂ substrates. Thickness values reported correspond just to the additional contribution from the polymer layer (i.e., for P(MPC) there is 25.1 nm of polymer on top of the original 9.0 nm of PDA). The values reported are the average and standard deviation for 3 replicate polymerizations.

An appreciable thickness increase for most of the polymers grown (except for P(MMA) and P(Styrene)) was observed, and the measured contact angles for each of these films was consistent with expected values. Each polymerization was conducted in triplicate to get a cursory idea for reproducibility. One of the striking features of this data is that P(MMA) and P(Styrene) did not grow very much, and were limited to extremely thin films. This may likely be due to chain ends becoming terminated prematurely, which is to be expected in an uncontrolled polymerization. This presented an opportunity to explore the capabilities of

PDA polymerizations further. Specifically, it was hypothesized that a molecule that reversibly comes on and off the growing polymer chain-end could prevent premature termination and enable thicker films to be polymerized. This is the basis of many controlled radical polymerizations. Indeed, inspired by reversible addition-fragmentation chain transfer (RAFT) polymerization, the chain transfer agents that are employed in RAFT were added to the PDA polymerizations. Specifically RAFT relies on the use of dithioesters as so-called RAFT agents that serve to reversibly cap chain-ends during the propagation of the polymerization. After termination of RAFT polymerizations the resultant polymers are then capped by the RAFT agent.

The present study employed 4-cyano-4-(((dodecylthio)carbonothioyl)thio)pentanoic acid (pictured below) as a RAFT agents, which will be referred to generically as a chain transfer agent (CTA).

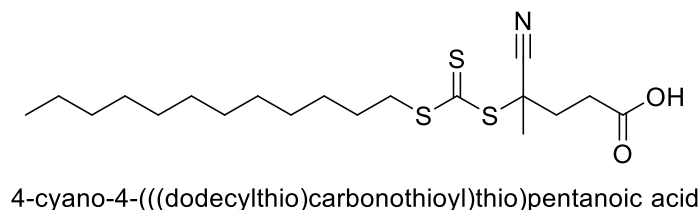


Figure 33: Chain transfer agent (CTA) used to cap the ends of polymer chains grown from a PDA initiating surface.

First, this hypothesis was tested with the polymerization of MMA by introducing small amounts of the CTA in with the monomer during polymerization. Using previously reported RAFT conditions as a starting point for appropriate concentrations,^{149,181–183} four concentrations of CTA were screened in order to determine the best concentration for this application.

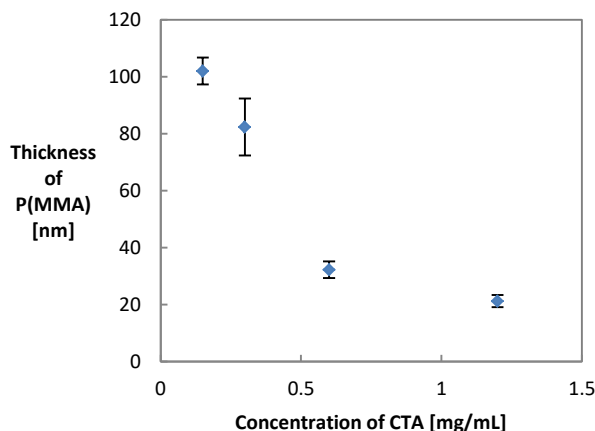


Figure 34: P(MMA) film thickness is inversely proportional to CTA concentration within the range of values tested. The data points reflect the average and standard deviation for 3 replicate experiments.

All the concentrations used showed a marked improvement in thickness of P(MMA) after 12 hours of irradiation (compared to 2.8 nm without CTA as shown in Figure 32 above). This is nice validation that for these reaction conditions the CTA enabled film thicknesses that were previously inaccessible. It was observed that lower concentrations seem to help achieve thicker films for P(MMA). Moving forward, the lowest concentration of 0.15 mg/mL was used to try to access the thickest films possible. MMA, SBMA, and Styrene were polymerized with the addition of the CTA to investigate its effect, the resulting polymer films are shown in Table 3 below.

Another benefit to the addition of the CTA is the potential for reactivation of chain ends after polymerization. To test this hypothesis MMA, SBMA and Styrene films polymerized with a CTA were submitted to a second polymerization step. In this second step both MMA and SBMA were used as the second blocks. For MMA, AIBN was used as a radical initiator to reactivate the CTA-capped polymer chains. The conditions employed were inspired by

previous reports.^{119,152,184} Similarly, with SBMA, ACVA was used as a radical initiator, with reaction conditions being inspired by previous work.^{184–186} Both the radical initiators were activated thermally, so these polymerizations were conducted at elevated temperature of 65 °C. Below the thickness data for the polymers grown using CTA, and their subsequent re-initiation is compiled.

Table 3: Polymer films grown from PDA in the presence of a chain transfer agent (CTA) compared to polymerizations without CTA. The CTA-terminated films were reinitiated and the subsequent diblock copolymer thicknesses are reported. Thickness values represent the average and standard deviation of 3 replicate polymerizations.

First Block	Starting Thickness [nm]	Second Block	Additional Thickness [nm]	Final Thickness [nm]
MMA	2.8 ± 1.6	MMA	-----	2.8 ± 1.6
MMA + CTA	74.1 ± 1.6	MMA	79.1 ± 9.2	153.2 ± 9.2
MMA + CTA	74.1 ± 1.6	SBMA	23.7 ± 1.7	97.8 ± 1.7
SBMA	18.9 ± 1.1	MMA	-----	18.9 ± 1.1
SBMA + CTA	13.1 ± 2.5	MMA	11.5 ± 1.9	24.6 ± 1.9
Styrene	6.9 ± 2.2	MMA	-----	6.9 ± 2.2
Styrene + CTA	28.2 ± 2.5	MMA	17.4 ± 1.8	45.6 ± 1.8

For MMA and Styrene monomers the CTA markedly improved the polymerization thickness after 12 hours irradiation. This is excellent confirmation of the hypothesis that CTA may help polymerize monomers that were previously inaccessible. For SBMA a similar thickness increase was not observed, however an appreciable polymer growth was still achieved.

When conducting a second polymerization on these surfaces, no additional polymer growth was observed for the control samples where CTA was not introduced. This is good

affirmation that the termini of these polymers cannot be reactivated. In contrast, the polymer films that were terminated with CTA showed noticeable growth of the second block after being submitted to subsequent polymerization. Specifically, a second block of MMA of approximately 79 nm was grown on top of a CTA-terminated P(MMA) block (originally 74 nm) to give a total of 153 nm. When SBMA was grown from this same starting P(MMA)-CTA film an additional 24 nm of polymer in the second block was observed. Similarly, successful re-initiation of P(SBMA)-CTA and P(Styrene)-CTA using MMA was achieved. To further support this thickness data, a few of these polymer films were analyzed by XPS as shown in Figure 35. Specifically, PDA-coated SiO₂, P(MMA)-CTA, P(SBMA)-CTA, P(SBMA-b-MMA), and P(MMA-SBMA) samples were all measured.

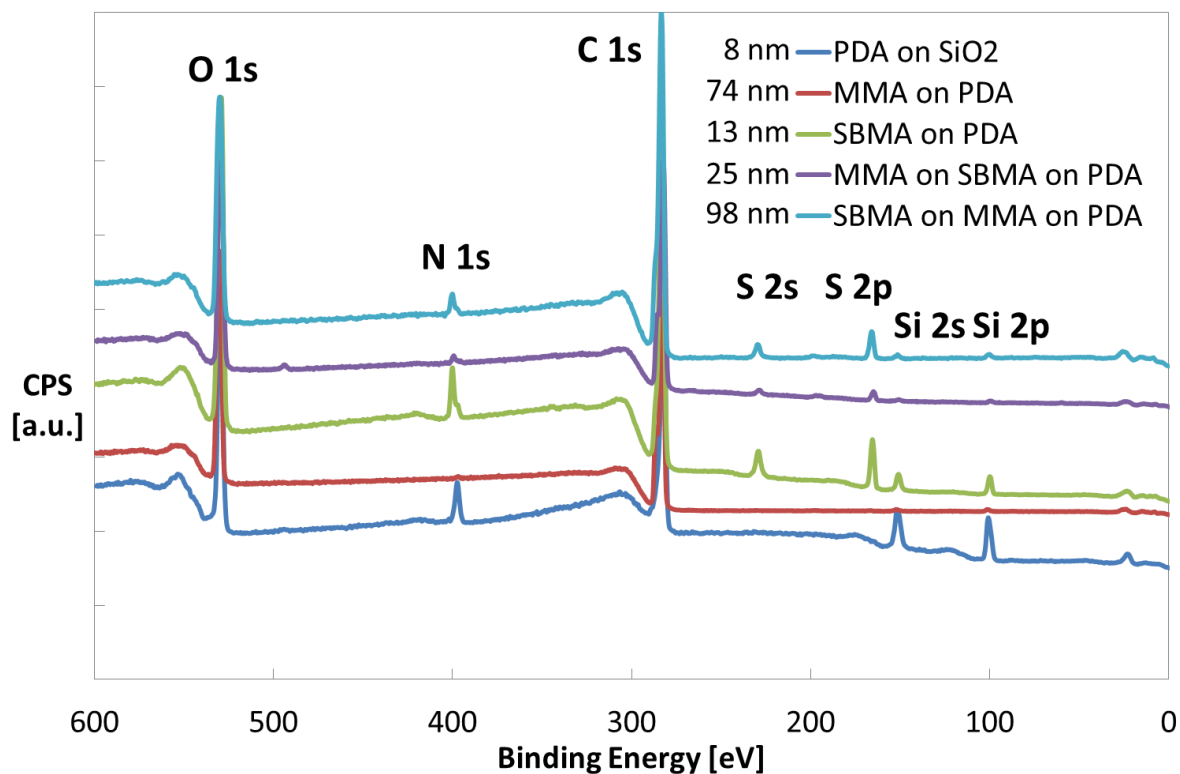


Figure 35: XPS data for various polymer films grown from PDA-coated SiO₂ substrates. O 1s, N1s, C 1s, S 2s, S 2p, Si 2s and Si 2p peaks are centered around 532, 399, 284, 229, 164, 149, and 100 eV respectively (as noted in the figure).

Each of the films studied shows the expected XPS signatures, helping to confirm the successful functionalization at each step. The PDA-coated substrate shows a prominent N 1s peak which disappears with subsequent polymerization of MMA. The SBMA film shows the presence of N 1s and S 2s / S 2p peaks to indicate the zwitterionic monomer on the surface. However, when MMA is polymerized on top of the SBMA the nitrogen and sulfur signals are significantly suppressed. This is expected as the typical penetration depth of XPS is approximately 10 nm, so the top layer of MMA masks the SBMA beneath. And lastly when SBMA is polymerized atop MMA, again the nitrogen and sulfur peaks are suppressed as expected.

D. Comparing ATRP- and PDA-Grown Polymer Films

PDA-initiated surface polymerizations are a promising method for facile, scalable, and versatile fabrication of polymer-coated substrates. However, this approach inherently generates a multi-layered structure with the PDA itself residing under the subsequent polymer that is grown from it. The effect that this underlying layer may have on the ultimate antifouling performance of the polymer-PDA multilayer is previously unknown and provides powerful insight into how these multi-component polymeric coatings behave. In addition, this sort of layered structure is very reminiscent of water filtration membranes. Specifically, RO membranes are composed of a multi-layered structure known as a thin-film composite. In the case of many commercial RO membranes, a negatively-charged, polyamide top layer performs the majority of the separation. This polyamide layer is then treated or coated with an antifouling coating to improve fouling resistance. The PDA-initiated polymer films serve as a convenient model system to explore the role of these multi-layered motifs. This study aimed to investigate two monomers, SBMA and Styrene grown with the two contrasting surface polymerizations to determine how the underlying PDA layer affected fouling performance.

1. Experimental Methods

Polymer brushes composed of SBMA and Styrene were grown using SI-ATRP and PDA methods separately. The synthesis of these materials is detailed in sections 4 and 3 of Materials and Methods, respectively. In order to enable direct comparison between the two different monomers, polymers of similar thickness were targeted, so thickness effects could be neglected. Polymer thicknesses of approximately 15-20 nm were targeted. For the purposes of optimization, the polymerization conditions were adjusted while monitoring the

thickness as grown on SiO₂ wafers. Once suitable reaction conditions were identified, the polymerizations were translated onto SiO₂-coated QCM-D sensors to then enable fouling experiments. Fouling experiments were conducted by mounting the QCM-D sensors, and allowing 48 hours of equilibration in pH 7.4 PBS buffer. After equilibration, the foulant is flown through the QCM-D sensor chamber for approximately 2 hours, after which the foulant solution is flushed out with PBS buffer.

One of the key elements to this study is knowing the surface charge of the polymer coatings generated. In order to measure the surface charge zeta potential measurements were conducted using a Zetasizer Nano ZS (Malvern, Worcestershire, UK). The best approach for measuring zeta potential is to coat nanoparticles with the material of interest. Then these coated nanoparticles can be measured in suspension. For these studies SiO₂ nanoparticles with an average diameter of approximately 140 nm were coated with PDA in an identical procedure as is used to coat planar substrates. These PDA-coated nanoparticles were also employed for surface polymerization and similarly analyzed by zeta potential.

2. Results and Discussion

The first pressing question was whether the PDA layer itself has a charged nature. Zeta potential is one method for addressing the question. Specifically, zeta potential measures the electrostatic potential at the electric double layer created around a charged surface in solution.^{92,187} These measurements are conducted by measuring light scattered from suspended particles, therefore, SiO₂ nanoparticles were coated with PDA. The particles were imaged by scanning electron microscopy (SEM) before and after coating, as shown below.

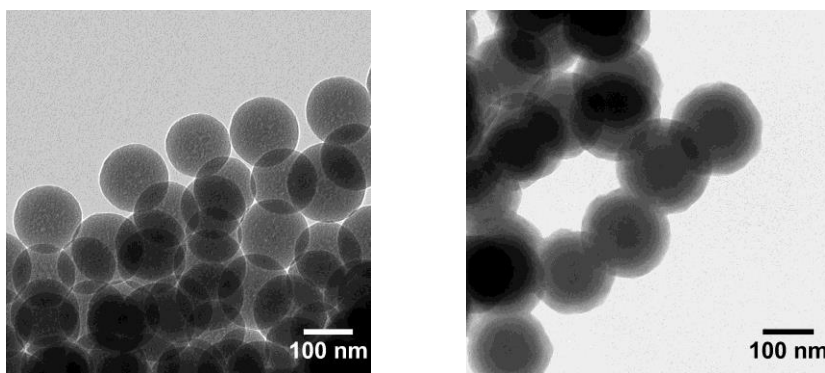


Figure 36: TEM images of SiO₂ nanoparticles before and after coating with PDA (left and right, respectively). The PDA “shell” is approximately 15 ± 2 nm in thickness.

The PDA-coated SiO₂ nanoparticles were suspended in a buffered solution of pH ranging from 3 – 11, and the zeta potential of this suspension was measured. These experiments were conducted at a concentration of 0.2 mg/mL of nanoparticles in buffer solution (concentration was determined as outlined in Supporting Info).

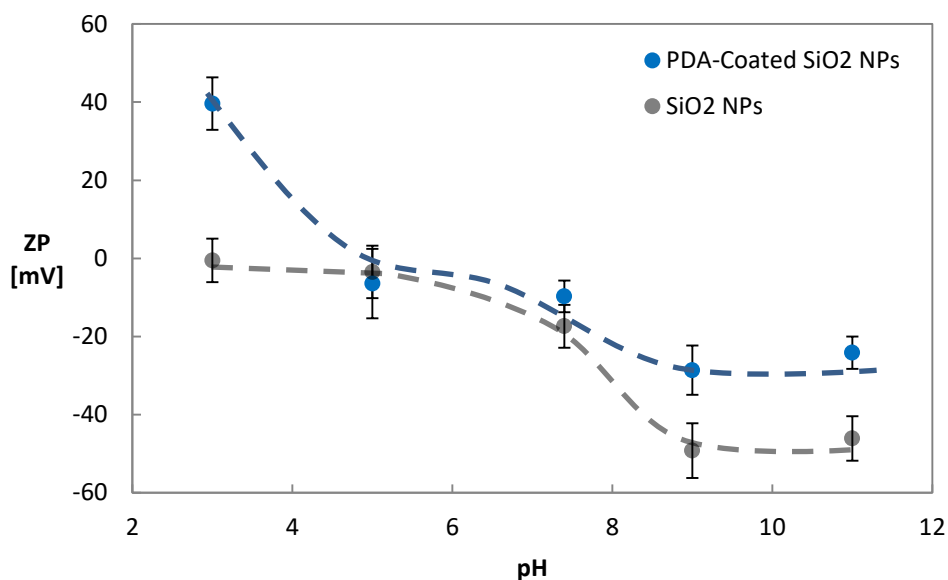


Figure 37: Zeta potential measurements for PDA-coated (blue) and bare SiO₂ (grey) nanoparticles across various pH. The average and standard deviation of five replicate runs are plotted. The dashed lines are added to guide the eye.

Zeta potential values from approximately -10 to 10 mV are considered neutrally charged, while values of higher absolute value are considered either negatively or positively charged according to the sign. This data shows that the SiO₂ nanoparticles are neutral in acidic pH, slightly negative at neutral pH, and become highly negatively charged at elevated pH. However, upon coating with PDA, the zeta potential was increased across the whole range of pH (except for pH 5). This is indication that the PDA layer is more positively charged, and results in a more neutral surface, particularly at moderate pH.

The PDA-coated SiO₂ surfaces were prepared from cleaved pieces from a larger wafer, and subsequently employed for surface-initiated polymerization. Specifically SBMA and Styrene monomers were grown from the PDA-coated substrates. The polymerization of each monomer was optimized to target a thickness of approximately 20 nm. Polymerizations were modified by changing the concentration of monomer, the polymerization time, and the irradiating light source. For these experiments two light sources were used: a 405 nm collimated LED, and a 100 W solar simulator. Each monomer was tested with both light sources, as well as at various concentrations, and using polymerization times ranging from 1 hour to 26 hours. The optimized conditions consisted of SBMA dissolved in water at a concentration of 0.5 g/mL using a 405 nm LED light, while Styrene was polymerized neat without solvent using a solar simulator. The resulting polymer thicknesses as measured by interferometry are shown in Table 4.

Table 4: Optimized reaction conditions for the polymerization of SBMA and Styrene from PDA-coated SiO₂ substrates. The values reported are the average and standard deviation of 20 thickness measurements taken over the entire surface area of the sample.

Monomer	Light Source	Time [hr]	Thickness [nm]
SBMA	405 nm	16	19.7 ± 1.2
			18.8 ± 1.5
			20.6 ± 0.5
			18.4 ± 0.9
			21.9 ± 1.0
			22.2 ± 0.9
Styrene	Solar Simulator	24	22.9 ± 1.4
			20.1 ± 2.5
			21.6 ± 2.5
			21.5 ± 2.0
			24.4 ± 1.5
			26.1 ± 2.3

The values reported in the table give an average and standard deviation of thickness over the surface, serving as a measure of the spatial homogeneity of each film. These conditions yielded P(SBMA) and P(Styrene) films of 20 ± 2 nm and 22 ± 2 nm respectively.

The next step is to synthesize P(SBMA) and P(Styrene) using SI-ATRP conditions to achieve film thicknesses of approximately 20 nm.

A. Fouling

In order to conduct QCM-D experiments with these films, SiO₂-coated QCM-D sensors were functionalized and polymers were grown from them to generate the desired surface chemistries. The amount of deposited foulant on each surface was calculated both while the foulant solution is in contact with the sensor, as well as after rinsing with buffer. In both cases the Sauerbrey Equation is employed. The computed foulant mass for all the foulants and surfaces tested are shown in Figure 38.

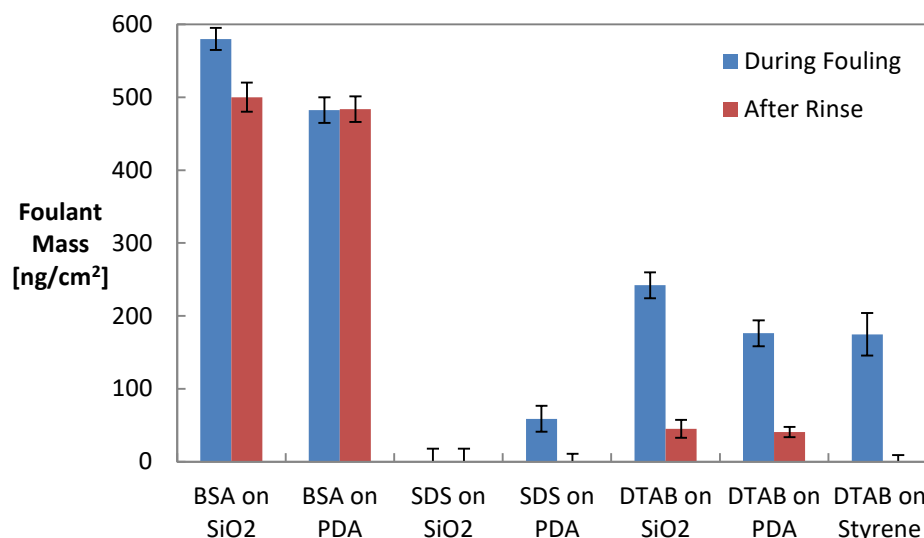


Figure 38: Foulant mass for BSA, SDS, and DTAB foulants as measured by QCM-D onto various surface chemistries. Values reported represent the average and standard deviation of 4 replicate experiments.

The fouling results indicate that SDS is significantly repelled from the negatively charged SiO₂ surface. In addition, the application of a PDA or another polymer layer seemed to universally improve fouling resistance after rinsing. This also translates to an improved ability to remove loosely adsorbed proteins, as evidenced by the ratio of foulant mass during and after fouling.

V. Thermally- and pH-responsive Polymers

The development of fouling resistant surfaces is crucial to many industries ranging from biomedicine to water filtration.^{78,90,172,188,189} Polymer-coated substrates are an attractive solution for fouling resistance as they serve as a versatile platform for modifying and controlling surface chemistry. One such approach to antifouling relies on employing

thermosensitive polymers to develop surfaces that can interact differently with foulants depending on temperature.^{190–195} For example, poly(N-isopropyl acrylamide) (PNIPAM), has been shown to release foulants when it undergoes the conformational change at the lower critical solution temperature (LCST).¹⁹¹ Below this temperature, the polymer is hydrophilic and well dissolved in water, however above the LCST, the polymer becomes hydrophobic, forms aggregates, and precipitates out of aqueous solution.¹⁹⁶ This same basic concept is used with a number of thermoresponsive, fouling resistant systems.^{197–200}

Poly(ethylene oxide) (PEO) also has an LCST and its intrinsic biocompatibility and protein resistant nature have made it prominent in biomedical and in antifouling applications.^{201–203} PEO's LCST has been extensively studied,^{201,204–209} however at around 100°C,²¹⁰ it is not practical for applications at body or room temperature.

Increasing the the polymer's molecular weight, or the polymer's hydrophobicity as well as increasing solution's ionic strength, have all been shown to suppress the LCST.^{198,204,211} However, these parameters can be inconvenient for certain applications. The goal was to suppress the LCST of PEO to a more accessible and application-relevant temperature range. This can be accomplished through copolymerization with a pH sensitive moiety, diisopropyl ethanolamine glycidyl ether (DEGE), that has a pK_a value of 9.²¹² As shown in Figure 39, protonation of the tertiary amine increases the hydrophilicity of the unit. Above the pK_a of DEGE, the amine is deprotonated and hydrophobic which should lower the LCST of the copolymer when pH is above the pK_a , whereas below the pK_a , DEGE moieties would be hydrophilic, achieving the inverse.

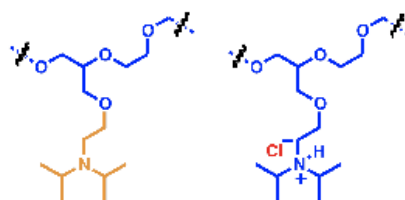


Figure 39: pH dependent protonation of DEGE.

These studies also seek to characterize how the ionic strength and pH of the solution, as well as the monomer ratio decrease the LCST of this copolymer, poly(ethylene oxide-co-diisopropyl ethanolamine glycidyl ether) (P(EO-co-DEGE)). Ultimately this would facilitate the development of fouling resistant and fouling releasing surfaces for a range of applications.

A. Materials and Methods

A number of techniques are used to determine the LCST. These include visual observation of the solution turning cloudy due to aggregate formation,^{204,213} small-angle neutron scattering,²¹⁴ and rheology,²¹⁵ though, light scattering methods are considered sufficiently accurate for LCST detection in most cases.^{204,211,213,214}

Using a Wyatt Dynapro Nanostar (Wyatt Technology Corporation, Santa Barbara, CA) dynamic light scattering instrument, the LCST of polymer solutions was determined. The LCST was identified from the temperature dependence plot as the point at which the aggregates' hydrodynamic radius began increasing drastically.

Samples were prepared with different monomer ratios, pH values, and increasing ionic strength. The monomer ratios tested were of 10, 20, 30, and 40 mol% DEGE content. The solutions for pH 6, 8, and pH 11 were respectively buffered with acetic acid and sodium acetate trihydrate, phosphate buffered saline tablet (Sigma Aldrich), and sodium hydroxide

with sodium bicarbonate. The ionic strength (μ) of the solutions were increased in a step-wise fashion by adding sodium sulfate salt.²⁰⁴

To validate this methodology, results from Pang & Englezos (2002) were replicated. Specifically, LCST values for PEO ($M_v = 5\text{MDa}$) were measured at different salt concentrations of sodium sulfate in aqueous solutions. Visual observations of the samples' cloud point temperature as well as DLS measurements were conducted for accurate measurements.

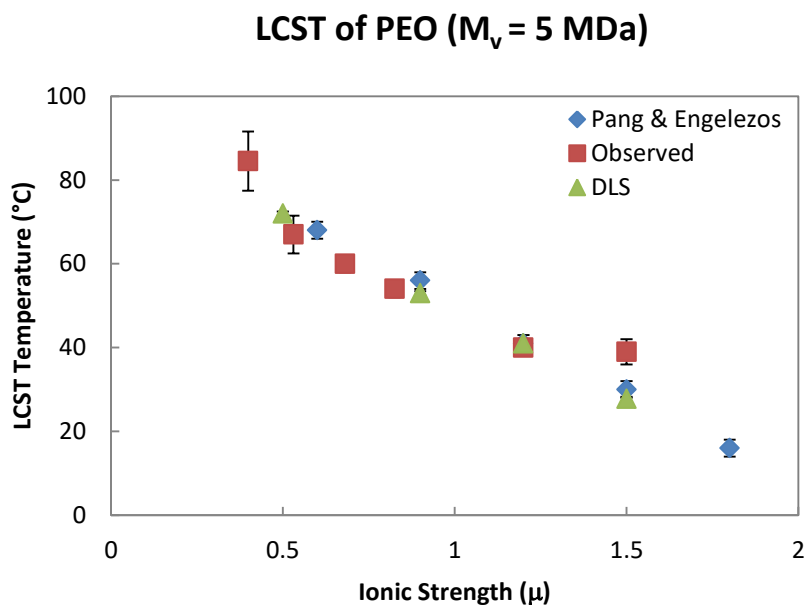


Figure 40: LCST of 5 kDa polyethylene oxide (PEO 5k) compared to previously published values to validate the proposed methodology.

The ability to measure the LCST of PEO in aqueous solutions using both visual inspection and by DLS measurements was in excellent accordance with the previously reported values.²⁰⁴ While visual evaluation of solution LCST was quite reliable, it remained a tedious task for large numbers of replicate experiments. Instead the DLS was utilized as a screening method for determining LCST.

B. Results and Discussion

Through DLS the LCST values for different copolymers, pH values, and ionic strengths were obtained. As shown in Figure 2A, ionic strength suppressed the LCST of these copolymers even at different pH. At a pH above DEGE's pK_a of 9, the LCST was significantly suppressed. This is as expected since above the pK_a , the DEGE is in a deprotonated hydrophobic state and thus less soluble.

For higher DEGE contents slightly different results were observed. While pH values higher than the pK_a still suppressed the LCST and increased DEGE content resulted in greater LCST suppression (Figure 3), the ionic strength dependence seems to be milder (Figure 2B). It seems as though the hydrophilicity of DEGE has a greater impact than the effect of ionic strength. This is clearest in Figure 2B, when the ionic strength dependence seems to be completely eliminated when the pH is significantly lower than the pK_a yet it is still present, although weakly, at higher pH values.

The LCST was successfully manipulated across a range from 96°C to as low as 12°C by adding a pH responsive moiety and increasing the pH. As seen in Figure 3, the suppression can be controlled by the amount of DEGE in the copolymer as well as by the pH of the solution.

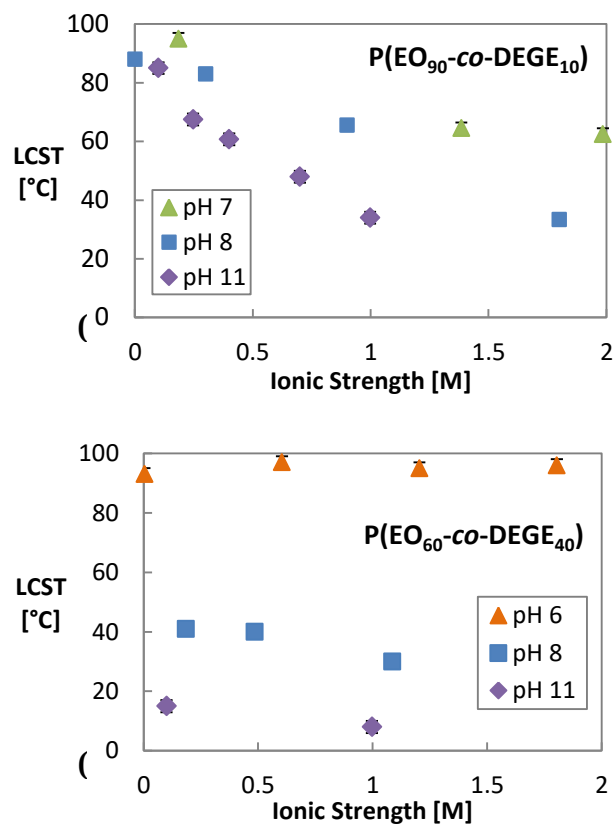


Figure 41: Effect of pH, ionic strength, LCST of two P(EO-co-DEGE) copolymers. Two separate copolymer compositions, P(EO₉₀-co-DEGE₁₀) (A) and P(EO₆₀-co-DEGE₄₀) (B) are shown.

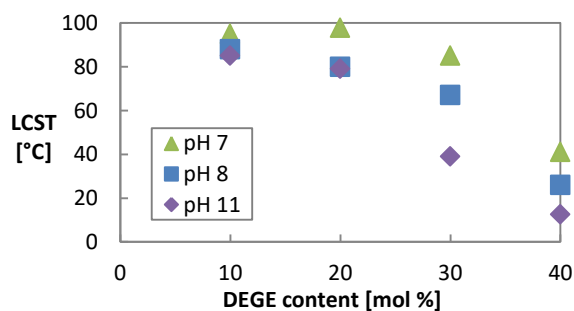


Figure 42: LCST response to increasing DEGE content. Ionic strength is minimized to that of the buffered solution used.

With this data copolymers with an LCST suitable for a given application can be precisely designed. In the case of antifouling, a surface could be refreshed without exposing

it to potentially damaging temperatures. This sort of “foulant release” has particular interest for antifouling, drug delivery, and sensing applications.

C. Conclusions and Future Work

Demonstrated herein is an approach for systematic and tunable modification of the lower critical solution temperature of polyethers. By introducing another co-monomer along with ethylene oxide the LCST of PEO was successfully, and predictably suppressed. Specifically the use of DEGE not only modified the LCST behavior of PEO, but also imparted pH responsiveness to the resulting copolymers. This enables two stimuli trigger the LCST transition of these materials. Moreover, this hydrophilic/hydrophobic transition that occurs at the LCST is a powerful property that can be used for a diversity of applications. One practical application for example would be in the release of encapsulated cargo embedded within a hydrogel composed of EO and DEGE copolymers. A hydrogel can be designed such that hydrophobic domains within a hydrogel containing a payload of interest can be triggered to cross the LCST. This transition would convert these hydrophobic domains to hydrophilic ones, effectively releasing the cargo contained, and dissolving the hydrogel. Since both pH and temperature can be employed for this, materials can be designed to have LCST behavior at appropriate physiological conditions.

Future studies of EO and DEGE hydrogels would enable further demonstration of the utility of temperature and pH responsive polymers. In particular an interesting study would be to observe the thermal behavior of single-component hydrogels composed of copolymers of EO and DEGE monomers. It would be an important observation to see that the polyethers retain their LCST behavior when incorporated into hydrogels or other constructions.

VII. Conclusions

Understanding the complex nature of fouling and how it can be mitigated requires a profound knowledge of the relevant foulants (and the associated deposition mechanisms), the surface chemistry, and the solution in which the two interact. These parameters are typically interconnected and need to be considered in tandem. There has been a plethora of research aimed at building more profound comprehension of how fouling and its resistance can be imparted to antifouling systems.¹⁻⁴ For most antifouling research two major components are typically studied in a laboratory setting: surface/interfacial properties, and fouling properties. Methods for modifying surface properties include: changing chemical composition, manipulating the conformation of surface-active components, and imparting surface roughness. In practice these surface properties can be achieved by a number of strategies, however, polymer-coatings have largely been at the forefront of this field in the past fifteen years.¹ Fouling performance can be measured as the adsorption of various foulants under different ambient conditions. However, due to the vast range of antifouling applications, there is no universal metric for evaluating fouling resistance. Both the synthesis of antifouling polymer films and the subsequent evaluation of their fouling resistance were conducted herein.

In this thesis the control and manipulation of surface properties was achieved by the use of surface-tethered polymers. A grafting-to technique, employing catecholic adhesive groups was used to make PEO-based polymer brushes. The resulting brush layers showed appreciable antifouling character, but were ultimately hindered by their lower grafting density. In order to improve the grafting density, surface-initiated ATRP was employed as a grafting-from technique to grow polymer brushes. This approach afforded higher grafting

density and better control over the brush thickness, but was limited to low throughput. A third technique was used to create polymer brushes using a polydopamine (PDA)-based surface polymerization. The PDA method enabled facile polymer brush synthesis with high throughput with moderate control of polymer thickness. In addition, it was demonstrated that the use of chain-transfer agents are an effective method for enabling block copolymerization from PDA-initiated polymer films. Moreover, the PDA-initiated films provided a convenient platform to test the effect of charged layers on the antifouling performance of polymer films. Comparing the fouling resistance of ATRP-grown and PDA-grown films revealed that the positively charged PDA layer does have an impact on fouling resistance. The PDA layer itself attracts oppositely charged (negative in this case) foulants to the surface, while comparatively repelling positively charged foulants.

The thermal and pH response of novel polyethers composed of ethylene oxide (EO) and diisopropyl ethanolamine glycidyl ether (DEGE) were explored, namely by investigating the ways in which the lower critical solution temperature (LCST) of such materials can be manipulated. It was determined that higher ionic strength and higher pH both suppress the LCST. Indeed, LCST values ranging from 12 – 96 °C were accessible using copolymers of EO and DEGE. This hydrophilic to hydrophobic transition that occurs at the LCST is a powerful way to manipulate the solution properties of polymers, and has particular use for fouling release applications of surface-tethered polymers. Specifically, the actuation of a polymer brush collapsing and extending can dislodge adsorbed foulants, as demonstrated herein by the thermally-triggered fouling release. While these initial experiments demonstrate fouling release, further investigation and optimization of this property is required moving forward. Based on these studies it was determined that a system with

short-term fouling resistance that also has thermal and/or pH responsive character would be the ideal design. This would enable surfaces to employ fouling release to refresh the surface as foulants accumulate over time, while also minimizing the frequency with which these surfaces need to be refreshed.

In this thesis the fouling performance of the polymer films synthesized was conducted using various strategies, most notably, quartz crystal microbalance with dissipation (QCM-D) and fluorescent microscopy to measure adsorption of fluorescently-labelled proteins. Both techniques enabled the direct quantification of foulants deposited onto a substrate, though QCM-D affords real-time monitoring of fouling. The quantification of fouling performance is paramount to being able to determine how modifications of surface properties affect the final characteristics of the synthesized polymer brushes. Although there exist several indirect methods for measuring fouling resistance, these techniques typically offer a poor representation of the extent of fouling. For this reason additional methodology to quantify fouling propensity under various conditions are in demand and should be explored further.

A. Future Work

This thesis encompasses an extensive library of polymer surface-coatings and accompanying antifouling performance quantification. Future work in this realm of antifouling requires careful control of surface chemistry and topology as was demonstrated with this body of work. While polymer coatings present a convenient method for manipulating surface properties, their ultimate utility as an investigative platform depends heavily on the reproducibility of polymer-modified surfaces. Many techniques explored herein have been well-documented in the literature, but inherently are difficult to obtain

consistent batch results. For this reason, developing carefully optimized reaction conditions that yield consistent surfaces with controllable film thickness (to within 1-2 nm), surface area, and roughness is paramount. While roughness as a variable was not explicitly explored in this thesis, it has been shown to be a critical component to fouling performance.⁵ Roughness in polymer films can be generated by several methods, including: 1) increasing the polydispersity of the polymers grafted to the substrate, 2) generating a surface with intrinsic surface roughness and subsequently functionalizing the surface with polymers, 3) designing surface-tethered polymers that will self-assemble or segregate on the surface to create topological roughness. Other approaches may include destructive techniques whereby a synthesized polymer surface can be intentionally roughened by etching away selective areas of a polymer brush. All these surface roughness techniques can then be quantified by AFM or other film thickness measurements such as XRR to ensure reproducibility and control.

As described previously, polymers with adhesive moieties at both termini have potential use in antifouling applications due to the steric constraints placed on these surface-tethered telechelic polymers.⁶⁻¹⁰ However, ascertaining the actual confirmation of this class of material is a practical challenge that has not been achieved experimentally. While there have been theoretical studies on how telechelic polymers deposit onto solid interfaces in a grafting-to procedure, understanding how this is practically manifested could enable insight into how surface conformation can be controlled. Collecting this data would require designing and synthesizing telechelic polymers that have adhesive moieties and a labelling moiety at both termini. This labelling moiety would be selected such that it allows the determination of the distance between it and the underlying substrate. Measuring this

distribution of distances of polymer termini from a surface would lend itself nicely to being paired with molecular dynamics simulations that would help inform these experiments. In doing so, the combination of experimental and simulation work would shed light on this long-standing curiosity in the field of polymer surface chemistry.

As demonstrated in this thesis, the ability of polydopamine (PDA) to deposit onto a variety of surfaces and subsequently initiate polymerization of common monomers is both versatile and practical. This method of grafting-from requires minimal preparation and can be conducted using basic lab equipment. Another potential application for this technology is for the polymer modification of nanoparticles. Polymer-coated nanoparticles have a number of applications including as contrast agents for medical imaging and as drug delivery vessels.¹¹⁻¹³ Currently these nanoparticle modifications necessitate complicated work-up and synthetic steps in order to attach polymerization initiators onto the surface and subsequently conduct polymerization. Using the PDA method developed herein the conformal coating of nanoparticles could be significantly simplified. The PDA coating of nanoparticles can be applied to number of different kinds of nanoparticles, independent of surface chemistry, in one step. This may also enable the “cloaking” of certain kinds of nanoparticles for biomedical applications. To highlight the versatility of the PDA approach, several nanoparticles chemistries (metals, oxides, and polymeric particles) and several monomer classes (such as methacrylates, acrylates, and styrenes) should be demonstrated.

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Appendices

Grafting-To: Amine-Epoxyde

Despite the inherent benefits of grafting previously-synthesized polymers to surfaces, grafting-to is traditionally limited to low grafting densities.^{1,2} In this context, grafting density describes the areal density of polymer chains attached to a surface. Among other factors, grafting density is strong predictor of antifouling performance for polymer surfaces, where higher grafting densities generally lead to better resistance to fouling.^{3,4} However, there have been recent experimental efforts that demonstrate methods that help overcome the grafting density limitations of grafting-to. Briefly, the poor grafting densities achieved by grafting-to are a result of the excluded volume of the individual, solvated polymers chains in solution. In their solvated state (specifically when in a good solvent), the polymers exist in a random coil configuration with a radius of gyration, R_g , that is proportional to $N^{3/5}$.^{5,6} However, when these same polymer chains are in the melt (in the absence of solvent), the R_g is proportional to $N^{1/2}$, occupying a smaller overall size than their solvated counterparts. Therefore, by conducting grafting-to deposition of polymers in conditions where their size is minimized can help improve the grafting densities achieved.^{7,8} For this reason a polymer/surface chemistry that would be amenable to grafting-to from the melt was devised, in an attempt to overcome grafting density limitations. Specifically, a surface was modified with amine groups, which were subsequently used to conduct a nucleophilic ring-opening of an epoxide group at the terminus of end-functionalized polymers. This strategy is commonly employed for grafting-to of polymer brushes.^{2,9,10}

Experimental Methods

This approach of preparing amine-coated substrates and subsequently grafting-to with epoxide terminated polymers was discussed previously in Materials and Methods. Briefly, silicon wafers with 100 nm of thermal wet oxide were cleaved into 1.5 x 1.5 cm squared using a diamond scribe. The wafer squares were sonicated for 10 minutes each in acetone then isopropanol, then dried in a nitrogen stream. Then UV-Oxone treatment was conducted for 20 minutes to remove residual organic films and to generate silanols at the oxide surface. The samples were then immersed in a solution of 1 % APTES in dry toluene overnight (at least 16 hours), rinsed with toluene, and dried with nitrogen. The APTES-coated samples were then annealed at 110 °C under vacuum for 30 minutes. Once annealed the APTES-functionalized substrates were then employed for grafting-to.

Epoxide-terminated polyethers were synthesized starting with hydroxyl-terminated polymers. The polymers were deprotonated using NaH and subsequently reacted with epichlorohydrin in THF under air-free conditions. After functionalization and work up, the dried polymers were melted onto the amine-coated substrates under elevated temperature (usually 80 °C). These surfaces were then rinsed with solvent and dried with nitrogen.

Fouling was conducted on these polymer-coated substrates using fluorescently-labelled bovine serum albumin (FITC-BSA). FITC-BSA was prepared at a concentration of 2 mg/mL in pH 7.4 PBS buffer solution. The polymer coated substrates were immersed in the FITC-BSA solution for 2 hours, then rinsed with PBS buffer solution. The samples were inspected under fluorescent microscopy to measure the fluorescence intensity, then the samples were immersed into a PBS solution and heated to 35 °C for 2 hours then the PBS solution was replaced and the solution was held at room temperature for 2 hours. The temperature treated samples were then rinsed with PBS buffer and once again analyzed by

fluorescent microscopy. This temperature cycle was chosen to cause the collapse of the polymer brush at elevated temperature above the LCST, and then rehydration when returned below the LCST. This actuation of the polymer brush should in theory expel adsorbed foulants.

Results and Discussion

The fouling release capability for polymers with an accessible LCST was tested. Drawing from previous work manipulating the thermal behavior of polyethers, a copolymer of EO and DEGE with 40 mol% of DEGE [P(EO₆₀-*co*-DEGE₄₀)] with a molecular weight of 7 kDa was targeted to achieve an LCST slightly above room temperature, around 26 °C. The objective of this study was to measure the fouling release ability of thermally responsive polymers. The behavior of this EO and DEGE copolymer (60/40 composition) was compared to two different molecular weights of PEO (7 and 20 kDa), as well as the bare SiO₂ substrate as shown in Figure 43.

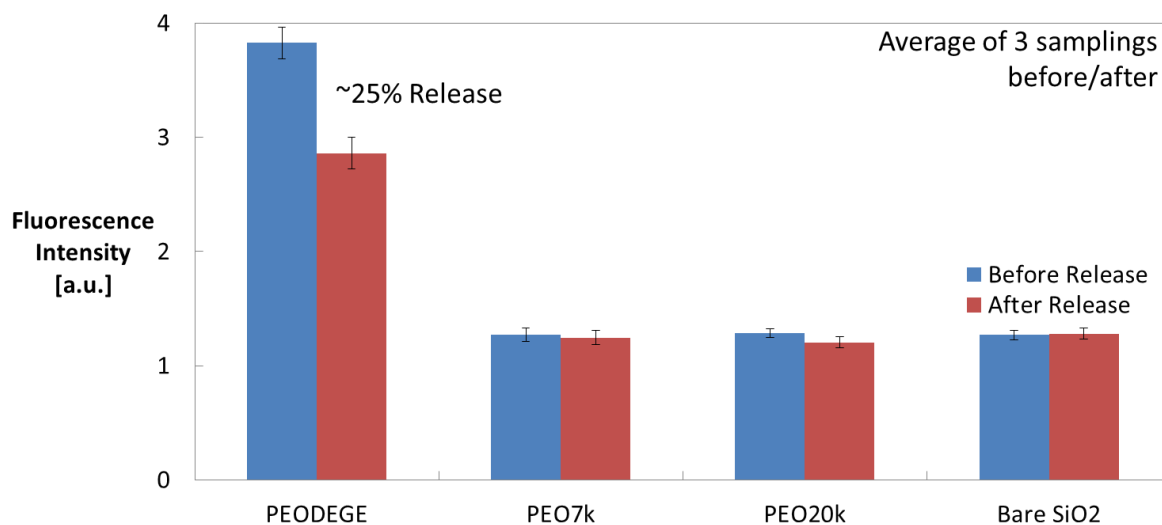


Figure 43: Fouling release data for various surface chemistries prepared by amine-epoxide mediated grafting-to on SiO₂. The PEODEGE sample is P(EO₆₀-co-DEGE₄₀) (40 mol% DEGE), and it is compared to 7k and 20k PEO as well as bare SiO₂. Fluorescence intensity of the FTIC-BSA fouled surfaces was measured by fluorescent microscopy, and the reported values were normalized by the fluorescence of the unfouled SiO₂ substrate. The values shown are the average and standard deviation for 3 replicate runs.

The fluorescence intensities measured were of arbitrary units, so these values were all normalized by the background fluorescence value measured for the bare SiO₂ substrate without any foulants. While the P(EO₆₀-co-DEGE₄₀) sample did show more fouling propensity than the other surfaces, the fouling release step successfully removed around 25 % of the originally deposited foulants. This sort of fouling release was not observed in the other samples. This serves as a conceptual proof that fouling release can be achieved using thermally responsive materials. However, in order to achieve both fouling resistance and fouling release would require further optimization of these materials.

Grafting-To: Telechelic Polymer Loops

Creating surface-tethered polymer “loops” has garnered a considerable amount of interest by both theorists and experimentalists.^{11–16} There are many interesting surface properties that may arise from looped polymer surfaces, with implications for applications such as lubricity and antifouling. In the context of antifouling, loops on a surface may result in a high effective surface coverage at a lower grafting density than could be achieved with linear analogues. This would enable fouling resistance in a synthetically accessible range, overcoming traditional limitations of grafting-to.

Catechols serve as a convenient moiety for grafting-to of polymers, however, they are also susceptible to oxidation, rendering them unable to bind to metal oxides. In the context of telechelic polymers where a catechol is present at both termini, the oxidation of one of the two catechol groups prevents polymer loops from forming on the surface. To overcome this limitation a new telechelic polymer was designed such that each polymer terminus had multiple catechol groups. Moreover, the synthetic strategy enabled these catechol groups to remain protected throughout so that the catechols could be de-protected immediately prior to deposition. Specifically, a triblock copolymer was synthesized with a mid-block composed of PEO, while the two end-blocks were copolymers composed of ethylene oxide (EO) and allyl glycidyl ether (AGE). A 20 kDa P[(EO-co-AGE)-*b*-EO-*b*-(EO-co-AGE)] block copolymer was synthesized and functionalized with protected catechols along the AGE units (Figure 44). In this instance the catechols were protected by tert-butyl dimethyl silane (TBDMS) groups which could subsequently be de-protected in acidic conditions.

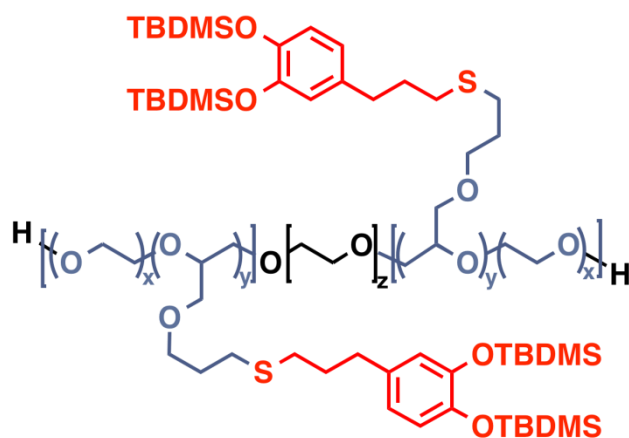


Figure 44: Chemical structure of telechelic P[(EO-co-AGE)-b-EO-b-(EO-co-AGE)] with the AGE units functionalized with TBDMS-protected catechols. The EO-co-AGE end-blocks are highlighted in blue while the attached, protected catechols are shown in red.

The synthesis of these materials has been discussed at length previously;¹⁷ in the interest of brevity only the deposition and fouling experiments will be discussed herein. The catechol groups were de-protected in a pH 2 acetate buffer solution (concentration 100 mg/mL) for 15 minutes, and then aliquots of this solution were diluted to 2 mg/mL using a buffered solution of varying pH. Several pH values between 2 and 12 were targeted in order to explore the pH behavior of these materials during deposition. The diluted solution was then introduced onto Ti-coated QCM-D sensors within the flow chambers, and the deposition was monitored in real-time.

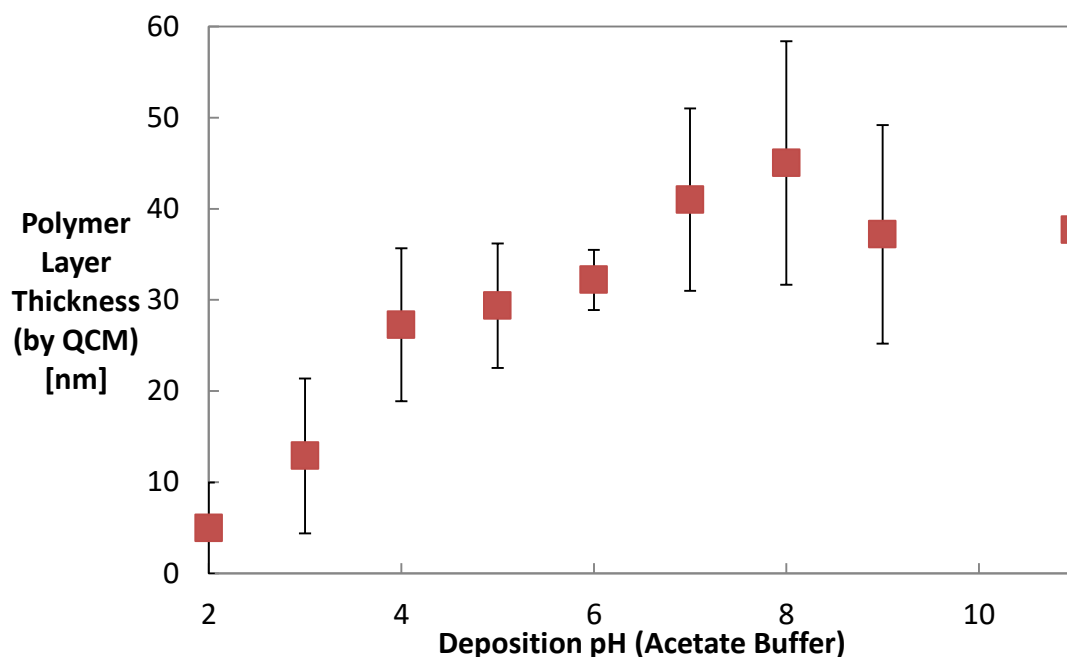


Figure 45: Polymer thickness after deposition onto Ti QCM-D sensors as a function of the pH used during deposition.

The polymer thickness achieved clearly has moderate dependence on the pH of the solution during deposition. Higher pH seems to promote more adsorption of the polymers to the surface, which plateaus above pH 7. This may possibly be due to the accelerated oxidation of catechols at higher pH, which may contribute to crosslinking events. Due to the multi-functional telechelic polyethers used, it is possible for cooperative binding, aided by crosslinking of catecholic moieties, to contribute to and improve surface deposition.

After the deposition was completed, the sensor surfaces were rinsed with pH 7.4 PBS buffer. Then the polymer-coated Ti sensors were submitted to fouling using BSA in a pH 7.4 PBS buffer (2 mg/mL).

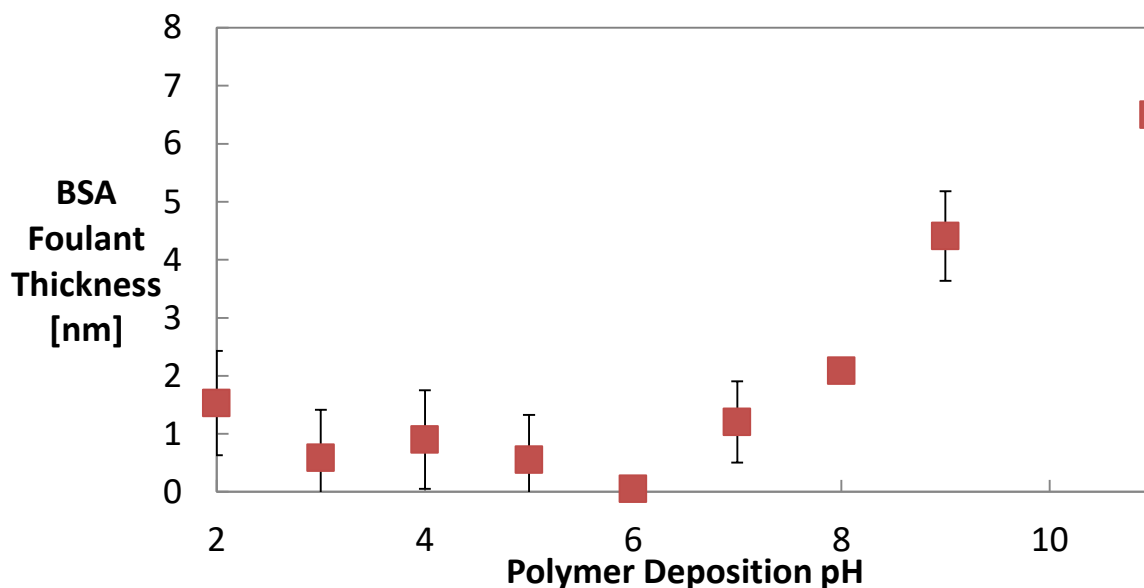


Figure 46: Fouling propensity as measured by thickness of BSA protein adsorbed onto polymer-coated substrates. The x-axis depicts the pH at which the polymers were deposited; the fouling experiments were conducted at pH 7.4 for all surfaces.

For these fouling experiments, viscoelastic modelling was employed to determine thickness of the BSA film that accumulated on the surface. To provide perspective, the BSA protein has an ellipsoidal shape with dimensions 14 x 4 x 4 nm.¹⁸ Therefore, according to this data, most of the polymer samples tested exhibited less than a full monolayer of adsorbed proteins. Specifically, the films deposited between pH 3-6 displayed the best fouling resistance. The conclusion therefore is that while more polymer is deposited using higher pH, the thicker films do not translate to better fouling resistance, but rather the opposite. There appears to be a sweet-spot around pH 5-6 where polymer deposition is still relatively high, without suffering on fouling resistance. However, it is clear that thickness alone does not fully describe the fouling trends observed, this may likely be due to complex nature of the catechol adhesive groups and the multi-functional nature of the telechelics used herein. Additional experiments would be necessary to further probe the conformation of

these surface-tethered polymers. Specifically, gleaning information on the distribution of distances of the polymer chains ends from the underlying substrate may give some insight into what portion of these telechelic polymer are actually in the “loop” conformation.

Supporting Information

QCM-D Validation:

In order to initially validate the reproducibility of protein adsorption conducted in-situ by QCM-D several test experiments were conducted. This consisted of measuring the frequency response of an unmodified SiO₂ QCM-D sensor when exposed to a solution of bovine serum albumin (BSA). For these experiments the QCM-D sensors were mounted in the flow module of the instrument, and the liquid chambers were filled with pH 7.4 PBS buffer. The frequency was allowed to stabilize over the course of approximately 10 minutes, at which point a solution of 2 mg/mL of BSA in PBS buffer was introduced into the flow chamber. The solution remained in contact with the sensor for over an hour to ensure that the frequency response again stabilized prior to rinsing the flow chamber again with PBS buffer. Four replicate experiments conducted with this identical procedure are plotted in Figure S47.

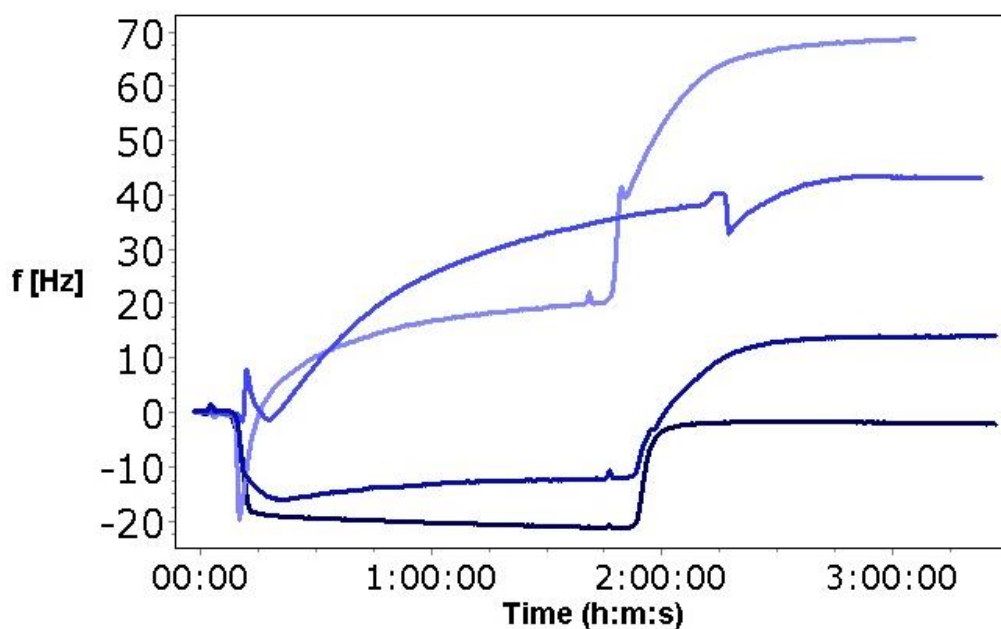


Figure S47: Frequency response of four identical depositions of BSA protein on SiO₂ QCM-D sensors. The vastly different frequency profiles for each run are indicative of the irreproducibility of this method.

Despite conducting these experiments under the exactly same conditions for four replicate runs, the frequency response were vastly different each time they were conducted. This sort of discrepancy was not expected and is indicative that this methodology is inherently irreproducible.

Validation of protein fouling in QCM-D

In order to improve the reproducibility of this technique, QCM-D sensors were mounted in the flow chambers, filled with PBS buffer solution, and left there for 48 hours, while monitoring the frequency. After the frequency stabilized after 48 hours fouling experiments

were conducted with BSA and fibronectin proteins.

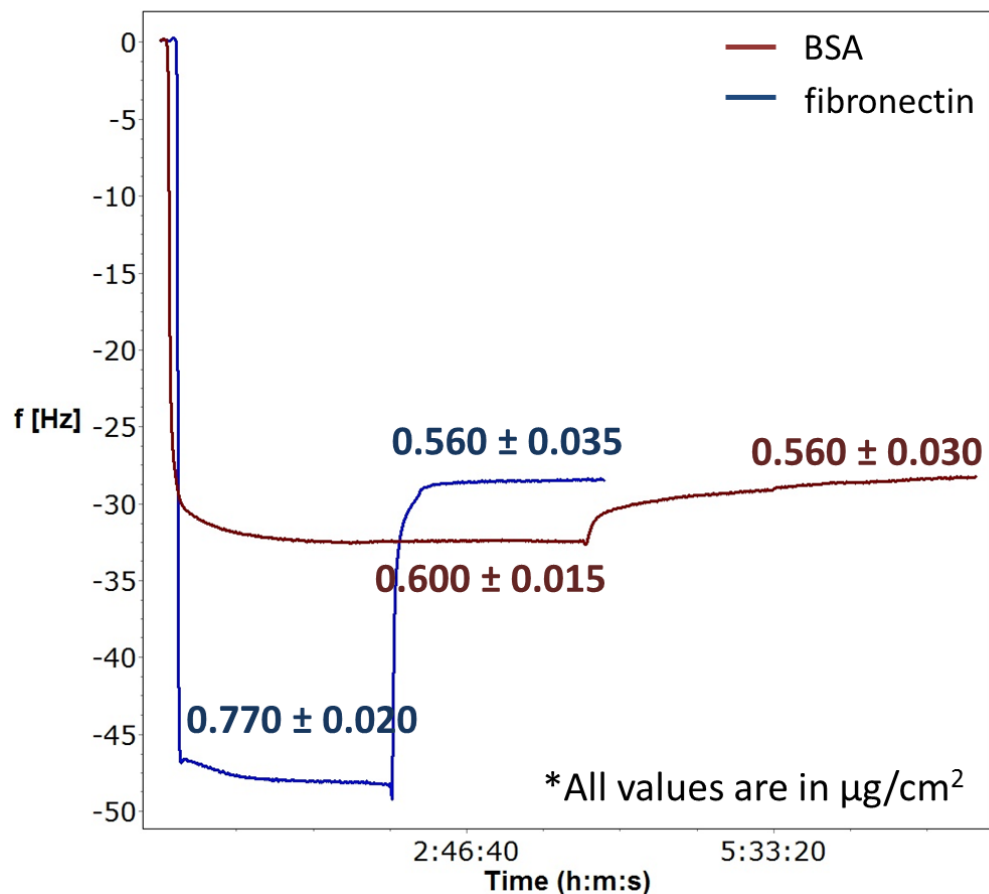


Figure S48: Frequency curves measured for SiO₂ QCM-D sensors for the third harmonic.

The frequency responses for BSA and fibronectin fouling experiments did not experience the same “drift” noticed in Figure S47, and more importantly, the mass of foulants matched nicely.

Table S5: Measured QCM-D values for protein fouling validated against previously reported values.

QCM-D values	BSA [$\mu\text{g}/\text{cm}^2$]	fibronectin [$\mu\text{g}/\text{cm}^2$]
Calculated Values	0.560 ± 0.030	0.560 ± 0.035
Literature Values*	0.600 ± 0.020	0.600 ± 0.020
* <i>Biomaterials</i> , 2006 , 27, 4856-4862.		

Cleaning of QCM-D sensors:

One important facet to conducting QCM-D experiments is the ability to reuse QCM-D sensors, particularly due to their relatively high cost. The most utilized sensors in these studies are the SiO_2 sensors, therefore a cleaning procedure was tested on these sensors. SiO_2 sensors were cleaned and coated with PDA using the usual procedure described in the Materials and Methods section. Then these PDA-coated sensors were cleaned by UV-Ozone (UVO) treatment for 10 minutes, followed by rinsing with deionized H_2O and rinsing with N_2 gas. The QCM-D sensor was then analyzed by x-ray photoelectron spectroscopy (XPS) before and after functionalization with PDA, as well as after cleaning. The XPS spectra are shown in Figure S49.

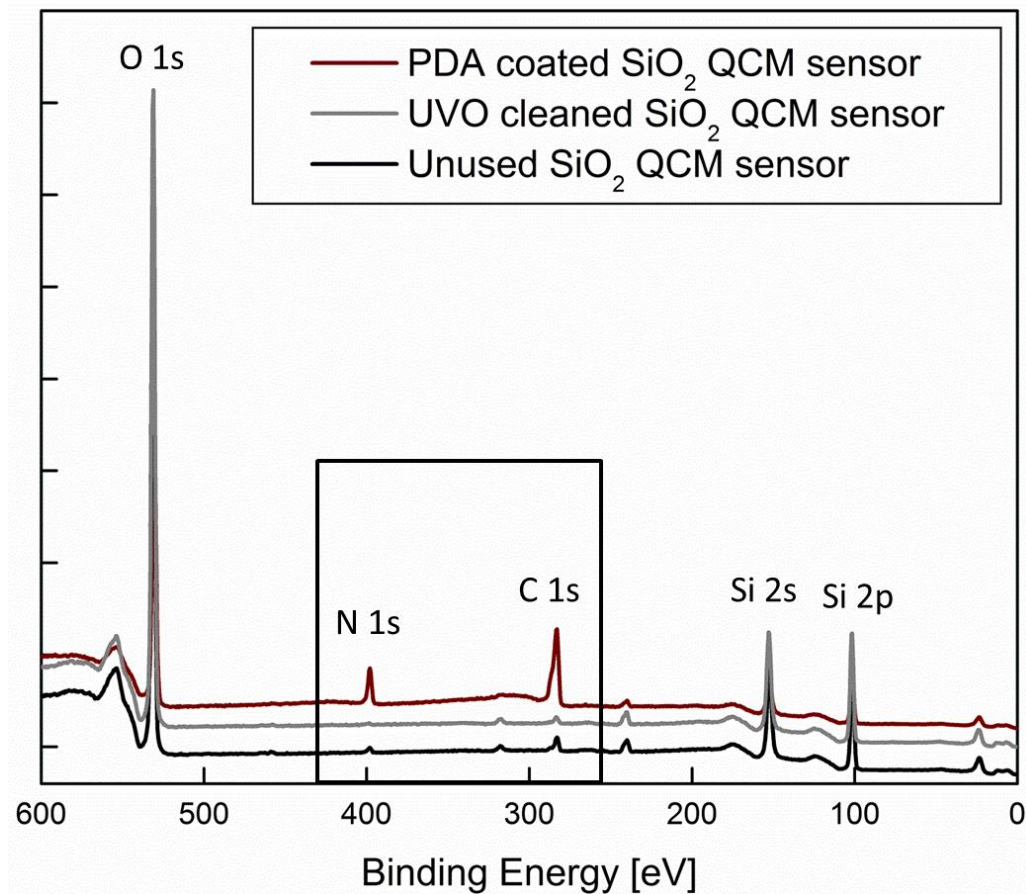


Figure S49: XPS spectra for SiO₂ QCM-D sensors before and after PDA-coating, compared to after cleaning with UVO treatment. The characteristic peaks for O 1s, N 1s, C1s, Si 2s and Si 2p are labelled accordingly. The carbon and nitrogen peaks are unique to the PDA layer, outlined by the black rectangle.

The peaks for carbon and nitrogen are highlighted as they are unique to the PDA layer and are used to validate the cleaning procedure. The bare SiO₂ sensor shows small amounts of adventitious carbon and nitrogen which are unavoidable. After coating with PDA the carbon and nitrogen signals increase accordingly, indicating successful functionalization. However, after cleaning by UVO treatment these peaks decrease, suggesting successful

removal of the PDA layer. After all the QCM-D experiments within this thesis, the used QCM-D sensors were cleaning using this UVO cleaning procedure.

PDA coating on organic substrates:

Polydopamine (PDA) is especially versatile due to its ability to deposit onto a variety of surfaces.¹ To verify the ability of PDA to deposit onto different surfaces, polymer substrates of Nylon and Teflon were tested. Small samples of Nylon and Teflon approximately 2 x 2 cm in size were cut from larger sheets. The samples were cleaned by rinsing both sides with isopropanol, and drying with N₂ gas. The polymer samples were then immersed in a 2 mg/mL solution of dopamine in pH 8.5 tris buffer solution.

Surface	Contact Angle (advancing) [°]	
Nylon	86 ± 1	
Nylon + PDA	64 ± 1	
Teflon	105 ± 1	
Teflon + PDA	61 ± 1	

Figure S50: Contact angle measurements of organic surfaces before and after coating with PDA.

Both Nylon and Teflon originally had high contact angles, suggesting hydrophobicity, however, after coating with PDA, the contact angle decreased to a more hydrophilic value, an indication that the surface chemistry has been altered. This is consistent with previously reported PDA-coating of organic substrates.¹

Determining Optimal Conditions for Zeta Potential Measurements:

Zeta potential measurements were conducted using a Zetasizer Nano ZS (Malvern, Worcestershire, UK). The measurements are conducted by dynamic light scattering (DLS), which also measures the particle size. However, this method is very sensitive to the concentration of particles used for analysis. Therefore the size measurements were used as a metric by which to evaluate appropriate concentrations for characterization. For this initial screening, bare SiO₂ nanoparticles with an approximate size of 140 nm dispersed in a pH 7.4 PBS buffer solution were employed. The results of the DLS measurements are shown in Figure S51.

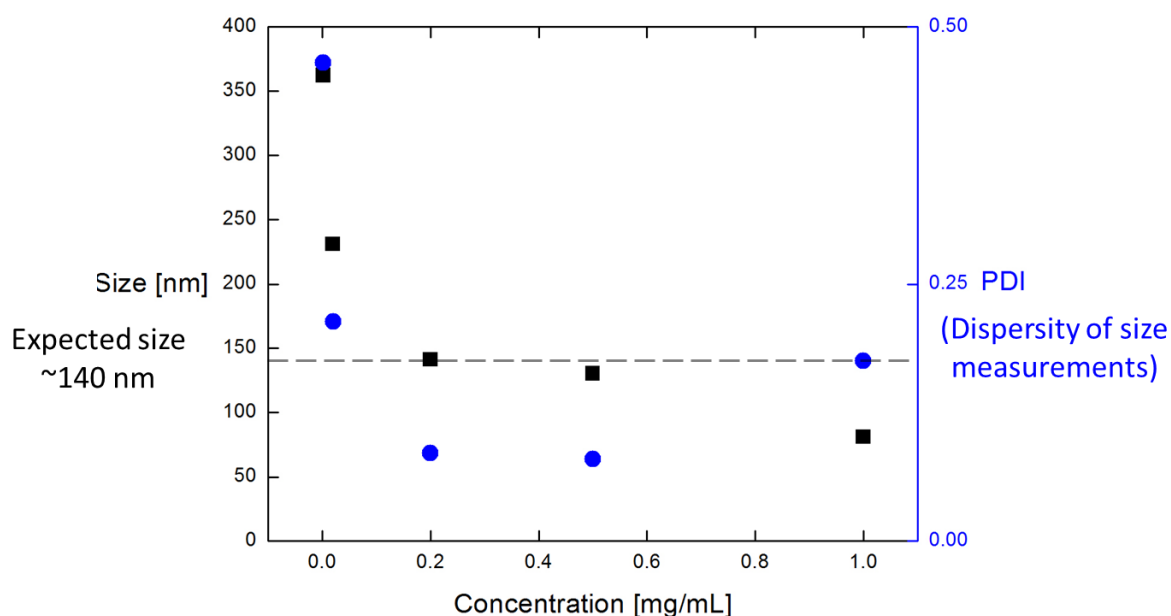


Figure S51: Size of SiO₂ nanoparticles measured by dynamic light scattering for various concentrations. The PDI (the dispersity of various replicate measurements, shown in blue) is an indication of how reproducible and reliable the measurement was.

Each size measurement was conducted five times to get an idea for the reproducibility of the DLS measurement. To validate the technique the target was to achieve the expected size

for the nanoparticles (about 140 nm), while minimizing the PDI. Of the concentrations tested, 0.2 mg/mL yielded a particle size that was commensurate with the expected value, and a relatively low dispersity. All subsequent experiments therefore used a concentration of 0.2 mg/mL in order to ensure accurate and reproducible measurements.